



Computational Stability of Cubical Homology

Orthogonal Controls for Fault Isolation in Retinal Diagnostics

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Abstract

Persistent homology is classically stable, yet its behaviour on grid-sampled medical images can appear unexpectedly brittle under acquisition shift. This thesis studies that discrepancy in the cubical setting and shows that topological instability under image acquisition can be decomposed into two physically distinct sources: mechanical perturbation from spatial misalignment and radiometric perturbation from sensor corruption.

Modelling acquisition as a sequential “Aim-and-Shoot” process yields a computable bound in which the mechanical term scales with spatial displacement and the discrete image gradient, while the radiometric term is uniformly controlled under standardised capture. This explains why cubical persistent homology can remain stable under quantisation and exposure drift yet fail sharply under interpolation-heavy transformations such as off-grid rotation, scaling, and blur.

The framework is tested on retinal fundus images using a constrained topological vector against a geometric baseline under a structured sensitivity audit. The resulting failure asymmetry is summarised by a Differential Failure Matrix, which uses the divergence between geometric and topological probes to diagnose whether degradation is primarily mechanical or radiometric.

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Introduction

Automated medical imaging depends not only on the quality of the classifier, but on the stability of the acquisition process itself. In retinal imaging, a diagnostic signal must first be spatially aligned with the sensor and only then radiometrically captured. This thesis studies how that two-stage physical process affects cubical persistent homology on sampled images.

The central claim is that the apparent instability of persistent homology in digital imaging is not a failure of the underlying stability theory, but a consequence of sampled acquisition. By modelling acquisition as a sequential “Aim-and-Shoot” process, the total topological error can be decomposed into a mechanical term, arising from spatial misalignment and interpolation, and a radiometric term, arising from sensor corruption. Under standardised capture, the radiometric contribution is bounded a priori, leaving mechanical perturbation as the dominant source of instability.

Rather than treating geometric and topological descriptors as competing methodologies, this thesis uses them as complementary probes of acquisition failure. Conventional image analysis relies on geometric features, such as moments and spatial integrals, which are often stable under rigid motion but brittle under radiometric drift. By contrast, topology measures global structural invariants via homology and, in our setting, exhibits a complementary sensitivity profile. This divergence is later formalised as the **Differential Failure Matrix**, in which geometry acts as a **Mechanical Control** and topology as a **Radiometric Control**.

The empirical study is designed to make that contrast visible rather than to maximise predictive performance. We use a structured sensitivity audit on the FIVES dataset [1], comparing a geometric baseline with a deliberately coarse topological probe. The point is to test a narrow question: whether relative connectivity survives acquisition degradations even when spatially integrated summaries fail.



Figure 1: Diabetic retinopathy (DR) as motivating case study. (Left) Normal vision. (Right) Vision affected by DR. The pathology is associated with structural retinal abnormalities that make fundus images a natural testbed for cubical filtrations [2, 3].

Diabetic retinopathy provides the validating case study. The International Diabetes Federation estimates that the global population living with diabetes will increase to 700 million by 2045 [4], and DR remains a leading cause of preventable vision loss, typically manifesting as retinal structural anomalies [5]. Retinal fundus images contain dark lesions, bright lesions, and vascular loops, making them a natural setting for cubical filtrations in dimensions 0 and 1. Prior work established the usefulness of sublevel filtration features for dark lesions [3]; here we extend that setting by incorporating superlevel filtrations to capture bright lesions explicitly.

Chapters 1–3 provide the mathematical runway: cubical homology, persistent homology, and the computational stability bound on sampled images. Chapters 4–6 turn that theory into the reliability-engineering framework, the experimental audit, and the resulting diagnostic interpretation. The appendices collect supplementary algebraic details and full numerical tables supporting the main argument.

Chapter 1

Cubical Homology

This chapter fixes the discrete topological language used throughout the thesis. Rather than reintroducing algebraic topology in full generality, it defines the cubical objects that later interact with sampled acquisition: pixels as cubes, filtrations induced by intensity, and homology in dimensions 0 and 1. Retinal fundus images provide the motivating examples because the clinically relevant lesions already appear as connected components, bright regions, and vascular loops [6]. Figure 1.1 gives the running visual context for the constructions that follow.

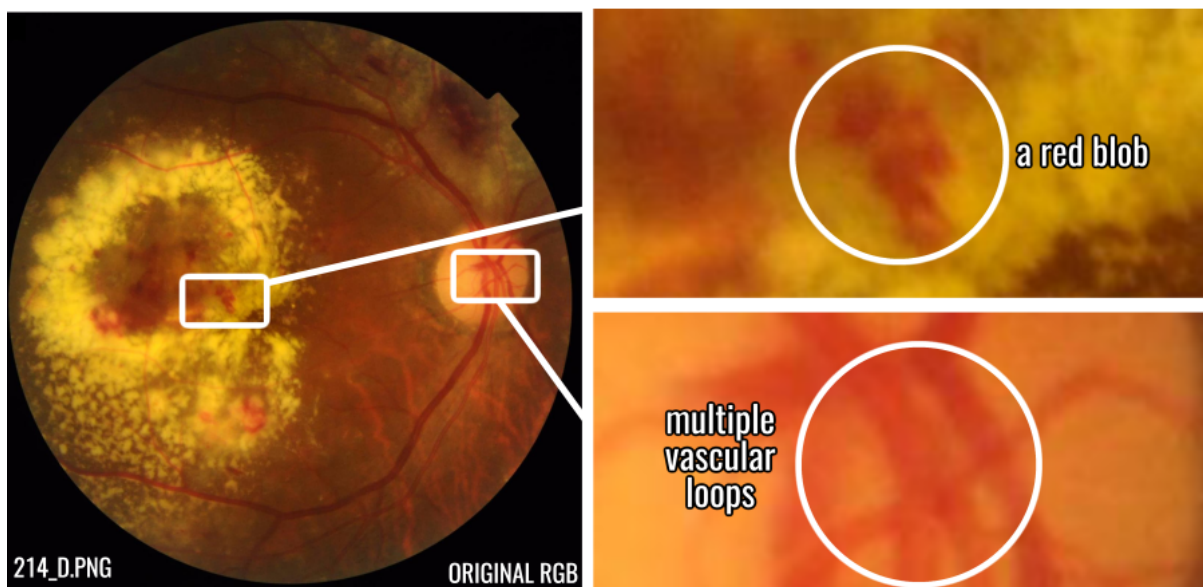


Figure 1.1: Original RGB fundus photo 214_D.png from the FIVES training set [1], with zoomed regions highlighting red blobs (topological components) and vascular loops (topological loops).

Figure 1.2 provides a schematic comparison of a normal and diabetic retinal fundus, whilst Table 1.1 classifies these pathologies by their strictly topological analogues.

Homology turns clinically legible shape cues into algebraic summaries of connectivity and voids. In the retinal setting, blob-like lesions and loop-like vessel patterns become features in dimensions 0 and 1, yielding numerical descriptors that can later be tracked across filtrations and used in the diagnostic pipeline. This chapter builds that discrete topological language rigorously from the cubical image model.

Remark 1.1 Throughout this thesis, we employ the fundus image 214_D.png from the FIVES training set [1] (shown in Figure 1.1) as a running example.

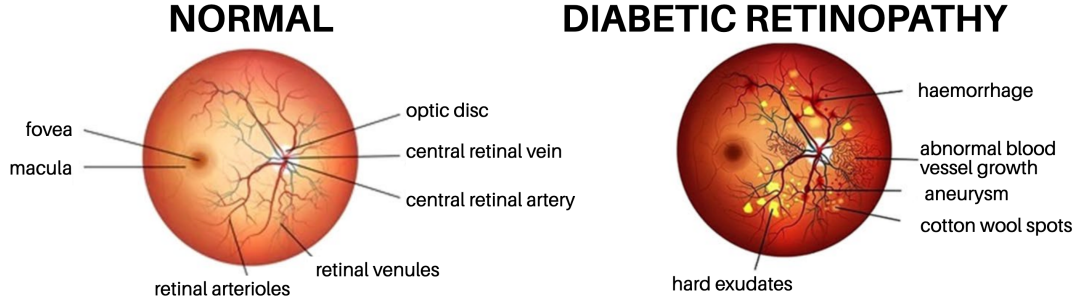


Figure 1.2: Schematic comparison of healthy (left) and diabetic (right) retinal anatomy [7]. Pathologies correspond to topological invariants: haemorrhages and exudates form connected components ($\mathbf{H}_0/\mathbf{H}_S$), while abnormal vascular growth forms 1-dimensional cycles ($\mathbf{H}_1$).

Lesion type	Brief description (visual cue)
<i>"Blob-like" findings (captured by $\mathbf{H}_0/\mathbf{H}_S$: connected components)</i>	
Microaneurysms	Tiny, round, well-defined dark dots
Intraretinal haemorrhages	Local dark patches (dot-blot or flame-shaped)
Hard exudates	Bright, waxy, well-circumscribed spots or plaques
Cotton-wool spots	Pale, fluffy patches with soft borders
<i>"Loop-like" vessel patterns (captured by $\mathbf{H}_1$: cycles)</i>	
Vessel tortuosity	Abnormally winding vessels
Intraretinal microvascular abnormalities (IRMA)	Fine, tortuous channels bridging vessels
Neovascularisation	Lace-like networks of new vessels

Table 1.1: Classification of DR pathologies by topological shape cue. Lesions are grouped by whether they manifest as components ($\mathbf{H}_0, \mathbf{H}_S$) or loops ($\mathbf{H}_1$). This correspondence justifies the restriction of our analysis to dimensions 0 and 1 [8].

1.1 Metric Topology

Lemma 1.2 (RIGID BODY ISOMETRY). *Let $\psi : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a rigid body transformation defined by $\psi(x) = Rx + \mathbf{t}$, where R is an orthogonal matrix and $\mathbf{t} \in \mathbb{R}^2$. Then ψ is an isometry with respect to the ℓ_2 -metric.*

Proof. Here, $d_2(\psi(x), \psi(y)) = \|(Rx + \mathbf{t}) - (Ry + \mathbf{t})\|_2 = \|R(x - y)\|_2$. Since R is orthogonal, it preserves the Euclidean norm. $\square$

Metric-induced topology [9]. A **topological space** $(X, \mathcal{T})$ is a set X paired with a collection $\mathcal{T}$ of subsets of X satisfying the following conditions:

- $X, \emptyset \in \mathcal{T}$
- $U \cap V \in \mathcal{T}$ for any $U, V \in \mathcal{T}$
- $\bigcup_{S \in \mathcal{W}} S \in \mathcal{T}$ for any $\mathcal{W} \subseteq \mathcal{T}$

The collection $\mathcal{T}$ is a **topology** on X , and its elements are called **open sets**.

Topological spaces are a generalisation of metric spaces. We restrict our attention to a specific type: the collection $\mathcal{T}_X$ of all open sets in (X, d_X) is called a **metric-induced topology**, and the pair $(X, \mathcal{T}_X)$ is a topological space.

Remark 1.3 We view each metric space X within its topological context: equipped with the metric-induced topology $\mathcal{T}_X$. We write id_X to denote the identity function $X \rightarrow X$. If $X \subseteq Y$, the **inclusion** $X \hookrightarrow Y$ follows the identity map $x \mapsto x$, differing from id_X solely by its codomain Y .

Homotopy [10]. Let $X, Y \subseteq \mathbb{R}^n$. Continuous functions $f, g : X \rightarrow Y$ are **homotopic** (written $f \sim g$) if there exists a continuous map $H : X \times [0, 1] \rightarrow Y$ such that for any $x \in X$, $H(x, 0) = f(x)$ and $H(x, 1) = g(x)$. Here, we say that f and g belong to the same **homotopy class**. Figure 1.3 illustrates the concept.

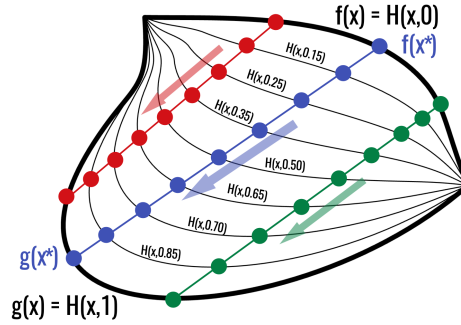


Figure 1.3: Homotopy between functions f and g . For a fixed $x^* \in X$, the path $t \mapsto H(x^*, t)$ traces a curve in Y from $f(x^*)$ to $g(x^*)$. The collection of these paths forms a continuous deformation (a "fabric") transforming f into g without tearing [11].

Indeed, the relation $\sim$ defined by $f \sim g$ is an equivalence relation which partitions the function space into homotopy classes. Intuitively, a homotopy represents a continuous deformation of one map into another. We extend this notion to define the similarity of shapes.

Homotopy Equivalence [10]. Topological spaces X and Y are **homotopy equivalent** (written $X \simeq Y$) if there exist maps $f : X \rightarrow Y$ and $g : Y \rightarrow X$ such that:

- $g \circ f \sim \text{id}_X$ (homotopic to the identity on X),
- $f \circ g \sim \text{id}_Y$ (homotopic to the identity on Y).

Here, we say such X and Y are of the same **homotopy type**, and that f and g are **homotopy inverses**.

Homotopy equivalence captures the "rubber-sheet" geometry of algebraic topology: two spaces are equivalent if one can be deformed into the other. Figure 1.4 illustrates this concept.



Figure 1.4: Glyphs of the letter "A". Despite variations in thickness, orientation, and style, the topological signature (one connected component, one hole) remains invariant.

Similar to homotopy, homotopy equivalence is also an *equivalence relation* on the set of topological spaces. Spaces belonging to the same homotopy equivalence are said to be of the same **homotopy type**. Properties shared by spaces within the same homotopy type are called **homotopy invariants** [12].

Proposition 1.4. Let $X, Y \subseteq \mathbb{R}^n$. If a mapping $f : X \rightarrow Y$ is a rigid body transformation (a surjective isometry), then X and Y are homotopy equivalent.

Proof. A rigid body transformation $f(x) = Rx + t$ is bijective and continuous, with a continuous inverse $f^{-1}(y) = R^T(y - t)$. Consequently, f is a homeomorphism. To show homotopy equivalence, we observe the compositions $f^{-1} \circ f = \text{id}_X$ and $f \circ f^{-1} = \text{id}_Y$. Since the compositions are strictly equal to the identity maps, they are trivially homotopic to the identity. Thus, $X \simeq Y$. $\square$

Homeomorphism vs Homotopy Equivalence. A **homeomorphism** requires continuous strict inverses ($f \circ g = \text{id}_Y$ and $g \circ f = \text{id}_X$), establishing a rigid topological equivalence that preserves properties such as dimension. While every homeomorphism constitutes a homotopy equivalence [12], the converse is false.

Restricting analysis to homotopy invariants prevents over-stratification. While homeomorphism strictly separates multi-pixel vascular loops from single-pixel curves, homotopy equivalence pools these variations, yielding structural signatures independent of local morphological thickness [3, 8].

1.2 Cubes and Filtrations

An **elementary interval** $I \subseteq \mathbb{R}$ is a closed interval of the form $I = [k, k+1]$ (a *non-degenerate* interval) or $I = [k, k]$ (a *degenerate* interval), for some $k \in \mathbb{Z}$. For brevity, we write $I = [k]$ to denote the degenerate interval $I = [k, k]$ [13]. An **elementary cube** κ in $X \subseteq \mathbb{R}^n$ is a finite product of elementary intervals

$$\kappa = I_1 \times \cdots \times I_n \subseteq X.$$

The *dimension* of κ , denoted $\dim \kappa$, is the number of non-degenerate factors in the product [13]. If κ has k non-degenerate factors, then we say that it is a ***k-cube***, a cube of dimension k .

We define the structural relationships between cubes via inclusion. A cube γ is a **face** of κ , denoted $\gamma \preceq \kappa$, if $\gamma \subseteq \kappa$. Conversely, κ is a **coface** of γ . The **codimension** of γ in κ is defined as $\dim \kappa - \dim \gamma$ [13]. If γ has codimension 1 as a face of κ , we say that γ is a **facet** of κ [14]. Figure 1.5 illustrates a 2-cube and its faces.

If the intervals of a k -cube $\kappa = I_1 \times \cdots \times I_k$ are all non-degenerate then we say that κ is a **non-degenerate cube**. We obtain a face of a κ by restricting an interval $I_i = [a_i, b_i]$ to one of its endpoints. The **upper i -th facet** of κ , denoted $\delta_i^+ \kappa$, is obtained by fixing the i -th interval at its maximum, whilst the **lower i -th facet** of κ , denoted $\delta_i^- \kappa$, fixes the i -th interval at its minimum:

$$\delta_i^+ \kappa = I_1 \times \cdots \times [\max I_i] \times \cdots \times I_n \quad \delta_i^- \kappa = I_1 \times \cdots \times [\min I_i] \times \cdots \times I_n$$

Figure 1.5 shows a 2-cube $\kappa_* := [0, 1] \times [0, 1]$, whose facets are

$$\delta_1^+ \kappa_* = [1] \times [0, 1] \quad \delta_1^- \kappa_* = [0] \times [0, 1] \quad \delta_2^+ \kappa_* = [0, 1] \times [1] \quad \delta_2^- \kappa_* = [0, 1] \times [0].$$

Cubical Complexes and Subcomplexes [15] A **cubical complex**¹ K in a bounded set $X \subseteq \mathbb{R}^n$ is a set of cubes in X satisfying the *face closure* condition, which requires that for any cube $\kappa \in K$:

$$\phi \preceq \kappa \implies \phi \in K.$$

The *face closure condition* ensures that the boundary of every cube is within the complex. A **subcomplex** L of K is a subset of K with face closure. Figure 1.6 shows examples of cubical complexes and subcomplexes. We write K_i for the set of i -cubes in K .

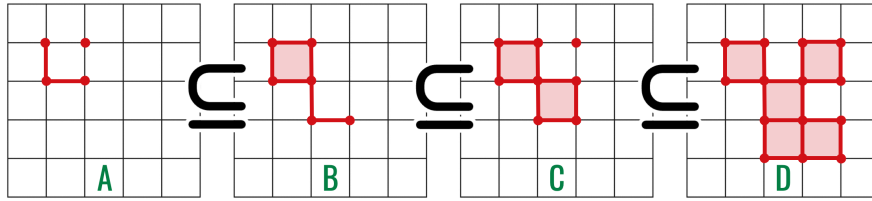


Figure 1.6: A nested sequence of cubical complexes $A \subset B \subset C \subset D$.

Cubical Map [13]. Let K and L be cubical complexes. A function $f : K \rightarrow L$ is a **cubical map** if it maps vertices to vertices and preserves face closure. In other words, $f(K_0) \subseteq L_0$ and for any $\sigma \in K$,

$$\gamma \preceq \sigma \implies f(\gamma) \preceq f(\sigma).$$

In Figure 1.6, such cubical maps are merely a sequence inclusion maps $A \hookrightarrow B \hookrightarrow C \hookrightarrow D$.

While standard texts prioritise simplicial homology, we follow Kaczynski et al., who posit that cubical complexes are not merely approximations of image data; rather, *they are the image itself* [13]. A digital image

¹We use Athanasiadis's definition of cubical complexes [15]. This definition is equivalent to the definition of *cubical sets* in Kaczynski et al [13], but aligns more closely to the approach of Nanda [10]. Kaczynski calls these structures cubical sets

consists of intensity values mapped onto a regular grid. Within this framework, a pixel at (x, y) is treated not as a dimensionless point, but as the geometric realisation of a closed elementary 2-cube encoding colour information. To constitute a valid cubical complex, this collection of pixels is augmented with the necessary lower-dimensional faces (edges and vertices).

The Digital Image. A **digital image** is formally defined as a pair $(\Omega, \mathcal{I})$ where $\Omega \subset \mathbb{Z}^2$ is a finite set of pixel coordinates and $\mathcal{I} : \Omega \rightarrow \mathbb{R}$ is a bounded intensity function. Each coordinate $z \in \Omega$ is associated with the elementary 2-cube κ_z centred at z . The **ambient cubical complex** $K(\Omega)$ is defined as

$$K(\Omega) := \{ \gamma \subset \mathbb{R}^n \mid \exists z \in \Omega \text{ such that } \gamma \preceq \kappa_z \}.$$

We identify the intensity of the top-dimensional 2-cube κ_z strictly with the pixel value $\mathcal{I}(z)$.

We apply this framework to digital fundus images, defined as discrete grids of pixels containing multi-channel data (x, y, R, G, B) . We restrict our analysis exclusively to the green channel, which provides the highest contrast for pathological features [16, 17]. Figure 1.7 demonstrates this contrast.



Figure 1.7: A magnified region of fundus image 214_D.png (FIVES [1]). The original RGB composite highlights yellow exudates and red haemorrhages. The extracted green channel demonstrates enhanced contrast, mapping exudates to high-intensity regions and haemorrhages to low-intensity regions.

Remark 1.5 For $n \in \mathbb{N}$, let $[n] := \{0, 1, \dots, n-1\}$. Because standard digital images operate on fixed bit-depths, the acquired intensity values are strictly finite and integer-valued. Throughout this work, we assume an 8-bit depth, restricting the intensity map to the form $\mathcal{I} : \Omega \rightarrow [\mathcal{M}]$, where the dynamic range is bounded by $\mathcal{M} = 256$.

Intensity Extension. The digital image provides intensity values $\mathcal{I}$ strictly for the top-dimensional 2-cubes (pixels). To construct a valid filtration, $\mathcal{I}$ must be extended to all lower-dimensional faces (edges and vertices). For a sublevel-set filtration, a lower-dimensional face inherits the intensity of its *lowest-value* coface. Let $\gamma \in K(\Omega)$ be a face of dimension $k < 2$. We define the extended intensity $\hat{\mathcal{I}}$ on the ambient complex as:

$$\hat{\mathcal{I}}(\gamma) := \begin{cases} \mathcal{I}(z) & \text{if } \gamma = \kappa_z \text{ (i.e., } \dim \gamma = 2), \\ \min\{\mathcal{I}(z) \mid z \in \Omega, \gamma \prec \kappa_z\} & \text{if } \dim \gamma < 2. \end{cases}$$

This extension guarantees that if a pixel κ_z satisfies the threshold condition $\mathcal{I}(z) \leq t$, all its constituent faces $\gamma \prec \kappa_z$ satisfy $\hat{\mathcal{I}}(\gamma) \leq \mathcal{I}(z) \leq t$. This ensures the face closure condition is strictly maintained at every step of the filtration.

Sublevel-set Filtration [18]. Let $(\Omega, \mathcal{I}^{\mathcal{M}})$ be a digital image with ambient complex $K(\Omega)$. For any threshold $t \in [\mathcal{M}]$, the t -th **sublevel set**, denoted $\mathbf{L}_t$, is defined using the extended intensity function $\hat{\mathcal{I}}^{\mathcal{M}}$:

$$\mathbf{L}_t = \left\{ \kappa \in K(\Omega) : \hat{\mathcal{I}}^{\mathcal{M}}(\kappa) \leq t \right\}.$$

The **sublevel-set filtration** (or cubical lower star filtration) is the nested sequence of subcomplexes:

$$\emptyset \subseteq \mathbf{L}_0 \subseteq \mathbf{L}_1 \subseteq \dots \subseteq \mathbf{L}_{\mathcal{M}} = K(\Omega).$$

Sublevel-set filtrations² naturally prioritise low-intensity pixels, allowing darker regions to enter the complex earlier. In retinas, this highlights features such as haemorrhages and vasculature, capturing them as connected components or loops born at low thresholds. Figure 1.8 (Top) visualises this progression.

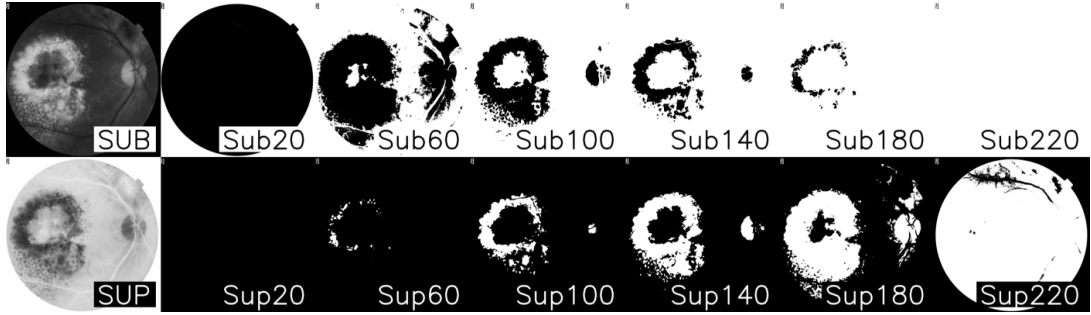


Figure 1.8: Filtration Duality. Filtration progression over the 8-bit intensity range $[0, 255]$ for image 214_D (FIVES [1]). **Top:** Standard sublevel sets. Darker features enter the filtration as the threshold rises. **Bottom:** Sublevel sets of the inverted image ($255 - \text{intensity}$), equivalent to superlevel sets. Brighter features appear early in this sequence.

Superlevel-set Filtrations (Bright Features). Conversely, identifying high-intensity pathologies such as exudates requires *superlevel-set filtrations* (Figure 1.8, Bottom). Topologically, this is the dual of the sublevel construction, equivalent to performing a sublevel filtration on the *inverted* image $\tilde{I}$:

$$\tilde{I}^{\mathcal{M}}(\kappa) = \mathcal{M} - I^{\mathcal{M}}(\kappa).$$

By inverting the pixel intensities, high-intensity regions become the minima, causing them to appear early in the filtration sequence. This enables the complex to detect bright lesions (exudates) and anatomical landmarks (the optic disc) that would otherwise be obscured in a standard sublevel analysis.

Remark 1.6 (NON-DEGENERATE DOMAIN, DEGENERATE IMAGE) We assume the *domain* of our digital images consists exclusively of non-degenerate cubes (pixels with non-zero area). However, we explicitly retain the concept of degenerate cubes in the *codomain* of cubical maps. This allows us to handle geometric collapse rigorously: if a map f projects a 2-cube onto a 1-edge (collapsing a dimension), the image is a degenerate 2-cube.

Remark 1.7 We fix our coefficient field to the field of two elements, denoted $\mathbb{F}_2$. This restriction is standard practice in computational topology (see **Appendix B**). Throughout this thesis, we write $\mathbb{F}$ and $\mathbb{F}_2$ interchangeably to denote the binary field.

1.3 Homology Groups

Distinct features such as disconnected components, loops, and enclosed voids are instances of topological *holes*. We formalise these invariants through **Cubical Homology**.

Definition 1.8 (BOUNDARY OPERATOR [13]). The *boundary operator* is the $\mathbb{F}_2$ -linear map defined by:

$$\partial(\kappa) := \sum_{i=1}^n (\delta_i^- \kappa + \delta_i^+ \kappa) \quad \text{for any cube } \kappa \subseteq \mathbb{R}^n.$$

The action of the boundary operator³ maps a cube to the formal sum of its constituent facets.

²This construction, which segregates features based on intensity thresholding, is also known as a *Lower Star Filtration* [19].

³Definition 1.8 is the restriction to $\mathbb{F}_2$ of Kaczynski et al.'s definition. See **Appendix B** for more information

Lemma 1.9 (FUNDAMENTAL LEMMA OF HOMOLOGY [13, 19]). *For any cube κ , $(\partial \circ \partial)(\kappa) = 0$.*

Proof. Let $\kappa \subseteq \mathbb{R}^n$ be a cube. Applying the boundary operator twice yields:

$$(\partial \circ \partial)(\kappa) = \sum_{i=1}^n \sum_{j=1}^n \sum_{\alpha, \beta \in \{+, -\}} \delta_j^\beta \delta_i^\alpha \kappa.$$

We evaluate this sum in two parts.

Case 1: If $i = j$, the operator δ_i^α fixes the i -th interval to a degenerate point; applying δ_i^β to this already degenerate interval yields the exact same face. Thus, pairs where $i = j$ sum to $2(\delta_i \kappa) = 0$ in $\mathbb{F}_2$.

Case 2: If $i \neq j$, the geometric operations of restricting the i -th and j -th axes are strictly independent and therefore commute: $\delta_j^\beta \delta_i^\alpha \kappa = \delta_i^\alpha \delta_j^\beta \kappa$. Consequently, each unique codimension-2 face τ generated by collapsing two distinct axes appears exactly twice in the summation (once when axis i is collapsed first, and once when axis j is collapsed first). Working over $\mathbb{F}_2$ where $1 + 1 = 0$, these identical pairs cancel. The result immediately follows. $\square$

The Fundamental Lemma of Homology (Lemma 1.9) is the pivotal result that enables the construction of homology, which relies on the formation of quotient vector spaces.

Chain Groups Let K be a cubical complex. The k -th **chain group** $\mathbf{C}_k(K)$, whose elements are k -**chains**, is the $\mathbb{F}$ -vector space⁴ spanned by the k -dimensional cubes of K . The k -th **boundary map** ∂_k is the restriction of the boundary operator ∂ to the k -chains of K , with the subscript indicating the domain dimension, i.e.,

$$\partial_k : \mathbf{C}_k(K) \longrightarrow \mathbf{C}_{k-1}(K) \quad \text{where } \gamma \mapsto \partial \gamma.$$

Chain Complex. [10] A **chain complex** $(\mathbf{V}_\bullet, \phi_\bullet)$ is a sequence of $\mathbb{F}$ -vector spaces $\mathbf{V}_k$ indexed by $k \in \mathbb{N}$, connected by $\mathbb{F}$ -linear maps $\phi_k : \mathbf{V}_k \rightarrow \mathbf{V}_{k-1}$ such that $\phi_{k-1} \circ \phi_k = 0$ for all k .

The boundary operator stratifies a cubical complex K can be conceptualised into $1 + \dim K$ distinct layers, where each layer aggregates cubes of uniform dimension. The construction of the chain complex $(\mathbf{C}_\bullet, \partial_\bullet^K)$ formalises this stratification by isolating these layers into a sequential array of vector spaces. For each dimension $k \in \{0, 1, \dots, \dim K\}$, the chain group $\mathbf{C}_k(K)$ is defined as the vector space of formal sums of k -cubes with coefficients in $\mathbb{F}$. The set of all k -cubes in K forms a natural basis for this space. The boundary map ∂_0 is the zero map by convention. Figure 1.9 illustrates such a stratification.

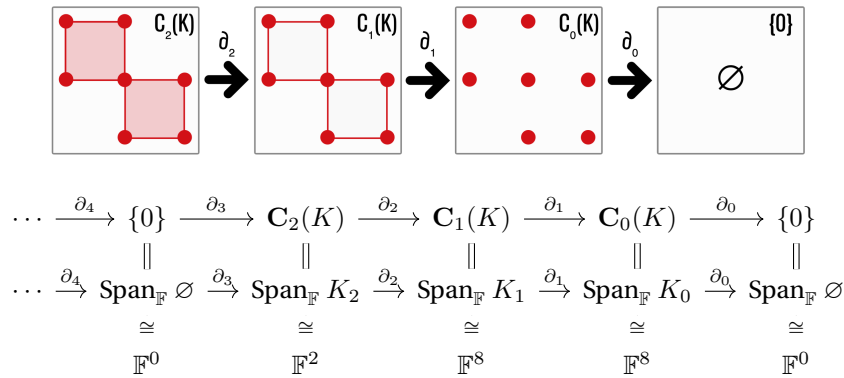


Figure 1.9: Stratification of a cubical complex K into a chain complex $(\mathbf{C}_\bullet, \partial_\bullet^K)$.

We previously distinguished topological features qualitatively by dimension (components, loops) that arise from the absence of "filling": removing an edge breaks a path, while removing a face leaves an empty cycle.

⁴While integral coefficients ($\mathbb{Z}$) generate free abelian groups, working over a field $\mathbb{F}$ yields vector spaces. This choice retains the geometric basis (cubes) while simplifying the algebraic structure, thereby enabling the application of standard linear-algebra algorithms rather than module-theory methods.

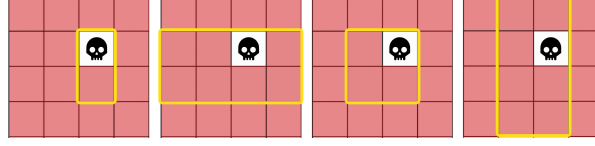


Figure 1.10: Four distinct 1-cycles surrounding the same hole (represented by the central "dead" pixel). These cycles differ in their edges but must be counted as a single feature.

A k -**dimensional hole** exists when a closed k -chain does *not* bound a collection of $(k + 1)$ -cubes. However, a simple enumeration of closed chains is insufficient, as multiple distinct cycles may correspond to the same hole. As illustrated in Figure 1.10, a raw count of cycles (1-holes) overestimates topological complexity.

Cycle subgroups and boundary subgroups [13, 20]. The k -**th cycle subgroup** of $(C_\bullet, \partial_\bullet)$ is the kernel of the k -th boundary map:

$$Z_k(K) = \ker \partial_k = \{\gamma \in C_k(K) \mid \partial_k(\gamma) = 0\}.$$

The k -**th boundary subgroup** of $(C_\bullet, \partial_\bullet)$ is the image of the $(k + 1)$ -th boundary map

$$B_k(K) := \text{Im } \partial_{k+1}.$$

Elements of Z_k are called k -**cycles** whilst elements of B_k are called k -**boundaries**. Geometrically, a cycle is a structure without a boundary (e.g., a closed loop of edges). Z_k enumerates all the possible cycles given the basis of k -cubes, with the possibility that these cycles may correspond to the boundary of some $(k + 1)$ -chain. In this case, the k -cycle is not a k -dimensional hole in K : higher-dimensional cubes fill it. When the context is clear, we write Z_k and B_k .

Proposition 1.10 ([10]). For all $k \in \mathbb{N}$, $B_k(K) \subseteq Z_k(K)$.

Proof. $\gamma \in B_k \Rightarrow (\exists \mu \in C_{k+1}, \gamma = \partial_{k+1}(\mu)) \Rightarrow \partial_k(\gamma) = (\partial_k \circ \partial_{k+1})(\mu) = 0 \Rightarrow \gamma \in \ker \partial_k$. $\square$

Proposition 1.10 guarantees boundaries are cycles (i.e., $\text{Im } \partial_{k+1} \subseteq \ker \partial_k$), allowing the formation of quotient vector spaces. Figure 1.11 illustrates this containment.

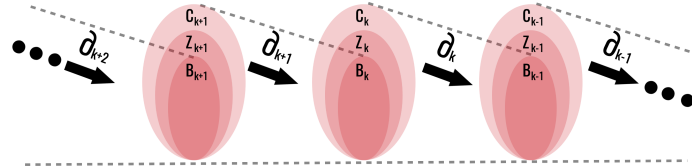


Figure 1.11: Venn diagram of the subspaces. Since $\partial^2 = 0$, the image of ∂_{k+1} (the boundaries B_k) is contained entirely within the kernel of ∂_k (the cycles Z_k) [20].

Homology groups [19]. Since B_k is a subspace of Z_k , we may form the quotient space. This quotient structures the "holes" by identifying cycles that differ only by a boundary. For $k \in \mathbb{N}_0$, the k -**th homology group** $H_k(K)$ is the quotient vector space $H_k(K) := Z_k(K)/B_k(K)$.

Elements of H_k are **homology classes**. For a cycle $z \in Z_k$, its class is denoted $[z] = z + B_k$. A non-zero element in H_k represents a non-trivial cycle—one that is not a boundary. Since the zero element of H_k is the subspace B_k , so it follows $[z] \in H_k$ represents a non-trivial homology class (a "hole") if and only if $[z] \neq 0$ in H_k , if and only if $z \notin B_k$.

The cycle group Z_k enumerates all potential k -dimensional features formed by closed chains of k -cubes. However, this collection includes both "genuine" holes and trivial cycles that merely bound a collection of $(k + 1)$ -cubes. We are interested exclusively in the former. To isolate these topological features, we perform a quotient operation of Z_k by the boundary subgroup B_k . This algebraic process partitions the k -cycles into equivalence classes, where each class represents a distinct topological hole, while simultaneously collapsing all trivial bounding cycles to the zero element. Two cycles are said to be **homologous** if they represent the

same topological feature (Figure 1.12). By elementary properties of quotient vector spaces, two cycles $z_1, z_2 \in Z_k$ are homologous if and only if $z_1 - z_2 \in B_k$.

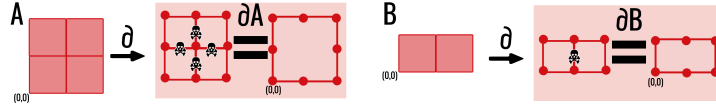


Figure 1.12: The action of the boundary operator on two different 2-chains A and B . Observe how edges (1-cubes) that appear twice cancel each other in the final sum.

We formalise the concept of "homology equivalence class count" via the k -th **Betti Number** β_k , defined as the dimension of the k -th homology group:

$$\beta_k(K) := \dim \mathbf{H}_k(K) = \dim Z_k - \dim B_k [10].$$

Betti numbers provide a discrete inventory of independent topological features: β_0 counts connected components (blobs), β_1 counts loops (vascular loops). Figure 1.13 illustrates Betti numbers.

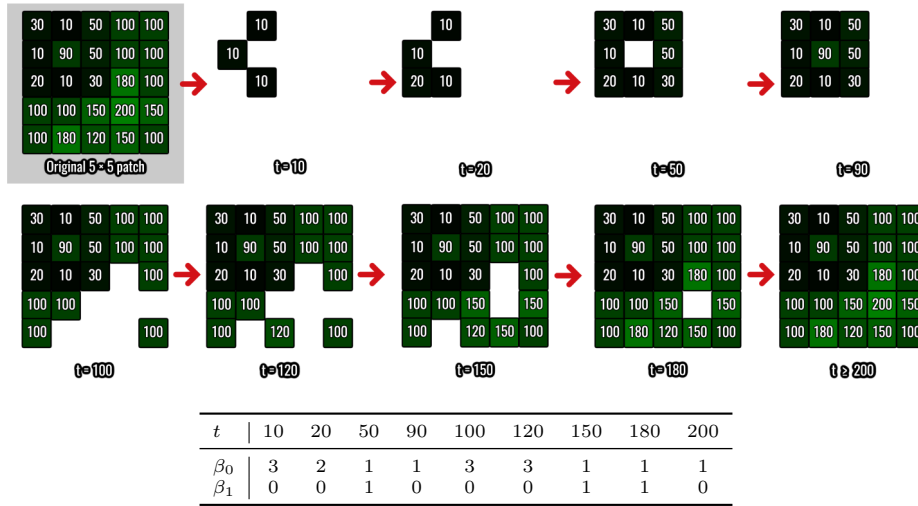


Figure 1.13: A 5×5 pixel patch, labelled by intensity. Sublevel sets are generated at thresholds $t \in \{10, \dots, 200\}$. The table lists the Betti numbers corresponding to the filtration sequence.

Computing these invariants yields a quantitative clinical baseline⁵. The underlying hypothesis posits that topological structure strictly correlates with pathology [3, 8]: comparable disease progression must yield comparable Betti numbers. This relies on the guarantee that Betti numbers are invariant under homotopy equivalence⁶. Table 1.2 details this specific mapping:

Clinical Feature	Geometry	Filtration Type	Invariant
Haemorrhages	Dark Components	Sublevel ($\mathbf{L}_\bullet$)	β_0
Neovascularisation	Vascular Loops	Sublevel ($\mathbf{L}_\bullet$)	β_1
Exudates	Bright Components	Superlevel ($\mathbf{L}_\bullet^*$)	β_0

Table 1.2: Mapping of DR lesions to corresponding geometric and topological invariants.

This static strategy presupposes the existence of an optimal threshold t capable of faithfully isolating the underlying anatomy from ambient noise. In practice, finding a single global threshold is fraught with ambiguity: a threshold isolating a haemorrhage in one region may simultaneously obscure vascular details in another. To resolve this parameter sensitivity, we abandon the search for a single static complex and instead evaluate the topological evolution of the image across *all* possible intensity thresholds. The formalisation of this dynamic framework—Persistent Homology—is the subject of Chapter 2.

⁵For matrix-based computation of Betti numbers, see Tinarrage [21, Chapter 6.3].

⁶Stated without proof in Tinarrage [21, Prop. 5.14]; see Hatcher [22, Chapter 2] for a formal derivation.

Chapter 2

Persistent Homology

The previous chapter defined the static homological invariants of a cubical image. We now pass from a single thresholded complex to a filtered family and ask how connected components and loops evolve across intensity levels. Persistent homology packages that evolution into persistence modules, barcodes, and diagrams; these are the quantitative objects that will later enter the stability analysis.

Remark 2.1 (CONVENTION: INDICES VS. THRESHOLDS) To avoid ambiguity between continuous pixel intensities and discrete algebraic steps, we distinguish **thresholds** ($t \in \mathbb{N}_0$), which refer to specific intensity values (e.g., the sublevel set L_t), from **indices** ($i \in \mathbb{N}_0$), which refer to discrete steps in a filtered sequence (e.g., the term V_i in filtration $\mathbf{F}_\bullet$). Unless stated otherwise, persistence modules are indexed by integers i, j , while filtrations may be defined over real thresholds t .

2.1 Births, Deaths, and Persistence

Persistence Modules and Persistent Homology Groups [10]. A *persistence module* $(V_\bullet, a_\bullet)$ over $\mathbb{F}$ is an $\mathbb{N}_0$ -indexed sequence of vector spaces $V_\bullet$ equipped with linear maps $a_\bullet : V_\bullet \rightarrow V_{\bullet+1}$, which we visualise in Figure 2.1. We may compose these linear maps to form $a_{i \rightarrow j} := a_{j-1} \circ a_{j-2} \circ \dots \circ a_{i+1} \circ a_i$.

$$V_0 \xrightarrow{a_0} V_1 \xrightarrow{a_1} \dots \xrightarrow{a_{k-1}} V_k \xrightarrow{a_k} V_{k+1} \xrightarrow{a_{k+1}} \dots$$

Figure 2.1: The diagram of the persistence module $(V_\bullet, a_\bullet)$ over $\mathbb{F}$.

For integers $i \leq j$, the *persistent homology group* $\mathbf{PH}_{i \rightarrow j}(V_\bullet, a_\bullet)$ of $(V_\bullet, a_\bullet)$ is defined as:

$$\mathbf{PH}_{i \rightarrow j}(V_\bullet, a_\bullet) = \text{im}(a_{i \rightarrow j}) \subseteq V_j \quad \text{with} \quad a_{i \rightarrow i} := \text{id}_{V_i}$$

We abbreviate this to $\mathbf{PH}_{i \rightarrow j}$. Here, $\mathbf{PH}_{i \rightarrow j}$ shows how much of V_i survives under $a_{i \rightarrow j}$.

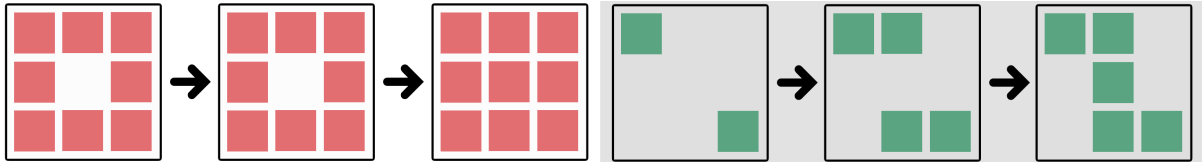


Figure 2.2: Visualisation of Homological Death. **Left pane:** An $\mathbf{H}_1$ feature (red loop) dies trivially as the enclosing cycle is filled. **Right pane:** Two $\mathbf{H}_0$ features (green components) merge.

We analyse the dimensions of these spaces to track topological evolution. If $\dim \mathbf{PH}_{i \rightarrow t} < \dim \mathbf{PH}_t$, new classes have appeared at step t that were absent at i . If $\dim \mathbf{PH}_t > \dim \mathbf{PH}_{t \rightarrow j}$, the map $a_{t \rightarrow j}$ has a non-trivial kernel on the image of V_t , indicating that features existing at t have died (vanished or merged) by step j . Figure 2.2 illustrates births of homological features at the beginning of a three-step filtration and their deaths at the final step.

The Elder Rule [19]. We need to introduce a slight nuance. Consider a class $\gamma \in V_t$. We assign a unique interval $[b, d]$ representing its lifespan. Then, a class is **born** at b if it is not in the image of the immediate predecessor ($\gamma \notin \text{Im } a_{b-1 \rightarrow b}$). Next, to determine its **death**, we track the forward evolution of γ . The class dies at d if d is the first index where γ becomes homologous to a cycle that existed *before* b . We seek the minimal $d > b$ such that:

$$a_{b \rightarrow d}(\gamma) \in \text{Im } a_{b-1 \rightarrow d}.$$

This condition captures both causes of death: the class may become trivial (homologous to zero), or it may merge with a non-trivial class born earlier. In either case, the feature represented by γ ceases to be an independent generator. The interval $[b, d]$ represents the persistence of γ . To derive a unique barcode from a filtration, we must resolve ambiguities when homology classes merge. The **Elder Rule** asserts that when two classes merge, the feature born *earlier* persists, while the younger feature dies (is mapped to the older class). Suppose both features were born at the same index, an arbitrary but fixed ordering is used to break the tie. The Elder Rule ensures that the feature with the deepest roots in the filtration is preserved as the "true" representative of the underlying topology.

Births, Deaths, Persistence [10]. Let $(V_\bullet, a_\bullet)$ be a persistence module.

- A non-zero $v \in V_i$ is **born** at index i if $v \notin \text{Im } a_{i-1 \rightarrow i}$.
- An element $v \in V_i$ born at i **dies** at index j ($j > i$) if j is the first index where v merges with the image of an older space:

$$a_{i \rightarrow j}(v) \in \text{Im } a_{i-1 \rightarrow j} \quad \text{but} \quad a_{i \rightarrow j-1}(v) \notin \text{Im } a_{i-1 \rightarrow j-1}.$$

A vector v has persistence $j - i$ corresponding to the interval $[b, d]$. If v never satisfies the death condition, it persists indefinitely (persistence $+\infty$).

2.2 Interval Modules

Interval modules. [10] The record of births and deaths describes the evolution of Betti numbers. To formalise this record, we decompose persistence modules into their atomic constituents. Let $b < d$ be elements in $\mathbb{Z}_{\geq 0} \cup \{\infty\}$. The **interval module** $I^{b,d}$ is the persistence module $(V_\bullet, \rho_\bullet)$ defined by:

$$V_t = \begin{cases} \mathbb{F} & \text{if } b \leq t < d \\ 0 & \text{otherwise} \end{cases} \quad \text{and} \quad \rho_{t \rightarrow t+1} = \begin{cases} \text{id}_{\mathbb{F}} & \text{if } b \leq t < d-1 \\ 0 & \text{otherwise} \end{cases}$$

Example 2.2. To illustrate the decomposition of a persistence module into its atomic interval modules, consider the filtration $\mathbf{F}_\bullet$ of length 6 shown in Figure 2.3. We track the zeroth persistent homology, which records the evolution of connected components.

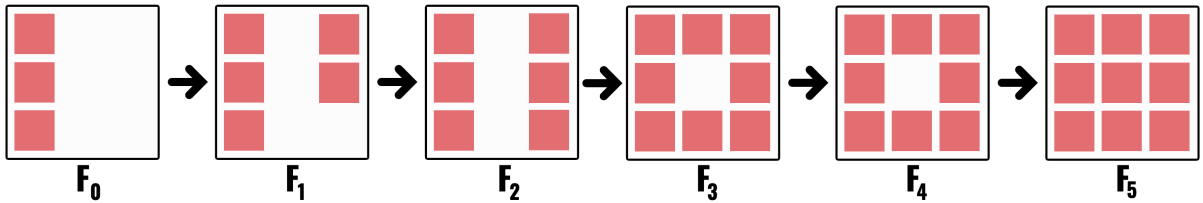


Figure 2.3: A geometric filtration $\mathbf{F}_\bullet$. The zeroth homology $\mathbf{H}_0 \mathbf{F}_\bullet$ records components: a new class appears at $t = 1$ and merges at $t = 3$. The first homology $\mathbf{H}_1 \mathbf{F}_\bullet$ records loops: a cycle appears at $t = 3$ and is filled at $t = 5$.

At $t = 0$, the complex contains a single connected component, represented by the homology class z_1 . At $t = 1$, a disjoint component appears, generating a new class z_2 . These components remain disjoint until $t = 3$, where a connecting edge forms. By the Elder Rule, the younger class z_2 merges into the older class z_1 , marking the death of z_2 . This evolution forms the zeroth persistence module $(\mathbf{H}_0 \mathbf{F}_\bullet, a_\bullet)$, visualised by the sequence of vector spaces and induced linear maps (inclusions) in Figure 2.4:

To track the appearance and disappearance of these independent features, we decompose this module. The vector space dimension drops at $t = 3$ because the transition map a_2 possesses a non-trivial kernel

$$\begin{array}{ccccccccc}
\mathbf{H}_0 \mathbf{F}_0 & \xrightarrow{a_0} & \mathbf{H}_0 \mathbf{F}_1 & \xrightarrow{a_1} & \mathbf{H}_0 \mathbf{F}_2 & \xrightarrow{a_2} & \mathbf{H}_0 \mathbf{F}_3 & \xrightarrow{a_3} & \mathbf{H}_0 \mathbf{F}_4 & \xrightarrow{a_4} & \mathbf{H}_0 \mathbf{F}_5 \\
\parallel & & \parallel & & \parallel & & \parallel & & \parallel & & \parallel \\
\text{Span}_{\mathbb{F}} \{z_1\} & \hookrightarrow & \text{Span}_{\mathbb{F}} \{z_1, z_2\} & \xrightarrow{\text{id}} & \text{Span}_{\mathbb{F}} \{z_1, z_2\} & \xrightarrow{z_2 \mapsto z_1} & \text{Span}_{\mathbb{F}} \{z_1\} & \xrightarrow{\text{id}} & \cdots & \xrightarrow{\text{id}} & \text{Span}_{\mathbb{F}} \{z_1\} \\
\cong & & \cong & & \cong & & \cong & & \cong & & \cong \\
\mathbb{F}^1 & & \mathbb{F}^2 & & \mathbb{F}^2 & & \mathbb{F}^1 & & \mathbb{F}^1 & & \mathbb{F}^1
\end{array}$$

Figure 2.4: Evolution of the zeroth persistence module tracking connected components.

(specifically, $z_1 - z_2 \mapsto 0$). The Structure Theorem guarantees that this interacting sequence can be decoupled into a direct sum of independent interval modules. These correspond to the lifespans of z_1 ($[0, \infty)$) and z_2 ($[1, 3]$), as shown in Figure 2.5:

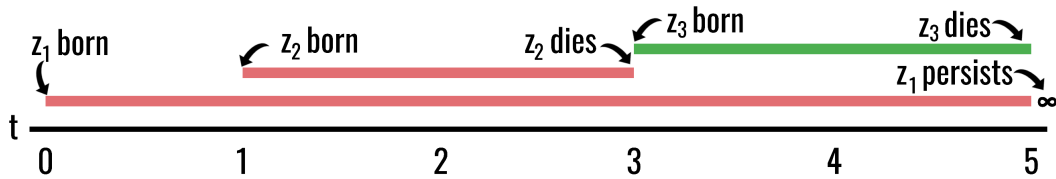
$$\begin{array}{rcl}
I_{\bullet}^{1,3} : & 0 & \longrightarrow \mathbb{F} \xrightarrow{\text{id}} \mathbb{F} \longrightarrow 0 \longrightarrow 0 \longrightarrow 0 \\
& \oplus & \oplus \quad \oplus \quad \oplus \quad \oplus \quad \oplus \\
I_{\bullet}^{0,\infty} : & \mathbb{F} & \xrightarrow{\text{id}} \mathbb{F} \xrightarrow{\text{id}} \mathbb{F} \xrightarrow{\text{id}} \mathbb{F} \xrightarrow{\text{id}} \mathbb{F} \xrightarrow{\text{id}} \mathbb{F} \\
& = & = \quad = \quad = \quad = \quad = \\
\mathbf{H}_0 \mathbf{F}_{\bullet} \cong & \mathbb{F}^1 & \longrightarrow \mathbb{F}^2 \longrightarrow \mathbb{F}^2 \longrightarrow \mathbb{F}^1 \longrightarrow \mathbb{F}^1 \longrightarrow \mathbb{F}^1
\end{array}$$

Figure 2.5: Decomposition of the zeroth persistence module into a direct sum of independent interval modules.

Regarding the first homology module ($\mathbf{H}_1 \mathbf{F}_{\bullet}, a_{\bullet}$), the decomposition is structurally simpler. Exactly one loop (denoted z_3) forms at $t = 3$ and is filled at $t = 5$. Because there are no interacting generators, the persistence module is directly isomorphic to the single interval module $I_{\bullet}^{3,5}$, as expressed in Equation 2.1:

$$0 \longrightarrow 0 \longrightarrow 0 \longrightarrow \text{Span}_{\mathbb{F}} \{z_3\} \xrightarrow{\text{id}} \text{Span}_{\mathbb{F}} \{z_3\} \longrightarrow 0 \quad (2.1)$$

The lifespans of the generators z_1, z_2 , and z_3 can be geometrically summarised as a collection of intervals. A bar spans the exact index range over which the corresponding homology class remains non-trivial. In Figure 2.6, the $\mathbf{H}_0$ generators (z_1, z_2) are represented by red intervals, and the $\mathbf{H}_1$ generator (z_3) by a green interval:

Figure 2.6: Persistence barcode summarising the lifespans of $\mathbf{H}_0$ generators z_1, z_2 (red) and $\mathbf{H}_1$ generator z_3 (green).

2.3 The Structure Theorem

Example 2.2 demonstrates the decomposition of a specific persistence module. Persistence modules derived from fundus images are substantially more complex, as represented by the barcodes in Figure 2.7.

To rigorously quantify topology, we must establish that *every* persistence module admits a decomposition, and that this decomposition yields a unique¹ signature termed a **persistence barcode**. Each horizontal bar within the barcode represents the lifespan of a single interval module. The persistence barcode, alongside

¹This uniqueness is strictly up to isomorphism. An interval module is merely *isomorphic* to the sub-module generated by a specific homology class, not the geometric feature itself. However, this algebraic equivalence captures the exact birth and death indices required to construct the barcode.

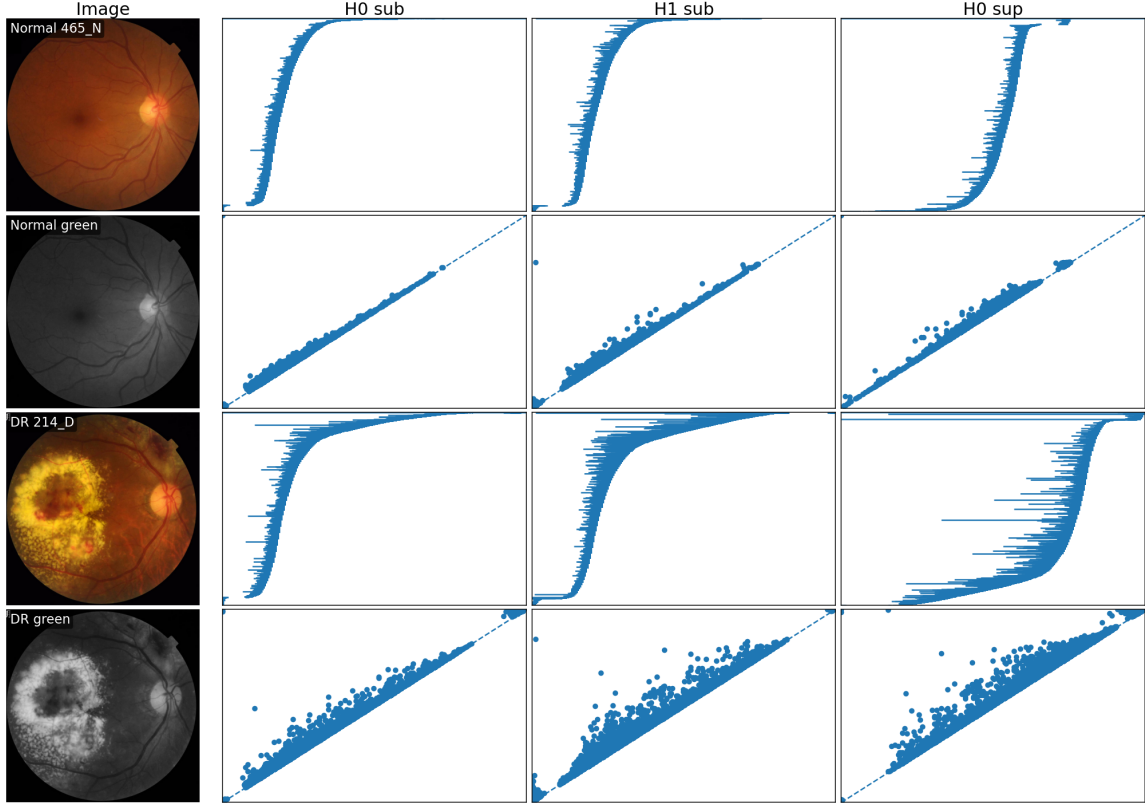


Figure 2.7: Topological signatures across channels and pathology. Comparison of a healthy subject (465_N, rows 1-2) and a diabetic subject (214_D, rows 3-4) from the FIVES dataset [1]. For each subject, analysis is performed separately on the original RGB image (rows 1 and 3) and the extracted green channel (rows 2 and 4). The leftmost column displays the source image. The subsequent three columns visualise specific homological features: the $\mathbf{H}_0$ sublevel, the $\mathbf{H}_1$ sublevel, and the $\mathbf{H}_S$ superlevel (the latter capturing bright lesions such as exudates). Within each topological column, the upper panel presents the persistence barcode, while the lower panel shows the corresponding persistence diagram.

its equivalent spatial representation, the *persistence diagram*, constitutes the primary operational output of the dynamic approach. Figure 2.7 illustrates this target output; the formal definitions of these structures follow subsequently. To mathematically justify the extraction of these signatures, we must establish three structural properties: (1) persistence modules can be decomposed into simple components via direct sums; (2) this decomposition is unique; and (3) the resulting multisets of intervals can be reliably vectorised.

Persistence Module Isomorphisms and Direct Sums. A morphism $\phi_\bullet$ is a *persistence module isomorphism* if each component map ϕ_t is an isomorphism of vector spaces [10]. We write $(V_\bullet, a_\bullet) \cong (W_\bullet, b_\bullet)$ to denote isomorphism. We define the *direct sum* of $(V_\bullet, a_\bullet)$ and $(W_\bullet, b_\bullet)$ as the module $(X_\bullet, c_\bullet) := (V_\bullet \oplus W_\bullet, a_\bullet \oplus b_\bullet)$, as illustrated in Figure 2.8:

$$\begin{array}{ccccccc}
 \cdots & \xrightarrow{c_{t-1}} & X_t & \xrightarrow{c_t} & X_{t+1} & \xrightarrow{c_{t+1}} & \cdots \\
 & & \parallel & & \parallel & & \\
 \cdots & \xrightarrow{a_{t-1} \oplus b_{t-1}} & V_t \oplus W_t & \xrightarrow{a_t \oplus b_t} & V_{t+1} \oplus W_{t+1} & \xrightarrow{a_{t+1} \oplus b_{t+1}} & \cdots
 \end{array}$$

Figure 2.8: The direct sum of persistence modules $(V_\bullet, a_\bullet)$ and $(W_\bullet, b_\bullet)$.

The linear map $c_t = a_t \oplus b_t$ acts component-wise and admits the block matrix representation $c_t = \begin{bmatrix} a_t & 0 \\ 0 & b_t \end{bmatrix}$.

Proposition 2.3 ([10]). Let $(V_\bullet, a_\bullet)$ and $(W_\bullet, b_\bullet)$ be persistence modules. For any indices $i \leq j$:

$$\mathbf{PH}_{i \rightarrow j}((V_\bullet, a_\bullet) \oplus (W_\bullet, b_\bullet)) \cong \mathbf{PH}_{i \rightarrow j}(V_\bullet, a_\bullet) \oplus \mathbf{PH}_{i \rightarrow j}(W_\bullet, b_\bullet).$$

Proof. Let $(X_\bullet, c_\bullet)$ be the direct sum. The transition map $c_{i \rightarrow j}$ is the composition of intermediate maps. By the block diagonal property:

$$c_{i \rightarrow j} = c_{j-1} \circ \dots \circ c_i = \begin{bmatrix} a_{i \rightarrow j} & 0 \\ 0 & b_{i \rightarrow j} \end{bmatrix}.$$

The persistent homology group $\mathbf{PH}_{i \rightarrow j}(X_\bullet)$ is the image of this map. For a block diagonal operator, the image decomposes directly:

$$\text{Im}(c_{i \rightarrow j}) = \text{Im}(a_{i \rightarrow j}) \oplus \text{Im}(b_{i \rightarrow j}).$$

Substituting the definitions of the persistent homology groups yields the result. $\square$

This proposition mathematically justifies the generalised decomposition of persistence modules in Example 2.2. It guarantees that topological features do not interact destructively when combined; the persistent homology of a complex system is strictly the direct sum of its constituent parts. Tracking the evolution of Betti numbers necessitates identifying the birth-death interval of each homology class. We view the sequence of k -th homology groups of a filtration $\mathbf{F}_\bullet$ as the sequence $(\mathbf{H}_k \mathbf{F}_\bullet, \mathbf{H}_k \iota_\bullet)$, shown in Figure 2.9:

$$\mathbf{H}_k \mathbf{F}_1 \xrightarrow{\mathbf{H}_k \iota_1} \dots \longrightarrow \mathbf{H}_k \mathbf{F}_t \xrightarrow{\mathbf{H}_k \iota_t} \mathbf{H}_k \mathbf{F}_{t+1} \xrightarrow{\mathbf{H}_k \iota_{t+1}} \dots \longrightarrow \mathbf{H}_k \mathbf{F}_m$$

Figure 2.9: The sequence of k -th homology groups induced by the filtration $\mathbf{F}_\bullet$.

Multiple homology classes appear and vanish throughout this sequence. To isolate the lifespan of individual classes, we seek a decomposition of the form:

$$(\mathbf{H}_k \mathbf{F}_\bullet, \mathbf{H}_k \iota_\bullet) \cong \bigoplus_{[b,d]} (I_\bullet^{b,d}, \varphi_\bullet^{b,d})^{\oplus \mu(b,d)}$$

where $\mu(b, d)$ denotes the multiplicity of the interval $[b, d)$. Here, $(I_\bullet^{b,d}, \varphi_\bullet^{b,d})$ represents the **interval module** supported on $[b, d)$. It captures a single topological feature that appears at step b and persists until step d , as visualised in Figure 2.10:

$$\begin{array}{ccccccccccccccc} \dots & \longrightarrow & I_{b-1} & \xrightarrow{\varphi_{b-1}} & I_b & \xrightarrow{\varphi_b} & \dots & \longrightarrow & I_d & \xrightarrow{\varphi_d} & I_{d+1} & \xrightarrow{\varphi_{d+1}} & \dots \\ & & \cong \downarrow & & \cong \downarrow & & & & \cong \downarrow & & \cong \downarrow & & \\ \dots & \xrightarrow{0} & 0 & \xrightarrow{0} & \mathbb{F} & \xrightarrow{1} & \dots & \xrightarrow{1} & \mathbb{F} & \xrightarrow{0} & 0 & \xrightarrow{0} & \dots \end{array}$$

Figure 2.10: Structure of the interval module $(I_\bullet^{b,d}, \varphi_\bullet^{b,d})$ supported on $[b, d)$.

A non-zero persistence module $(I_\bullet, \varphi_\bullet)$ is **indecomposable** if it cannot be written as a direct sum of non-trivial modules. That is, if $(I_\bullet, \varphi_\bullet) \cong (V_\bullet, a_\bullet) \oplus (W_\bullet, b_\bullet)$, then either $(V_\bullet, a_\bullet) \cong 0$ or $(W_\bullet, b_\bullet) \cong 0$.

We formalise the birth-death persistence module as follows.

Proposition 2.4 ([10]). Given $0 \leq b \leq d \leq \infty$, a nonzero persistence module $(I_\bullet, \varphi_\bullet)$ is indecomposable if

$$\dim I_t = \mathbb{I}(b \leq t < d) \quad \text{and} \quad \text{rank}(\varphi_t : I_t \rightarrow I_{t+1}) = \mathbb{I}(b \leq t < d).$$

Proof. Suppose $(I_\bullet, \varphi_\bullet) \cong (V_\bullet, a_\bullet) \oplus (W_\bullet, b_\bullet)$. Since rank is additive over direct sums:

$$\text{rank } a_t + \text{rank } b_t = \text{rank}(a_t \oplus b_t) = \text{rank } \varphi_t = \mathbb{I}(b \leq t < d).$$

We observe that

- For $t \notin [b, d]$, the total rank is zero, implying $V_t \cong W_t \cong 0$.
- For $t \in [b, d]$, the sum of ranks is 1.

Since $\dim I_t = 1$, we must have $\dim V_t + \dim W_t = 1$. This forces a partition: at any index t , exactly one summand is non-zero (dimension 1). Assume for contradiction that neither summand is the zero module. Let t^* be an index where a switch occurs, such that $V_{t^*} \cong \mathbb{F}$ but $V_{t^*+1} \cong 0$. Since $V_{t^*} \neq 0$, we must have $W_{t^*} = 0$. Then, $a_{t^*} : V_{t^*} \rightarrow V_{t^*+1}$ and $b_{t^*} : W_{t^*} \rightarrow W_{t^*+1}$ are both rank 0 maps. This yields

$$\text{rank}(a_{t^*} \oplus b_{t^*}) = 0,$$

contradicting the hypothesis that the rank is 1 for $b \leq t^* < d$. $\square$

Proposition 2.4 establishes interval modules as the atomic constituents of persistence modules. Consequently, we must demonstrate that every persistence module decomposes into a combination of these structures.

Theorem 2.5 (STRUCTURE THEOREM [23]). *Let $(V_\bullet, a_\bullet)$ be a persistence module with $\dim(V_t) < \infty$ for all t . There exists a multiset of intervals $\mathcal{B}$, unique up to isomorphism, such that:*

$$(V_\bullet, a_\bullet) \cong \bigoplus_{[b,d]} (I_\bullet^{b,d})^{\oplus \mu(b,d)} \quad \text{where } \mu(b,d) \text{ denotes the multiplicity of the interval } [b,d].$$

Proof Sketch. By imposing a compatible ordering, the filtration is serialised such that exactly one elementary cube is added at each step. This guarantees that every transition triggers exactly one topological event (a birth or a death), allowing the unique multiset of intervals to be constructed by strict induction. For the complete derivation, refer to **Appendix D**. $\square$

The Structure Theorem allows us to condense persistence modules—which detail the evolution of Betti numbers through births and deaths—into a concise multiset of intervals termed the barcode.

Definition 2.6 (BARCODE [10]). Let $(V_\bullet, a_\bullet)$ be a persistence module. The *barcode*, denoted $\text{Bar}(V_\bullet, a_\bullet)$, is the unique multiset of intervals $\mathcal{B}$ defined by the decomposition in Theorem 2.5.

The barcode constitutes the *topological numerical signature*: a computable fingerprint encoding the evolution of homological features.

Example 2.7 (Barcodes and the Elder Rule). We illustrate the construction of a persistence barcode using a synthetic 3×3 dataset, as shown in Figure 2.11.

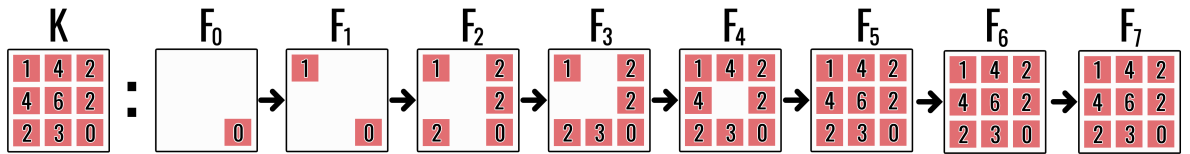


Figure 2.11: A synthetic 3×3 digital image and its induced sublevel-set filtration $\mathbf{F}_\bullet$.

The filtration in Figure 2.11 becomes stationary for $t \geq 7$. We track $\mathbf{H}_0$ features, labelling generators a, b, c by birth order:

- Components a, b , and c are born at $t = 0, 1, 2$, respectively.
- At $t = 3$, a and c merge; by the Elder Rule, the older component a persists whilst c dies, yielding the interval $c : [2, 3)$.
- At $t = 4$, a and b merge. As a is older than b , a persists; b dies, yielding $b : [1, 4)$.

The resulting multiset $\text{Bar}(\bullet) = \{[0, \infty), [1, 4), [2, 3)\}$ constitutes the barcode for the 0-th homology of K , as illustrated in Figure 2.12.

Persistence Diagram [19] Given a barcode $\text{Bar}(V_\bullet, a_\bullet) = \{[b_i, d_i)\}_{i \in I}$, its *persistence diagram* $\mathcal{D}(V_\bullet, a_\bullet)$ is the multiset of points in the extended real plane:

$$\mathcal{D}(V_\bullet, a_\bullet) = \{(b_i, d_i) \mid i \in I\} \cup \Delta, \quad \text{where } \Delta = \{(x, x) \mid x \in \mathbb{R}_{\geq 0}\}.$$

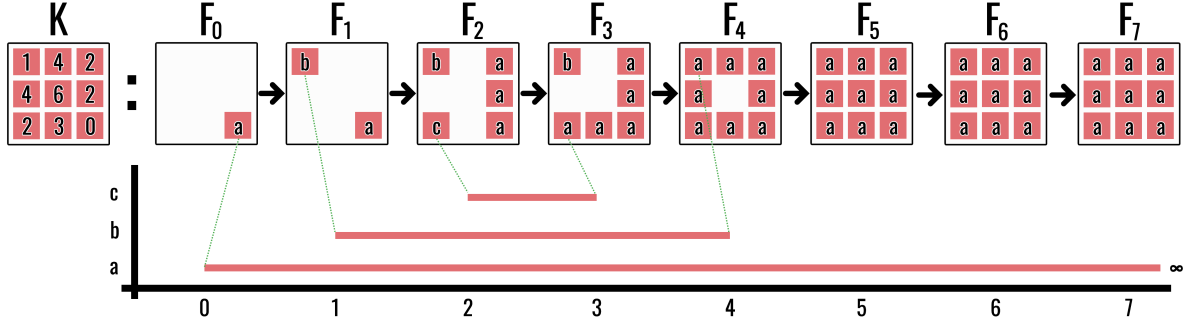


Figure 2.12: The persistence barcode for the 0-th homology of the synthetic complex, demonstrating the application of the Elder Rule.

While intuitive for small datasets, barcodes become visually uninterpretable for high-resolution fundus images due to the sheer volume of homological features. Figure 2.7 resolves this by applying persistence diagrams to clinical data. Compared to the normal retina (465_N.png), the pathological diagram (214_D.png) exhibits a higher density of points away from the main diagonal, directly capturing the persistent H_0 and H_1 lesions characteristic of diabetic retinopathy.

Remark 2.8 Given the natural bijection $(b_i, d_i) \leftrightarrow [b_i, d_i]$, we use $\mathcal{D}$ and Bar to denote the same mathematical object.

Because these signatures encode the underlying retinal anatomy rather than mere digital artefacts, they must function as robust diagnostic biomarkers. This necessitates a formal guarantee of **stability**: minor variations in the physical acquisition process must yield commensurately small perturbations in the resulting barcode. To mathematically quantify this stability and rigorously bound the operational limits of the diagnostic pipeline (precisely identifying where the sensor succeeds and where it fails), we must first establish a formal distance metric on persistence modules.

2.4 Discrete Interleaving

We now formalise the notion of distance between persistence modules, which quantifies how barcodes fluctuate under distortions. Figure 2.13 demonstrates that while the resulting barcodes differ numerically, they are structurally equivalent: if we shift the indices of the baseline barcode one unit to the right, we recover the perturbed barcode exactly. This implies a persistence module isomorphism induced by mapping the i -th subcomplex of the baseline filtration to the $(i + 1)$ -th subcomplex of the perturbed filtration. However, our standard framework restricts filtrations to a fixed range $0 \leq n \leq m$ (where $m = 255$).

Remark 2.9 A simple index shift could push features beyond the maximum intensity, as digital images possess a finite intensity range, typically $0 \leq t \leq 255$. However, the stability analysis in this section requires comparing filtrations that are shifted relative to one another. Hence, we adopt the standard convention that all filtrations and persistence modules are **constantly extended** to infinity.

Discrete interleaving [10]. We say that two persistence modules $(V_\bullet, a_\bullet)$ and $(W_\bullet, b_\bullet)$ are k -**interleaved** (for $k \in \mathbb{N}_{\geq 0}$) if there exist families of linear maps $\phi_\bullet : V_\bullet \rightarrow W_{\bullet+k}$ and $\psi_\bullet : W_\bullet \rightarrow V_{\bullet+k}$ such that for all $t \in \mathbb{N}_0$, the following conditions (**M1**, **M2**, **T1**, and **T2**) hold:

Parallelogram Relations		Triangle Relations	
M1:	$b_{t+k} \circ \phi_t = \phi_{t+1} \circ a_t$	T1:	$\psi_{t+k} \circ \phi_t = a_{t \rightarrow t+2k}$
M2:	$a_{t+k} \circ \psi_t = \psi_{t+1} \circ b_t$	T2:	$\phi_{t+k} \circ \psi_t = b_{t \rightarrow t+2k}$

Parallelogram Relations. The maps ϕ and ψ must commute with the internal structure maps of $(V_\bullet, a_\bullet)$ and $(W_\bullet, b_\bullet)$ (Figure 2.14 and Figure 2.15).

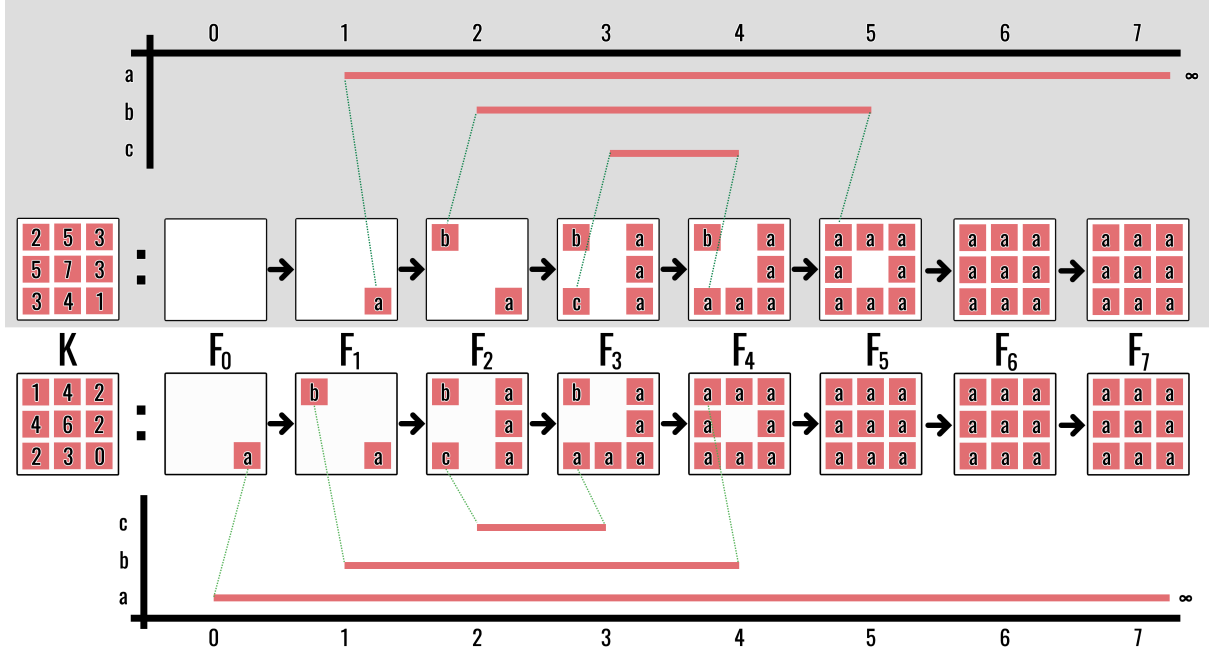


Figure 2.13: Invariance under uniform intensity shift. The baseline image $\mathcal{I}$ (bottom barcode, from Example 2.7) is perturbed by a $+1$ intensity increase to simulate lighting variation, yielding $\mathcal{I}'$ (top barcode). While all birth and death indices shift right by exactly 1, the topological feature counts and *persistence* (bar lengths) remain strictly invariant.

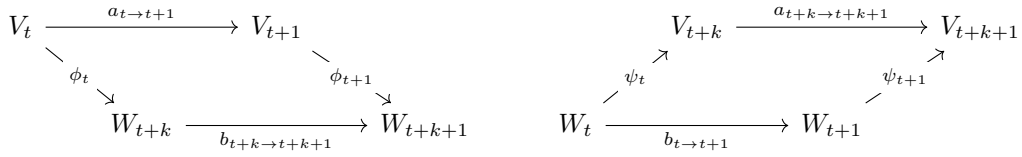


Figure 2.14: Commutative diagrams illustrating the Parallelogram Relations for a k -interleaving.

Triangle Relations. The compositions $\psi \circ \phi$ and $\phi \circ \psi$ must recover the standard internal shift maps of $(V_\bullet, a_\bullet)$ and $(W_\bullet, b_\bullet)$ respectively, encompassing a total shift of $2k$, as illustrated in Figure 2.16.

The triangle relation **T1** functions as a rigorous "persistence conservation" test: it necessitates that the composition of maps sending a feature on a round trip $(V \rightarrow W \rightarrow V)$ be algebraically indistinguishable from retaining it within the original module $(V \rightarrow V)$ for the corresponding duration [10, Section 6.4]. The triangle relation **T1** enforces persistence conservation. The composite map $(V \rightarrow W \rightarrow V)$ must equal the internal structure map $(V \rightarrow V)$ over a $2k$ shift. This structural constraint strictly precludes the mapping of persistent topological features to transient noise.

Triangle Relations effectively prevent the mapping of a significant topological structure to noise. Should one attempt to map a persistent feature in V to a transient "gap" or noise feature in W , the round trip breaks: the feature "falls out" of the bottom comb (as the noise decays) and cannot climb back up to V_{t+2k} . This

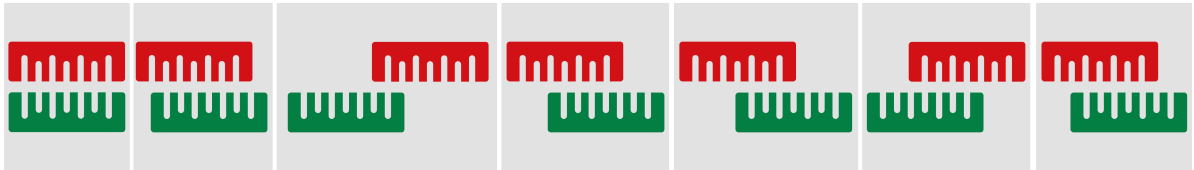


Figure 2.15: Visualisation of Parallelogram Relations as a shift operation. We represent the persistence modules $(V_\bullet, a_\bullet)$ and $(W_\bullet, b_\bullet)$ as discrete combs (red and green, respectively), where the teeth represent the vector spaces and the spine represents the internal morphisms. The algebraic condition **M1** is geometrically equivalent to mechanically shifting the bottom (green) comb to the left by k units, vertically aligning W_{t+k} with V_t . Conversely, condition **M2** corresponds to a symmetric leftward shift of the top (red) comb.

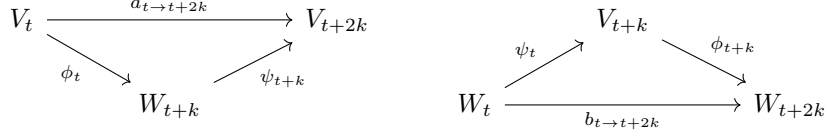


Figure 2.16: Commutative diagrams illustrating the Triangle Relations (T1 and T2) for a k -interleaving.

contradiction ($0 \neq$ surviving feature) strictly enforces that significant topological signals must correspond to other significant signals, permitting only sufficiently short-lived features to map to trivial noise. The symmetric triangle relation **T2** imposes the analogous constraint in the reverse direction. Figure 2.17 shows a 1-interleaving of the persistence modules in Figure 2.13.

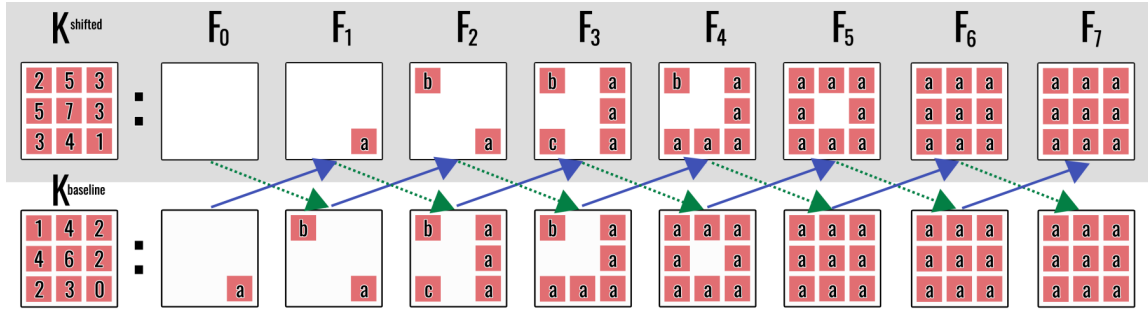


Figure 2.17: In Figure 2.13, the baseline filtration was perturbed by a uniform intensity shift of 1. The resulting persistence modules are 1-interleaved. The arrows (colour-coded by map family) demonstrate that the algebraic structure of the original module is recoverable from the perturbed module within a shift of $k = 1$.

Interleavings establish the mechanism for matching two persistence modules at a specific scale.

Interleaving Distance [10] Let $(V_\bullet, a_\bullet)$ and $(W_\bullet, b_\bullet)$ be persistence modules. The *interleaving distance* between $(V_\bullet, a_\bullet)$ and $(W_\bullet, b_\bullet)$ is defined as:

$$d_I((V_\bullet, a_\bullet), (W_\bullet, b_\bullet)) = \inf\{k \in \mathbb{N}_{\geq 0} \mid (V_\bullet, a_\bullet) \text{ and } (W_\bullet, b_\bullet) \text{ are } k\text{-interleaved}\}.$$

If no such k exists, we set the distance to ∞ .

The interleaving distance provides a measure of the *dissimilarity between two persistence modules*². Note that we explicitly allow for an infinite distance. Triangle Relations (**T1** and **T2**) impose strict structural limitations on valid mappings; consequently, if no k -interleaving exists for any finite k , the interleaving distance is defined as ∞ . We clarify the notion of interleaving distance through the following corollary to the Cubical Structure Theorem.

Corollary 2.10. *Let $V_\bullet$ and $W_\bullet$ be persistence modules. The following are equivalent:*

- (i) $V_\bullet$ and $W_\bullet$ have a 0-interleaving.
- (ii) $V_\bullet$ and $W_\bullet$ are isomorphic.
- (iii) $V_\bullet$ and $W_\bullet$ have the same barcode.
- (iv) $V_\bullet$ and $W_\bullet$ have isomorphic persistent homology groups for all $i \leq j$.

Proof. Let the interval decompositions be $V_\bullet \cong \bigoplus (I_\bullet^{b,d})^{\oplus \mu(b,d)}$ and $W_\bullet \cong \bigoplus (I_\bullet^{b',d'})^{\oplus \mu(b',d')}$.

(i $\Rightarrow$ ii) Assume a 0-interleaving exists via $\phi_\bullet, \psi_\bullet$. By Triangle Relations, for any t , $\psi_t \circ \phi_t = a_{t \rightarrow t} = \text{id}_{V_t}$ and $\phi_t \circ \psi_t = b_{t \rightarrow t} = \text{id}_{W_t}$. Thus, ϕ_t and ψ_t are strict inverses, implying $V_\bullet \cong W_\bullet$.

(ii $\Rightarrow$ iii) If $V_\bullet \cong W_\bullet$, then by the uniqueness of the decomposition (Theorem 2.5), their multisets of intervals must be identical.

²Formally, d_I defines an extended pseudometric on the category **PM** of persistence modules [24]. Unlike a standard metric, a pseudometric allows distinct objects to have zero distance; specifically, a distance of zero implies that the modules are isomorphic, not necessarily identical. For our purposes, this notion of distance is sufficient.

(iii $\Rightarrow$ iv) Fix $i \leq j$. The persistent homology group $\mathbf{PH}_{i \rightarrow j}(V_\bullet)$ is formed by basis vectors corresponding to intervals $[b, d]$ such that $[i, j] \subset [b, d]$. If the barcodes are identical, these direct sum decompositions are identical:

$$\mathbf{PH}_{i \rightarrow j}(V_\bullet) \cong \bigoplus_{[i, j] \subset [b, d]} (I_\bullet^{b, d}) \cong \mathbf{PH}_{i \rightarrow j}(W_\bullet).$$

(iv $\Rightarrow$ i) Since $\mathbf{PH}_{t \rightarrow t}(V_\bullet) = V_t$, identical persistent homology groups imply identical rank invariants. By the Structure Theorem, this implies $V_\bullet \cong W_\bullet$. Let $\phi_\bullet$ be this isomorphism and take $\psi_\bullet = \phi_\bullet^{-1}$. The interleaving conditions are trivially satisfied. $\square$

Connection to the Fundamental Lemma. The argument for (iii) $\Rightarrow$ (iv) reconstructs the Fundamental Lemma of Persistent Homology.

Lemma 2.11 (FUNDAMENTAL LEMMA OF PERSISTENT HOMOLOGY [19]). *For any pair of indices $i \leq j$, the persistent Betti number $\beta_{i \rightarrow j}(V_\bullet)$ equals the number of intervals in the barcode containing $[i, j]$:*

$$\beta_{i \rightarrow j}(V_\bullet) = |\{[b, d] \in \mathbf{Bar}(V_\bullet) \mid b \leq i \leq j < d\}|.$$

Edelsbrunner and Harer state that this lemma "encodes the entire information about persistent homology groups" [19], aligning with our result that the barcode uniquely characterises the isomorphism class.

2.5 The Algebraic Stability Theorem

Ideally, barcodes derived from the same subject should be identical. However, discrete sampling artefacts make strict isomorphism rare. Hence, a realistic requirement is *stability*: minor input variations should yield small deviations in the topological signature. While minimising the interleaving distance provides a rigorous framework for this, computing d_I directly is NP-hard in the general case [25]. This complexity motivates a faithful, efficiently computable proxy: the **bottleneck distance** d_B . Aided by graph theory, we derive d_B via optimal pairings on a specialised bipartite structure³.

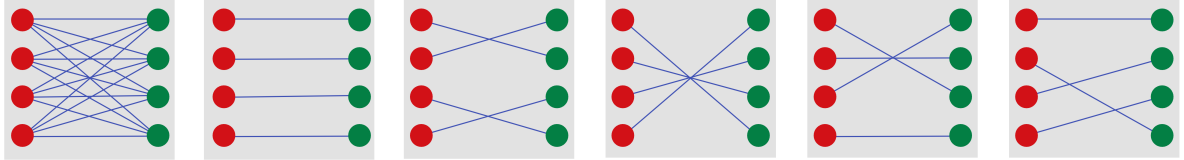


Figure 2.18: Bipartite matching of barcodes. Two barcodes $\mathcal{B}_1$ (red) and $\mathcal{B}_2$ (green). The first panel displays the complete bipartite graph on $\mathcal{B}_1 \cup \mathcal{B}_2$. Subsequent panels illustrate matching strategies. To achieve a *perfect matching*, we augment the vertex sets with projections to the diagonal Δ . Unpaired off-diagonal features are formally paired with a trivial interval on the diagonal.

To compare persistence diagrams, we must account for features of differing cardinalities. We achieve this by augmenting each diagram $\mathcal{D}$ with the diagonal $\Delta = \{(x, x) \mid x \in \mathbb{R}\}$ with infinite multiplicity, creating the augmented set $\mathcal{D}^* = \mathcal{D} \cup \Delta$.

Diagram Pairing Graph. Let $\mathcal{D}_1$ and $\mathcal{D}_2$ be persistence diagrams. We define the *diagram pairing graph* $\mathcal{G}[\mathcal{D}_1, \mathcal{D}_2]$ as a *weighted* bipartite graph with vertex partitions $X = \mathcal{D}_1^*$ and $Y = \mathcal{D}_2^*$. The edge set E allows for two types of pairings, weighted by the ℓ_∞ -distance:

- **Feature-to-Feature:** Edges (u, v) linking $u \in \mathcal{D}_1$ to $v \in \mathcal{D}_2$ with weight $\|u - v\|_\infty$;
- **Feature-to-Noise:** Edges linking a feature u to its projection $\pi(u)$ on Δ with weight $\|u - \pi(u)\|_\infty$.

³Our formulation follows the algorithmic construction of Edelsbrunner and Harer [19, Section VIII.4], who define the distance via *perfect matchings* on bipartite graphs augmented with diagonal projections. This contrasts with the formulation of Nanda [10, Def. 6.18], who employs *partial matchings*. These definitions are mathematically equivalent: the inclusion of the diagonal Δ with infinite multiplicity ensures that any partial matching can be extended to a perfect matching by linking unpaired points to the diagonal (Figure 2.18).

Intuitively, this places the augmented diagrams on opposing sides of a bipartite graph. A matching strategy M covers all vertices, and its *cost* is the weight of the most expensive edge:

$$\text{cost}(M) = \sup_{(u,v) \in M} \|u - v\|_\infty.$$

Bottleneck Distance. The *bottleneck distance* is the cost of the optimal matching strategy. Formally, it is the infimum of costs over all perfect matchings M in $\mathcal{G}[\mathcal{D}_1, \mathcal{D}_2]$:

$$d_B(\mathcal{D}_1, \mathcal{D}_2) = \inf_{M \in \mathcal{M}} \left(\sup_{(u,v) \in M} \|u - v\|_\infty \right).$$

Since points corresponding to noise are nearer to the diagonal Δ , we expect such points to be more likely to be paired to points on Δ to minimise the total sup-norm distance of the matching strategy. The pseudometric d_B is not merely a heuristic, but a faithful calculation of the interleaving distance.

Theorem 2.12 (ISOMETRY THEOREM [24]). *Let $\mathbf{V}$ and $\mathbf{W}$ be persistence modules decomposable into intervals. Then, the algebraic interleaving distance is isometric to the combinatorial bottleneck distance:*

$$d_I(\mathbf{V}, \mathbf{W}) = d_B(\text{Bar}(\mathbf{V}), \text{Bar}(\mathbf{W})) = d_B(\mathcal{D}(\mathbf{V}), \mathcal{D}(\mathbf{W})).$$

Proof. The complete algebraic proof relies on the Induced Matching Theorem for interval-decomposable modules. As establishing that interleaving morphisms strictly induce a valid bipartite matching requires advanced representation theory, we defer the rigorous result to Lesnick [24]. $\square$

The Isometry theorem bridges theory and practice. While computing d_I directly is often intractable (and NP-hard in multiparameter settings [25]), d_B offers an efficient alternative⁴. While we defer the formal proof of the Isometry Theorem to established literature, the explicit construction of the diagram pairing graph $\mathcal{G}[\mathcal{D}_1, \mathcal{D}_2]$ remains operationally essential for this work. The Isometry Theorem acts as a theoretical bridge, translating the abstract algebraic stability of persistence modules (d_I) into the concrete, computable combinatorial space of barcodes (d_B). Once in this domain, optimal perfect matchings provide the exact mechanical engine required to establish our physical engineering bounds.

Theorem 2.13 (ALGEBRAIC STABILITY THEOREM [28]). *Let $\mathcal{I}, \mathcal{J} : \Omega \rightarrow \{0, \dots, m\}$ be two integer-valued intensity functions such that $\|\mathcal{I} - \mathcal{J}\|_\infty \leq k$ for some integer $k \geq 0$. Then, $d_B(\mathcal{D}(\mathcal{I}), \mathcal{D}(\mathcal{J})) \leq \|\mathcal{I} - \mathcal{J}\|_\infty$.*

Proof Sketch. The ℓ_∞ -bound ensures that the sublevel sets of the two images are geometrically nested with a relative shift of k indices. These asynchronous geometric inclusions induce corresponding linear maps between the vector spaces of their respective persistence modules. This network of shifted maps satisfies the strict structural requirements of an algebraic k -interleaving, which the Isometry Theorem subsequently translates into the combinatorial bottleneck distance. The generalised proof is established by Lesnick [24], but a self-contained cubical derivation is provided in **Appendix A**. $\square$

The Algebraic Stability Theorem establishes that the persistence diagram operator $\mathcal{D}(\bullet)$ is 1-Lipschitz with respect to the supremum norm. Thus, local perturbations between intensity functions $\mathcal{I}$ and $\mathcal{I}'$ on a fixed domain Ω (e.g., a unit intensity shift at a single pixel) strictly bound the bottleneck distance:

$$d_B(\text{Bar}(\mathcal{I}), \text{Bar}(\mathcal{I}')) \leq \|\mathcal{I} - \mathcal{I}'\|_\infty \leq 1.$$

Despite this theoretical guarantee, empirical studies frequently report that persistent homology lacks robustness when applied to real-world digital images [29–31]. This discrepancy arises not because the theorem is false, but because its bound becomes **operationally trivial** in high-contrast imaging.

⁴Computational implementations in this work employ the GUDHI library [26], which uses the Efrat-Itai-Katz algorithm [27] with complexity $\mathcal{O}(n^{1.5} \log n)$.

Sharp contrast boundaries, such as vessels against the retinal background characterise medical images. For a fundus image $\mathcal{I} : \Omega \rightarrow [0, \mathcal{M}]$, intensities span the full dynamic range:

$$0 \approx \min_{\kappa} \mathcal{I}(\kappa) \ll \max_{\kappa} \mathcal{I}(\kappa) \approx \mathcal{M}.$$

Consequently, a geometric shift of a single pixel across a vessel edge forces the intensity to swing from the minimum (0) to the maximum ($\mathcal{M}$). Under a spatial perturbation $\mathcal{I}^*$, this yields a maximal residual:

$$\max_{\mathbf{x} \in \Omega} |\mathcal{I}(\mathbf{x}) - \mathcal{I}^*(\mathbf{x})| \approx \mathcal{M} \implies d_B \leq \mathcal{M}.$$

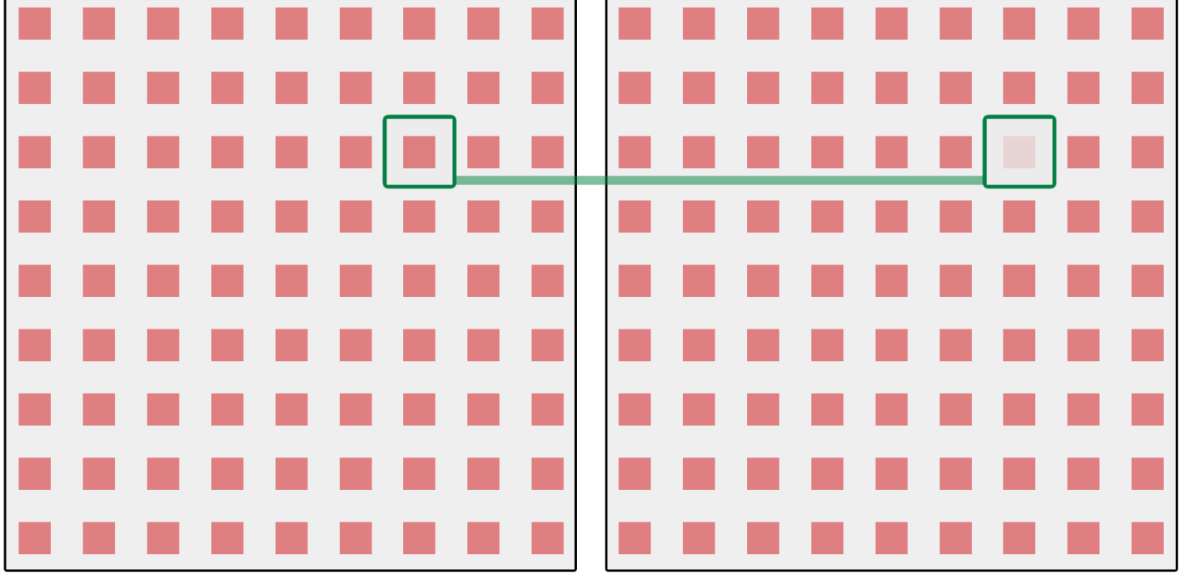


Figure 2.19: (Left) A uniform pixel array at maximum intensity $\mathcal{M}$. (Right) A single pixel is inverted to intensity 0. Although the macroscopic topology remains largely intact, the Algebraic Stability Theorem dictates a bottleneck bound of $\mathcal{M}$.

Figure 2.19 illustrates this phenomenon. In finite cubical filtrations, the maximum possible persistence of any feature is bounded by the dynamic range $\mathcal{M}$. Thus, an error bound of $\mathcal{M}$ is *effectively infinite*. This provides zero diagnostic utility, as it renders minor sensor noise mathematically indistinguishable from catastrophic structural failure. Because sub-pixel interpolation across high-contrast edges physically guarantees this maximal residual, seeking a tighter monolithic bound is futile. To extract actionable diagnostic limits, Chapter 3 mechanistically decomposes the Algebraic Stability Bound into its isolated physical constituents.

Chapter 3

Computational Stability

The previous chapter established stability at the algebraic level. This chapter asks what that result means when the input is a sampled image on a fixed lattice. The key obstruction is that continuous geometric perturbations do not in general respect the pixel grid. As a result, the classical stability theorem becomes operational only after separating grid-preserving transformations from those that force interpolation.

We say that a transformation is **grid-aligned** if its restriction to $\mathbb{Z}^2$ injectively maps pixel coordinates to pixel coordinates without interpolation.

Proposition 3.1 (DISCRETE ISOMETRIC INVARIANCE). *Let $(\Omega, \mathcal{I})$ be a digital image and let T_A be a grid-aligned transformation producing $(\Omega, \mathcal{I}_A)$. Then, $d_B(\mathcal{D}(\mathcal{I}), \mathcal{D}(\mathcal{I}_A)) = 0$.*

Proof. A grid-aligned transformation preserves the pixel adjacency graph. Consequently, the induced cubical complex is isomorphic to the baseline complex, differing only by a relabelling of vertices. By Corollary 2.10, isomorphic complexes generate isomorphic persistence modules and identical barcodes. $\square$



Figure 3.1: Approximation of continuous rotation at high resolution. A black bar is rotated from the baseline (0°) by $18^\circ, \dots, 90^\circ$. At this scale, aliasing artefacts are visually negligible.

Discrete isometric invariance is an idealisation: it may hold largely at high resolutions (Figure 3.1), where sampling density renders artefacts negligible, but as feature size approaches the pixel scale, **aliasing** (interpolation) becomes the dominant error source. For any rotation $\theta \notin \{k \cdot 90^\circ\}_{k \in \mathbb{Z}}$, the system must interpolate off-grid signals. This introduces a **discretisation gap** where the continuous and discrete rotation operators fail to commute. We illustrate this failure via two stress tests targeting connected components (β_0) and cycles (β_1). Consider the two images in Figure 3.2.

Figure 3.3 presents the "String of Pearls" test: a linear array of 30 disjoint pixels separated by 1-pixel gaps. While the topology should remain invariant ($\beta_0 \equiv 30$), interpolation at oblique angles smears intensity across the gaps. This causes spurious merging and periodic collapses in the Betti count.

Similarly, Figure 3.4 examines 1-dimensional cycles using rectangular loops with 1-pixel voids. Non-aligned angles induce two failure modes: *closure*, where interpolation fills the void ($\beta_1 \rightarrow 0$), and *fragmentation*, where interference patterns split the loop ($\beta_1 > \text{Truth}$)¹. These results confirm that the grid lacks isotropic topological resolution. Once we exit the grid-aligned regime ($\theta \notin 90\mathbb{Z}$), the protection of Proposition 3.1 vanishes, and we are left with the operationally trivial bound of Theorem 2.13.

Remark 3.2 We now assume a fixed domain $\Omega \subset \mathbb{Z}^2$, allowing us to represent images solely by their intensity functions $\mathcal{I}$. This reflects the standardised resolution of retinal imaging protocols. Also, graph-theoretic concepts follow Diestel [32].

¹Code for Figures 3.3 and 3.4 available at: <https://github.com/rejosephnieto/ph>

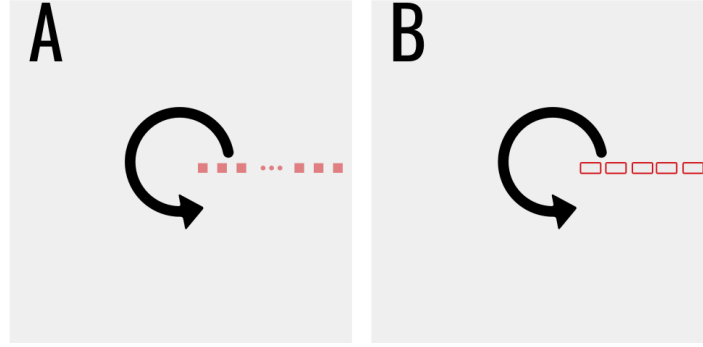


Figure 3.2: Schematic Representation of the Rotational Stress Test. To ensure visual clarity, this idealised schematic is presented instead of the raw experimental pixel grids, which are optically dense and difficult to interpret at this scale. **(A)** A linear array of 30 disjoint pixels (separated by 1-pixel gaps). **(B)** A linear array of 5 rectangular loops (with an internal void width $\delta = 1$ pixel). Both will be subjected to a continuous affine rotation $\theta \in [0, 360^\circ]$.

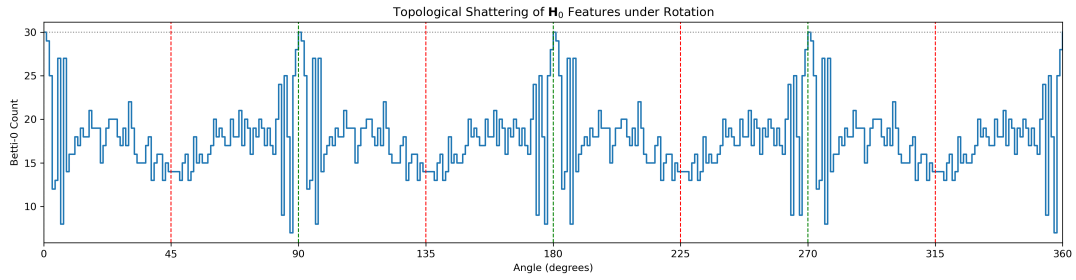


Figure 3.3: Numerical evaluation of β_0 for a linear array of 30 disjoint pixels (separated by 1 pixel) subject to affine rotation $\theta \in [0, 360^\circ]$ (Image A in Figure 3.2). The ground truth topology ($\beta_0 = 30$) is recovered only at $\theta \in 90\mathbb{Z}$. At oblique angles (red dashed lines), interpolation fills gaps, leading to spurious merging.

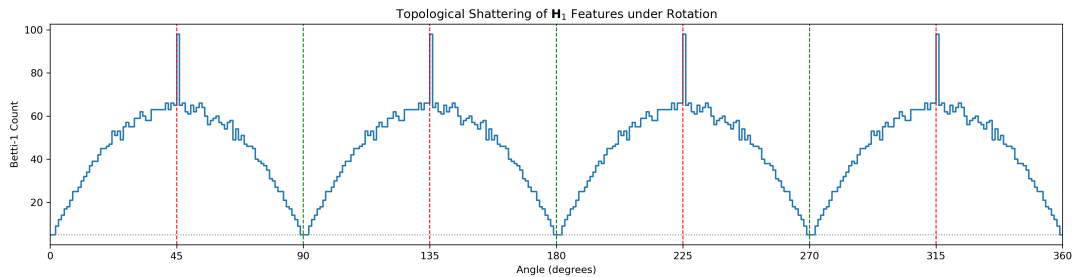


Figure 3.4: Evolution of β_1 for a linear array of 5 rectangular loops (internal void width $\delta = 1$ pixel) under rotation (Image B in Figure 3.2). The ground truth ($\beta_1 = 5$) is recovered only at grid-aligned angles (green dashed lines). At oblique angles (red dashed lines), interpolation bridges the internal void (closure) or fragments the feature.

3.1 The "Aim-and-Shoot" Acquisition Model

Pixel connectivity. Let $(\Omega, \mathcal{I})$ be a digital image t an intensity value. The *foreground* of the t -th subcomplex consists of all pixels p where $\mathcal{I}(p) \leq t$. If we view these foreground pixels as vertices of a graph G_t , connectivity is determined by the chosen adjacency metric. Under **von Neumann 4-connectivity** [33], edges connect pixels sharing a side ($\|x - y\|_1 = 1$). Under **Moore 8-connectivity** [34], edges connect pixels sharing a side or a corner ($\|x - y\|_\infty = 1$). Figure 3.5 illustrates these two paradigms.

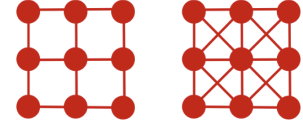


Figure 3.5: (Left) 4-connectivity; (Right) 8-connectivity.

Recall that we model a digital image as a cubical complex, where each pixel is a closed unit square. Consider an image where the foreground forms a "diamond" configuration of four pixels surrounding a central hole (Figure 3.6).

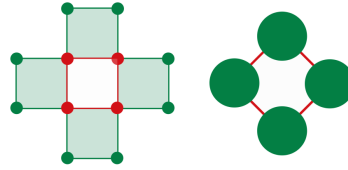


Figure 3.6: Connectivity in Cubical Complexes. (Left) A "diamond" configuration of four foreground pixels (light green squares). The *inner boundary* (red edges) and *outer boundary* (green edges) both form 1-cycles. (Right) The dual graph representation. Since the pixels share vertices in the cubical complex, the corresponding graph nodes must be connected via diagonal edges (red lines), necessitating 8-connectivity.

For the inner 1-cycle γ_{in} to exist as a continuous loop bounding the void, the chain of edges must be unbroken. This continuity is physically mediated by the shared vertices between the diagonally adjacent pixels. If we map this structure to a dual graph where pixels are nodes (Figure 3.6, Right), the shared vertex in the complex mandates a connection between the diagonal nodes. Observe that enforcing 4-connectivity (von Neumann) would artificially remove these diagonal edges, severing the loop and contracting the homology class to zero, which contradicts the algebraic reality of the underlying cubical complex. Thus, Cubical Homology intrinsically mandates **Moore neighbourhoods** (8-connectivity) to preserve the validity of 1-cycles across shared vertices. Consequently, we quantify intensity fluctuations using the ℓ_∞ -logic that recognises diagonal adjacency. This perfectly aligns with ℓ_∞ -metric used in the Algebraic Stability Theorem.

The Moore-based Graph G_∞ . We model spatial misalignment on the discrete lattice using the corresponding Moore neighbourhood². Let $\Omega \subset \mathbb{Z}^2$ be the set of pixel coordinates. We define the **Moore-based Graph**³ $G_\infty = (\Omega, E_\infty)$ where edges connect pixels adjacent in the ℓ_∞ -norm: $uv \in E_\infty \Leftrightarrow \|u - v\|_\infty = 1$.

Definition 3.3 (GRAPH LIPSCHITZ CONSTANT). We view the intensity function $\mathcal{I}$ as a signal on the vertices of G_∞ . The *Graph Lipschitz Constant*, denoted $L_{\mathcal{I}}$, is the maximum local gradient:

$$L_{\mathcal{I}} = \max_{uv \in E_\infty} |\mathcal{I}(u) - \mathcal{I}(v)|.$$

Remark 3.4 We rely on the discrete Lipschitz constant $L_{\mathcal{I}}$ to bound the mechanical error. While a tighter bound might theoretically be derived using operator theory, such analysis exceeds the scope of this work. Furthermore, introducing such concepts here would obscure the spatial intuition required for the "Aim-and-Shoot" engineering model. We therefore retain the Lipschitz heuristic as a conservative, spatially localised upper bound that remains operationally actionable.

The Aim-and-Shoot Acquisition Model. To derive an operationally actionable stability framework, we must acknowledge this saturation and decompose the monolithic error term $\|\mathcal{I} - \mathcal{I}'\|_\infty$ into its physical con-

²Under a **Moore Neighborhood**, two pixels at $\mathbf{x}, \mathbf{y} \in \mathbb{Z}^2$ are adjacent if and only if $\|\mathbf{x} - \mathbf{y}\|_\infty = 1$. Hence, a pixel is connected to all 8 surrounding pixels.

³The choice of nomenclature stems from the choice of Moore 8-connectivity: it must not be confused with Moore Graphs.

stituents. We achieve this by introducing the **"Aim-and-Shoot" Acquisition Model**, which formalises clinical error as a sequential composition of two distinct operators: **Mechanical Alignment** (T_A) and **Radiometric Capture** (T_S). This separation is motivated by physical causality: the optical system must be spatially oriented before the sensor activates.

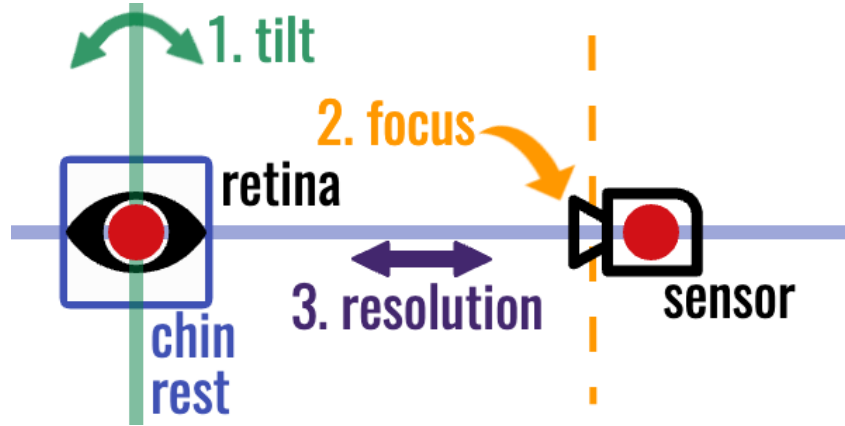


Figure 3.7: The **Aim-and-Shoot** Acquisition Model. **Phase I (Aim):** The patient stabilises their head on a chin rest (blue), aligning the retina (left red dot) with the camera sensor (right red dot) along the optical axis (purple line). **Mechanical** noises are modelled as *in-plane rotation* (green arrow), **resolution scaling** via axial distance (purple arrow), or **variations of focus** (blue). **Phase II (Shoot):** Once aligned, the sensor captures the light. All **Radiometric** corruptions are modelled as sensor perturbations distinct from the spatial alignment. Intuitively, all mechanical noises occur to the left of the orange dashed line, while all radiometric noises occur at the sensor array itself.

Figure 3.7 illustrates the physical setup. We assume that the physical constraint (a chin rest) eliminates lateral translation, leaving three primary degrees of freedom in the **Aim** phase: "Head Tilt" (rotation relative to the vertical axis), "Resolution" (determined by the physical distance between the eye and the camera), and "Blur" (determined by the camera's focus). Once the aim is fixed, the **Shoot** phase introduces errors that are strictly related to the sensor's light sensitivity and electronic noise.

Aim: Mechanical Noise The protocol requires the patient to fixate on an internal target to align the eye's optical axis with the sensor plane. Ideally, this results in the retinal surface being orthogonal to the optical axis at a fixed focal distance. In practice, motor instability introduces *Mechanical Noise*. We model these as an operator T_A acting on the intensity function, which combines all the rigid or affine transformations of the domain Ω . Further, since the sensor grid Ω is fixed (e.g., 1024×1024 px), we model resolution loss not as a changing grid size, but as changes in pixel intensities in the sampling density via optical scaling:

Perturbation	Physical Origin (Clinical)	Model
In-Plane Rotation (Head Tilt)	The patient's head rotates relative to the sensor's horizon. By the principle of relative motion, we treat the fundoscope as the fixed reference frame.	Orthogonal Rotation $x \mapsto R_\theta x$
Axial Translation (Resolution)	Variations in eye-to-sensor distance (z -axis) alter magnification. A low-resolution capture is functionally equivalent to an image taken from a greater distance (optical scaling), then upsampled to the baseline grid.	Scaling $x \mapsto \lambda x$
Optical Defocus	The dispersion of a point source into a patch due to lens imperfections or focus error (Blur).	Convolution $\mathcal{I} \mapsto \mathcal{I} * K$

These transformations, illustrated in Figure 3.8, typically necessitate **Interpolation**, which inevitably results in **Mechanical Noise**. That is, while the "canvas" Ω remains fixed, the physical misalignments modify the intensity function $\mathcal{I}$ by mapping integer lattice points to non-integer coordinates. In particular, we explicitly restrict the mechanical operator T_A to **bilinear interpolation**. While higher-order methods may preserve structures more effectively, they introduce additional artefacts that are even harder to predict [35]. Because $u, v \in [0, 1]$, the bilinear coefficients are strictly non-negative and sum to 1. This guarantees that the interpolated intensity is strictly a convex combination of the local vertices [35]. This convexity precludes overshoot, ensuring the local bounds required to derive the discrete gradient penalty in Lemma 3.11.

Definition 3.5 (BILINEAR INTERPOLATION [35]). Let $\mathcal{I} : \Omega \rightarrow \mathbb{Z}_{\geq 0}$ be a discrete, integer-valued intensity function. For any continuous coordinate (x, y) residing within the unit grid cell bounded by the lattice vertices (x_0, y_0) , (x_1, y_0) , (x_0, y_1) , and (x_1, y_1) , the *bilinearly interpolated intensity* is defined as:

$$\mathcal{I}(x, y) = \lfloor (1-u)(1-v)\mathcal{I}(x_0, y_0) + u(1-v)\mathcal{I}(x_1, y_0) + (1-u)v\mathcal{I}(x_0, y_1) + uv\mathcal{I}(x_1, y_1) \rfloor$$

where u and v are the fractional spatial offsets in $[0, 1]$, and $\lfloor \cdot \rfloor$ denotes rounding to the nearest integer. The standard continuous formulation is adapted here using the operator $\lfloor \cdot \rfloor$, which rounds to the nearest integer, to strictly preserve the integer-valued codomain required for algebraic stability.

Figure 3.8 illustrates how interpolation occurs under mechanical perturbations.

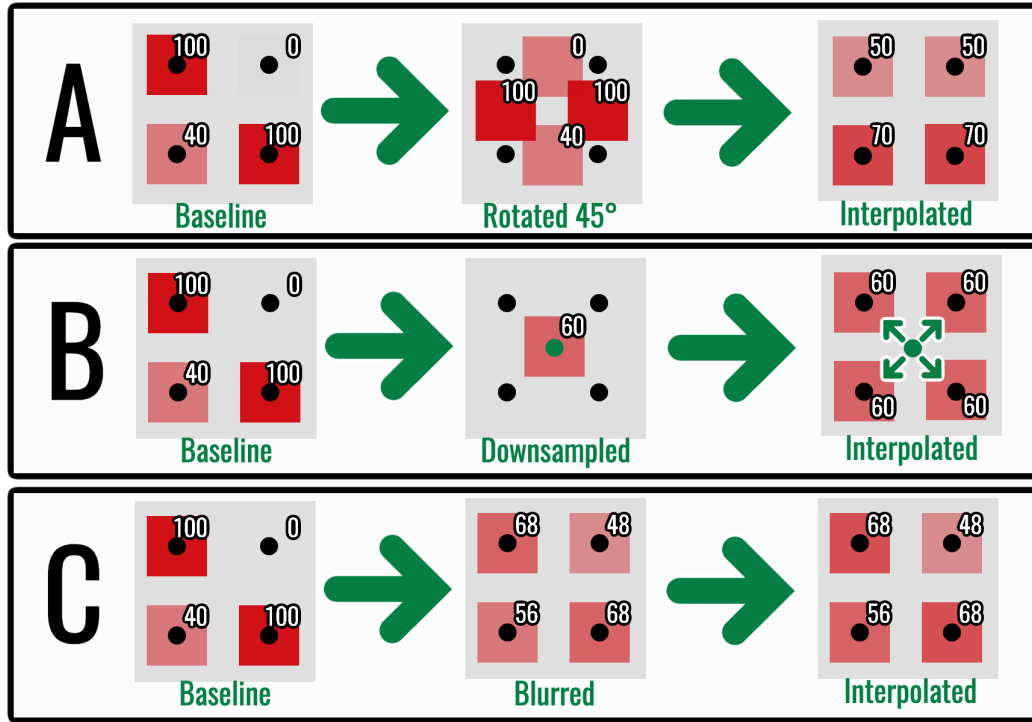


Figure 3.8: Mechanical perturbations of the discrete intensity function $\mathcal{I}$. A 2×2 baseline image $(\Omega, \mathcal{I})$ with intensities $\{100, 0, 100, 40\}$ (clockwise from top-left) subjected to spatial errors. **(A) Rotation:** A 45° shift forces off-grid interpolation, smearing sharp gradients across adjacent pixels (heuristic intensities shown). **(B) Resolution:** Reduced sampling density integrates the 2×2 area into a single measurement (intensity 60), which is mapped back onto the baseline grid Ω to evaluate stability. **(C) Blur:** Defocus acts as a spatial average; here, applying a 40% primary and 20% neighbour weighting yields $\{68, 48, 68, 56\}$.

Remark 3.6 We define these transformations as *Mechanical Noise*, realised strictly as a transformation of the intensity function $\mathcal{I}$. The domain Ω remains constant, and the physical misalignments translate into pixel intensity fluctuations on that fixed domain.

Shoot: Radiometric Noise Once aligned, the shutter opens, and the sensor integrates the incident photon flux over an exposure time t . This process introduces **Radiometric Noise**, a broad category of photometric perturbations arising from hardware limitations rather than spatial orientation. This category encompasses all errors that corrupt the intensity signal without warping the grid. Unlike the "Aim" phase, which involves resampling, the "Shoot" phase corrupts the measurements *at* the aligned coordinates. These cause deterministic or stochastic intensity fluctuations relative to the baseline image. Intuitively, radiometric noise may be viewed as mechanical noise without interpolation.

Formalising the Acquisition Process To formalise the acquisition process, we model the perturbations as functional operators acting directly on the discrete intensity function defined on the fixed grid Ω . This formalisation reflects real-world scenarios: the location of each "lattice point" on the sensor remains fixed, but the intensities at those points vary with the perturbation.

Aim (T_A): We define T_A as the operator that spatially registers the anatomical scene to the sensor grid. Physical transformations map integer pixel coordinates to non-integer locations or blur their separation; T_A inherently performs **interpolation**. Functionally, for a pixel $x \in \Omega$ and a geometric map g , the aligned intensity is the interpolated value of the source signal at the coordinate $g^{-1}(x)$.

Shoot (T_S): The sensor operator acts on the aligned intensities, introducing photometric corruptions that directly modify the intensity map $\mathcal{I}$.

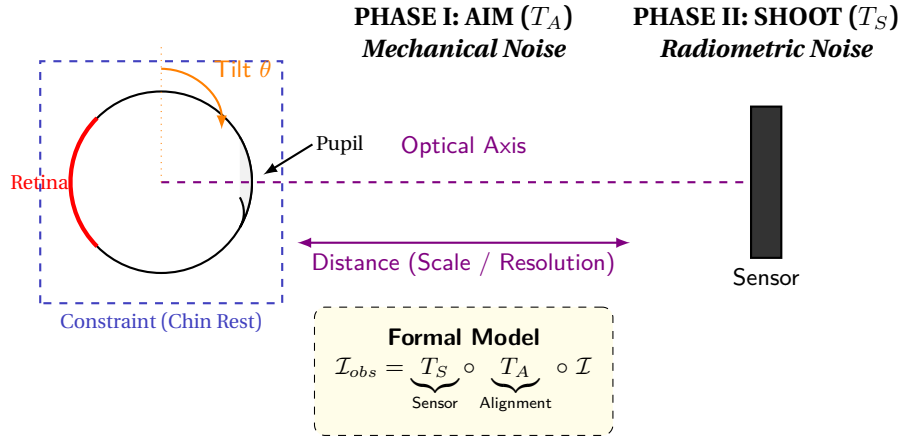


Figure 3.9: The Formalised Acquisition Model. The pipeline corresponds to the sequential composition of operators. **Phase I (Aim):** First, the alignment operator T_A transforms the intensity function $\mathcal{I}$. The mechanical constraint (blue box) fixes x, y , leaving only rotational tilt (θ) and axial distance (scale/resolution) as sources of Mechanical Noise. **Phase II (Shoot):** The sensor operator T_S then acts on the aligned intensities ($T_A \circ \mathcal{I}$), introducing Radiometric Noise. The retina (red arc) is the signal source, and the pupil acts as the optical aperture.

The formalised **Aim-and-Shoot** model (Figure 3.9) restricts all transformations to a fixed spatial domain Ω . This satisfies the prerequisite for the d_∞ and d_B metrics used in the Algebraic Stability Theorem by projecting all physical perturbations as **sequential intensity shifts**.

Defining the composite degradation operator $T = T_S \circ T_A$, we factorise the total error to explicitly separate mechanical and radiometric noise. Let $\mathbf{U} := \mathbf{Bar}(\mathcal{I})$ denote the baseline barcode, $\mathbf{V} := \mathbf{Bar}(T_A \circ \mathcal{I})$ the mechanically perturbed intermediate, and $\mathbf{W} := \mathbf{Bar}(T_S \circ T_A \circ \mathcal{I})$ the final observed barcode. Applying the Triangle Inequality to the bottleneck distance alongside the Algebraic Stability Theorem yields:

$$d_B(\mathbf{U}, \mathbf{W}) \leq d_B(\mathbf{U}, \mathbf{V}) + d_B(\mathbf{V}, \mathbf{W}) \leq \underbrace{\|\mathcal{I} - T_A \circ \mathcal{I}\|_\infty}_{\text{Mechanical Stability}} + \underbrace{\|(T_A \circ \mathcal{I}) - T_S \circ (T_A \circ \mathcal{I})\|_\infty}_{\text{Radiometric Stability}} \quad (3.1)$$

3.2 Shoot: Radiometric Stability Bound

The second term quantifies the magnitude of photometric corruption. Since Original Equipment Manufacturers (OEM) typically provide rigorous specifications for sensor noise floors and linearity, this term is theoretically tractable.

Proposition 3.7 (RADIOMETRIC STABILITY). *Let T_S be the sensor operator acting on the aligned image $T_A \circ \mathcal{I}$. We model the capture process as a superposition of a deterministic transfer function ρ and stochastic perturbations η , i.e., $T_S(T_A \circ \mathcal{I}) = \rho \circ (T_A \circ \mathcal{I}) + \eta$. The topological error introduced during the "Shoot" phase is strictly bounded by the deviation of the transfer function and the peak amplitude of the noise:*

$$d_B(\mathbf{Bar}(T_A \circ \mathcal{I}), \mathbf{Bar}(T_S(T_A \circ \mathcal{I}))) \leq \|(T_A \circ \mathcal{I}) - \rho \circ (T_A \circ \mathcal{I})\|_\infty + \|\eta\|_\infty.$$

Proof. This readily follows from the triangle inequality on d_B . □

Tractability via Protocol Standardisation. In regulated diagnostic environments, strict adherence to Standard Operating Procedures (SOPs) is intended to enforce two constraints. First, it fixes the transfer function ρ (ideally $\rho \approx \text{id}$), effectively eliminating the deterministic residual. Second, it minimises operator variance, causing the stochastic term $\|\eta\|_\infty$ to converge to the OEM Sensor Noise Floor. When these conditions are met, the Radiometric Stability term becomes bounded *a priori*.

Corollary 3.8 (THE STANDARDISED BASELINE). *Under strict protocol standardisation and where ϵ_{rad} is the OEM-provided specification, the radiometric error is bounded by the sensor noise floor:*

$$d_B(\text{Bar}(T_A \circ \mathcal{I}), \text{Bar}(T_S(T_A \circ \mathcal{I}))) \leq \|\eta\|_\infty \leq \epsilon_{\text{rad}}.$$

Corollary 3.8 establishes the theoretical safety limit, but real-world acquisition may violate this assumption. The empirical audit in Chapter 5 implements these violations within the diagnostic framework developed in Chapter 4. This bound covers all standard radiometric distortions listed in Table 3.1:

Distortion Type	Intensity Mapping	Stability Bound ($d_B \leq \dots$)	Clinical Manifestation
Functional Mapping ($\eta = 0$)			
Gamma	$\rho(v) = 255(v/255)^\gamma$	$\max_v v - \rho(v) $	Exposure Response
Contrast	$\rho(v) = \alpha(v - 127.5) + 127.5$	$ 1 - \alpha \times 127.5$	Media Opacity (Cataracts)
Noise Injection ($\rho = \text{id}$)			
Drift	$\eta(x) \equiv \delta$	$ \delta $	Vignetting / Misalignment
Gaussian Noise	$\eta \sim \mathcal{N}(0, \sigma^2)$	$\ \eta\ _\infty$	Thermal Noise (High ISO)
Poisson Noise	$\eta(x) \sim \text{Poisson}(\mathcal{I}(x)) - \mathcal{I}(x)$	$\ \eta\ _\infty$	Photon Noise (Low Light)
Salt & Pepper	$\eta(x) \in \{0 - \mathcal{I}, 255 - \mathcal{I}\}$	255 (Worst Case)	Dead Pixels / Dust
Bit Depth	$\eta(x) = \Delta \lfloor \mathcal{I}(x) / \Delta \rfloor - \mathcal{I}(x)$	2^{8-k}	Compression Banding

Table 3.1: Analytical stability bounds for radiometric perturbations.

Among the radiometric perturbations, bit-depth reduction (quantisation) merits specific formalisation, as it is a deterministic functional projection rather than additive noise.

Proposition 3.9 (QUANTISATION STABILITY BOUND). *Let $\mathcal{I} : \Omega \rightarrow [0, \mathcal{M}]$ be a bounded intensity function defined on the discrete grid Ω . Let Q_k be a quantisation operator reducing the dynamic range to k bits (step size $\Delta = \mathcal{M}/2^k$). The bit resolution strictly bounds the topological error induced by quantisation:*

$$d_B(\mathcal{D}(\mathcal{I}), \mathcal{D}(Q_k \circ \mathcal{I})) \leq \Delta.$$

Proof. The quantisation operator $Q_k(v) = \Delta \lfloor v/\Delta \rfloor$ approximates the intensity v . Hence,

$$|\mathcal{I}(x) - (Q_k \circ \mathcal{I})(x)| = \left| \mathcal{I}(x) - \Delta \left\lfloor \frac{\mathcal{I}(x)}{\Delta} \right\rfloor \right| < \Delta \quad \text{for all } x \in \Omega.$$

The sup-norm satisfies $\|\mathcal{I} - Q_k \circ \mathcal{I}\|_\infty < \Delta$. Applying the Algebraic Stability Theorem yields the result. $\square$

Corollary 3.10 (THE TERRACE THRESHOLD). *If the persistence $\pi(z)$ of a pathological feature z satisfies $\pi(z) > 2\Delta$, the feature is topologically guaranteed to survive quantisation.*

Proof. Let $z = (b, d) \in \mathcal{D}(\mathcal{I})$ with persistence $\pi(z) = d - b$. By Proposition 3.9,

$$d_B(\mathcal{D}(\mathcal{I}), \mathcal{D}(Q_k \circ \mathcal{I})) \leq \Delta.$$

Let $z' \in \mathcal{D}(Q_k \circ \mathcal{I})$ be its counterpart under an optimal bottleneck matching. Then, $\|z - z'\|_\infty \leq \Delta$. Now suppose, for contradiction, that the feature z does not survive quantisation. In the context of bottleneck matching, this means z is paired with a trivial feature on the diagonal $\Delta_\infty = \{(x, x) \mid x \in \mathbb{R}\}$.

The ℓ_∞ -distance from a point $z = (b, d)$ (where $b < d$) to the diagonal is determined by minimising the maximum coordinate-wise difference:

$$\inf_{x \in \mathbb{R}} \|(b, d) - (x, x)\|_\infty = \inf_{x \in \mathbb{R}} \max(|b - x|, |d - x|).$$

This infimum is achieved at the midpoint $x = \frac{b+d}{2}$. Substituting this value yields exactly half the persistence:

$$\max\left(\left|b - \frac{b+d}{2}\right|, \left|d - \frac{b+d}{2}\right|\right) = \left|\frac{b-d}{2}\right| = \frac{d-b}{2} = \frac{\pi(z)}{2}.$$

If z is matched to the diagonal, the matching cost is $\frac{\pi(z)}{2}$. This contradicts $\pi(z) > 2\Delta$, which strictly implies $\frac{\pi(z)}{2} > \Delta$. Hence, z cannot be mapped to the diagonal. That is, z corresponds to a strictly off-diagonal (surviving) feature $z' \in \mathcal{D}(Q_k \circ \mathcal{I})$. $\square$

Corollary 3.10 guarantees that sufficiently persistent features survive bit-depth reduction. A topological discriminator tracking dominant features z_i theoretically preserves its diagnostic signal provided $\pi(z_i) > 2\Delta$. This manifests empirically as the **Terrace Effect**: rather than degrading linearly with information loss, classifier accuracy forms a stable plateau. Performance remains invariant under progressive quantisation until the step size Δ eclipses the persistence of the defining clinical lesions. Crossing this critical threshold ($\pi(z_i) \leq 2\Delta$) maps these features directly to the diagonal, inducing an abrupt collapse in discriminative power.

3.3 Aim: Mechanical Stability Bound

The first term of Equation 3.1, $\|\mathcal{I} - T_A \circ \mathcal{I}\|_\infty$, measures the persistence of topological features under geometric warping. This factor is determined by the field-acquisition environment rather than hardware specifications. Having established the Moore neighbourhood (G_∞) as the underlying pixel adjacency structure, we leverage the **Graph Lipschitz Constant** $L_{\mathcal{I}}$ (Definition 3.3) to quantify stability. We model geometric misalignment as a traversal on this graph.

Lemma 3.11. *Let $(\Omega, \mathcal{I})$ be a digital image on the Moore graph G_∞ with signal volatility $L_{\mathcal{I}}$. Let T_A be a mechanical operator defined by $(T_A \circ \mathcal{I})(x) = \text{Interp}(\mathcal{I}, A(x))$, where $A : \Omega \rightarrow \mathbb{R}^2$ is the coordinate map. If the global maximum coordinate displacement is bounded by $\sup_{x \in \Omega} \|A(x) - x\|_\infty \leq \lambda$, then the topological stability is strictly bounded by:*

$$d_B(\mathcal{D}(\mathcal{I}), \mathcal{D}(T_A \circ \mathcal{I})) \leq \lceil \lambda \rceil L_{\mathcal{I}}.$$

Proof. Let $x \in \Omega$ and $x' := A(x)$ its mapped coordinate. The global displacement constraint ensures that for any spatial axis $j \in \{1, 2\}$, the coordinate-wise distance is bounded by $|x_j - x'_j| \leq \lambda$. Bilinear interpolation (Definition 3.5) constructs the signal at x' using the four bounding lattice vertices v_i of the unit cell containing x' . For any such vertex, its coordinate on axis j must satisfy $v_{i,j} \in \{\lfloor x'_j \rfloor, \lceil x'_j \rceil\}$.

We bound the grid distance from the source pixel x to any interpolation vertex v_i . If $v_{i,j} = \lfloor x'_j \rfloor$, the distance is bounded by:

$$x_j - \lfloor x'_j \rfloor \leq (x'_j + \lambda) - \lfloor x'_j \rfloor < \lambda + 1$$

Because x_j and $\lfloor x'_j \rfloor$ are both integers, their difference must be an integer. The maximum integer strictly less than $\lambda + 1$ is universally $\lceil \lambda \rceil$. By symmetry, if $v_{i,j} = \lceil x'_j \rceil$, the bound is:

$$\lceil x'_j \rceil - x_j \leq \lceil x'_j \rceil - (x'_j - \lambda) < \lambda + 1 \implies |x_j - v_{i,j}| \leq \lceil \lambda \rceil$$

Since this bound holds independently for all axes, the ℓ_∞ -distance on the Moore graph G_∞ satisfies $\|x - v_i\|_\infty \leq \lceil \lambda \rceil$. The discrete gradient constrains the intensity deviation for any bounding vertex:

$$|\mathcal{I}(x) - \mathcal{I}(v_i)| \leq \lceil \lambda \rceil L_{\mathcal{I}} \quad \text{for all } v_i.$$

By Definition 3.5, the final intensity $(T_A \circ \mathcal{I})(x)$ is the rounded result of a convex combination of these vertex intensities $\{\mathcal{I}(v_i)\}$. The continuous convex combination necessarily lies within the closed real interval $[\min \mathcal{I}(v_i), \max \mathcal{I}(v_i)]$. Because the bounding intensities $\mathcal{I}(v_i)$ are themselves integers, applying the round-

ing operator $\lfloor \cdot \rfloor$ guarantees the final discrete value remains strictly within this exact integer interval. Thus, the maximum possible deviation from $\mathcal{I}(x)$ cannot exceed the maximum deviation of any single vertex:

$$|\mathcal{I}(x) - (T_A \circ \mathcal{I})(x)| \leq \max_{v_i} |\mathcal{I}(x) - \mathcal{I}(v_i)| \leq \lceil \lambda \rceil L_{\mathcal{I}}.$$

The result follows from the Algebraic Stability Theorem. $\square$

Ideally, a rigid transformation is an isometry ($d_B = 0$). However, Lemma 3.11 predicts an error bounded by $\lceil \lambda \rceil L_{\mathcal{I}}$. This arises strictly from the **Algebraic Stability Theorem's** reliance on the ℓ_{∞} -norm, which evaluates spatial shifts as pointwise intensity changes. This discrepancy defines the **Discretisation Gap**: on a fixed sensor grid, mechanical misalignments (λ) inevitably convert into functional residuals scaled by the image gradient ($L_{\mathcal{I}}$).

Spatial Scaling of Affine Misalignment The parameter λ bounds the *global maximum* displacement. While λ is constant under uniform translation, affine transformations such as rotation induce a spatial scaling effect in which the local displacement $\lambda(\mathbf{x})$ is proportional to the distance from the origin. For a discrete lattice of radius R , a rotation θ induces a maximum peripheral displacement

$$\lambda_{\max} \approx R \sin \theta.$$

Thus, in a high-resolution clinical array with $R \approx 1024$ px, even a rotation of 3° can force a peripheral displacement $\lambda_{\max} > 50$. The consequence is that the global supremum $\lceil \lambda \rceil$ can leave the perturbative regime even when the visual misalignment appears small near the centre of the image.

A Local Bound. We note that the bound in Lemma 3.11 is *local*. For large grid-aligned transformations such as a 90° rotation, λ becomes large, so the bound $\lceil \lambda \rceil L_{\mathcal{I}}$ may become vacuous despite zero topological error (Proposition 3.1). In reliability-engineering settings, however, acquisition hardware constrains unintended misalignments to a perturbative envelope. Within that regime, λ remains small enough for registration error to couple meaningfully to the image gradient.

Remark 3.12 (CONTINUOUS DISPLACEMENT ON DISCRETE GRIDS) Throughout this work, we denote the magnitude of geometric perturbation by λ . Strictly speaking, geometric transformations operate on $\mathbb{R}^2$. However, on the discrete lattice $\mathbb{Z}^2$, error is fundamentally quantised. The term $\lceil \lambda \rceil$ captures this worst-case cost: a sub-pixel shift ($\lambda < 1$) incurs the full penalty of a pixel transition ($1 \cdot L_{\mathcal{I}}$).

3.4 A Computational Stability Bound

We now unify the mechanical and radiometric bounds into a single, computable theorem. In a real-world clinical audit, an image is never subjected to purely mechanical or purely radiometric distortion; it is affected by both simultaneously. We model this as the composite degradation operator $T = T_S \circ T_A$.

The Observed Realisation To analyse the stability of this composite system, we rely on the **Aim-and-Shoot** definitions established in Section 3.1. Physically, the acquisition process is sequential: the anatomical scene is first spatially transformed (aimed), and only then is the projected light field discretised and corrupted by sensor noise (shot). Let $(\Omega, \mathcal{I})$ be the Baseline Image. We define the **Observed Realisation**, denoted

$$\mathcal{I}_{obs} : \Omega \rightarrow \mathbb{R},$$

as the final discrete intensity function obtained by the sequential application of the mechanical and radiometric operators

$$\mathcal{I}_{obs} := T_S \circ (T_A \circ \mathcal{I}).$$

Hence, if $\mathcal{I}$ is the Baseline Image and let $\mathcal{I}_{obs}$ is the Observed Realisation, it readily follows from the the Algebraic Stability Theorem (Theorem 2.13) that the functional residual strictly bounds the topological error:

$$d_B(\mathcal{D}(\mathcal{I}), \mathcal{D}(\mathcal{I}_{obs})) \leq \|\mathcal{I} - \mathcal{I}_{obs}\|_{\infty}.$$

Decoupling the bounds using the triangle inequality yields

$$d_B(\mathcal{D}(\mathcal{I}), \mathcal{D}(\mathcal{I}_{obs})) \leq \|\mathcal{I} - \mathcal{I}_{obs}\|_\infty \leq \underbrace{\|\mathcal{I} - \mathcal{I}_{aim}\|_\infty}_{\text{Spatial Shift}} + \underbrace{\|\mathcal{I}_{aim} - T_S(\mathcal{I}_{aim})\|_\infty}_{\text{Radiometric Deviation}}.$$

Then, applying 3.11 and Corollary 3.8 to these summands yields the following proposition.

Proposition 3.13 (A COMPUTATIONAL STABILITY BOUND). *Let $L_{\mathcal{I}}$ be the Graph Lipschitz constant (signal volatility), let λ be the maximum spatial displacement (from alignment or blur), and let ϵ_{rad} be the aggregate radiometric error (linearity deviation + noise). The total topological instability under the aim-and-shoot model with protocol standardisation is bounded by:*

$$d_B(\mathcal{D}(\mathcal{I}), \mathcal{D}(\mathcal{I}_{obs})) \leq \underbrace{\lceil \lambda \rceil L_{\mathcal{I}}}_{\text{Mechanical Residual}} + \underbrace{\epsilon_{rad}}_{\text{Radiometric Bound}}.$$

Mechanistic Parameterisation of Failure. Proposition 3.13 shows that topological error on a discrete lattice is controlled by the interaction between spatial interpolation ($\lceil \lambda \rceil$) and the discrete image gradient ($L_{\mathcal{I}}$). The point is not that algebraic stability fails, but that sampled acquisition converts geometric misalignment into functional residual on the grid. Under grid-aligned isometries ($\lambda = 0$), the mechanical term vanishes and the bound reduces to $d_B \leq \epsilon_{rad}$. When sharp gradients are present, however, even sub-pixel misalignment can dominate the mechanical residual. This explains why a complementary geometric control is useful, while leaving the actual operational thresholds to the empirical audit in Chapter 5.

Chapter 4

Reliability Engineering

This chapter turns the computational stability analysis into an operational diagnostic framework. If Chapter 3 separates topological error into mechanical and radiometric terms, the present chapter asks how that separation can be detected when a deployed retinal imaging system begins to fail.

The motivating setting is domain shift: changes in acquisition hardware, alignment, or illumination that break the assumptions under which a model was trained [36–39]. Rather than treating such failure as a single drop in predictive accuracy, we recast the problem as one of *reliability engineering* [40–42]: the aim is to infer whether degraded performance is primarily mechanical, arising from misalignment and interpolation, or radiometric, arising from illumination drift or sensor noise.

We do this by pairing descriptors with complementary failure envelopes. Geometric summaries, such as moments and spatial integrals, are often comparatively stable under rigid motion but brittle under radiometric drift [43]. Topological summaries behave differently: in our setting they are comparatively stable under some radiometric distortions yet fragile under interpolation-heavy transformations. Their agreement and disagreement are organised into the **Differential Failure Matrix**, where geometry acts as a **Mechanical Control** and topology as a **Radiometric Control**. The feature design is intentionally constrained, because the goal is not maximal predictive accuracy but an interpretable low-dimensional summary whose failure remains legible under controlled perturbation.

4.1 Vectorisations and Theoretical Failure Modes

Standard statistical classifiers require fixed-length feature vectors in $\mathbb{R}^n$, rather than multisets of intervals. Where $(\mathcal{D}, d_B)$ is the space of persistence diagrams, a **vectorisation map** is a function¹

$$\Phi : (\mathcal{D}, d_B) \rightarrow (\mathbb{R}^n, d_2).$$

To preserve the stability guarantees derived in Proposition 3.13, Φ must be Lipschitz continuous with respect to the bottleneck distance. Specifically, there must exist a constant L_Φ such that for any two diagrams $\mathcal{D}, \mathcal{D}'$:

$$\|\Phi(\mathcal{D}) - \Phi(\mathcal{D}')\|_2 \leq L_\Phi \cdot d_B(\mathcal{D}, \mathcal{D}').$$

Then, applying the computational stability bound (Proposition 3.13) yields the following theorem.

Theorem 4.1 (TOTAL COMPUTATIONAL STABILITY). *Let Φ be an L_Φ -Lipschitz vectorisation map. Then,*

$$\|\Phi(\mathcal{D}(\mathcal{I})) - \Phi(\mathcal{D}(\mathcal{I}_{obs}))\|_2 \leq L_\Phi (\lceil \lambda \rceil L_{\mathcal{I}} + \epsilon_{rad}).$$

Theorem 4.1 formalises the trade-off introduced by the vectorisation step. High-fidelity vectorisations (for example, Persistence Landscapes [44]) preserve more structural detail, but a large Lipschitz constant² ($L_\Phi \gg 1$) can amplify the mechanical residual $\lceil \lambda \rceil L_{\mathcal{I}}$. Coarser vectorisations reduce that amplification by

¹In finite-dimensional vector spaces, the norms induced by ℓ_1 , ℓ_2 , and ℓ_∞ are topologically equivalent. We select the ℓ_2 -metric to align with the Tikhonov regularisation employed in Chapter 5.

²The precise derivation of amplified Lipschitz constants under dense discretisation grids is beyond the scope of this thesis.

bounding the effective complexity of the summary, either combinatorially (as with Top- k summaries) or through truncation of low-persistence features.

On a discrete grid there is no single context-free notion of a "small" perturbation. Theorem 4.1 instead shows that allowable output variation depends jointly on the discrete image gradient $L_{\mathcal{I}}$, the interpolation scale λ , and the radiometric term ϵ_{rad} . Table 4.1 summarises representative failure modes suggested by this bound.

Table 4.1: Representative failure modes suggested by the computational stability bound.

Failure Mode	Description	Potential Mitigation
Resolution Trade-off	Downsampling acts as a low-pass filter (reducing $L_{\mathcal{I}}$ and mechanical error), but excessive reduction aliases physical features (e.g., microaneurysms) into the background.	Select an intermediate operating resolution that balances spatial discretisation against suppression of high-frequency sensor noise.
Vignetting	Radial illumination decay introduces a spatially dependent drift $\eta(x)$, causing a location bias where peripheral features prematurely fall below the filtration threshold.	Apply flat-field normalisation (e.g., CLAHE) to enforce a consistent physiological intensity threshold across all spatial coordinates.
Compression Artefacts	Lossy compression (JPEG) introduces grid-aligned block artefacts. The filtration may misinterpret these artificial gradients as genuine local extrema.	Prefer lossless formats (PNG, TIFF, DICOM). If compressed historical archives must be analysed, apply mild smoothing and interpret the resulting topology cautiously.
Homological Shattering	High-gradient anatomical structures ($L_{\mathcal{I}} \gg 0$), such as vessels, fragment under sub-pixel interpolation, reducing β_1 and artificially inflating β_0 .	Integrate topological descriptors with orthogonal geometric controls (e.g., Hu Moments) and employ coarse vectorisation strategies.

4.2 The Differential Failure Matrix

The theoretical failure modes above describe what can go wrong, but not yet how a deployed system should distinguish one mode from another. The practical problem is domain shift: performance drops after deployment, and the question becomes whether the underlying fault is primarily mechanical or radiometric.

To make that distinction, we evaluate two auxiliary probes trained on the clean baseline. The first is the topological vector $\mathbf{H}_0\mathbf{H}_1\mathbf{H}_S$, which acts as a **Radiometric Control**: by Proposition 3.13, it is comparatively stable under some radiometric perturbations but fragile under mechanical misalignment and interpolation. The second is the Log-Hu vector, which acts as a **Mechanical Control**: as a globally integrated geometric summary, it is comparatively stable under rigid spatial perturbations but vulnerable to radiometric washout. Their agreement and disagreement organise the fault-isolation rule summarised in the Differential Failure Matrix below.

	Topology Pass ($\mathbf{H}_0\mathbf{H}_1\mathbf{H}_S$)	Topology Fail ($\mathbf{H}_0\mathbf{H}_1\mathbf{H}_S$)
Geometry Pass (Log-Hu)	Operational Envelope: Macroscopic integrity remains intact. The primary classifier's failure stems from a limit in diagnostic resolution or an unmodelled covariate orthogonal to both controls.	Mechanical Issue (Aim): Topological shattering alongside geometric survival isolates a mechanical fault.
Geometry Fail (Log-Hu)	Radiometric Issue (Shoot): Topological survival alongside geometric washout indicates distorted absolute intensity integrals but preserved structural order. This isolates a radiometric fault.	Catastrophic Failure: Mutual collapse violates the single-mode failure assumption, indicating simultaneous mechanical and radiometric corruption or a severe optical anomaly.

4.3 Probe Construction

To build the differential framework, we now define the two vectorisations and explain why their stability profiles are deliberately complementary.

Mechanical Control: Log-Hu Invariant Moments

We employ **Hu Invariant Moments** [45, 46] as the mechanical control vector ($\mathbf{v}^{\text{geo}} \in \mathbb{R}^7$). While the computer vision literature offers numerous shape descriptors invariant to rigid motion (such as Zernike moments [47] or Fourier descriptors [48]), Hu Moments are uniquely suited to this differential framework because they operate directly on the global intensity function with minimal preprocessing.

Let $(\Omega, \mathcal{I})$ be a digital image defined on the finite grid $\Omega \subset \mathbb{Z}^2$. The *raw image moment* of order (p, q) is the spatial integral of the intensity:

$$M_{pq} := \sum_{(x,y) \in \Omega} x^p y^q \mathcal{I}(x, y).$$

The coordinates of the spatial centroid are given by $(\bar{x}, \bar{y}) = (M_{10}/M_{00}, M_{01}/M_{00})$. To establish translation invariance, the *central moments* are defined by shifting the origin to the centroid:

$$\mu_{pq} := \sum_{(x,y) \in \Omega} (x - \bar{x})^p (y - \bar{y})^q \mathcal{I}(x, y).$$

To establish scale invariance, the *scale-normalised central moments* are defined for $p + q \geq 2$ as:

$$\eta_{pq} := \frac{\mu_{pq}}{\mu_{00}^\gamma}, \quad \text{where } \gamma = 1 + \frac{p+q}{2}.$$

The **Hu Moments** $(\phi_1, \dots, \phi_7) \in \mathbb{R}^7$ are a set of seven distinct polynomial combinations of the normalised central moments η_{pq} up to order three. By construction, these integral descriptors guarantee global invariance under continuous translation, scale, and rigid rotation. However, this precise reliance on global integration renders them acutely susceptible to radiometric variance. This radiometric fragility, coupled with mechanical invariance, yields a stability profile that is functionally orthogonal to persistent homology.

Remark 4.2 The algebraic expansion of the seven Hu moment polynomials is standard in the computer vision literature. To ensure numerical stability and reproducibility, we delegate their computation to the optimised `cv2.HuMoments` function from the OpenCV library [46].

Log-Hu Moments [49] Raw Hu moments exhibit extreme variance in magnitude, so to ensure numerical stability for the linear probe, we define the geometric feature vector $\mathbf{v}_{\text{geo}} \in \mathbb{R}^7$ via a log-modulus transformation. We define the *ϵ -stabilised log-transformed Hu Moments* for $i \in \{1, \dots, 7\}$ as:

$$h_i := -\text{sgn}(\phi_i) \log_{10}(|\phi_i| + \epsilon)$$

where $\text{sgn}(\cdot)$ is the signum function and $\epsilon > 0$ is a strictly positive stabilisation constant.

Remark 4.3 (ANALYTICAL PROPERTIES OF THE GEOMETRIC CONTROL) Because the geometric map Φ_{geo} operates directly on the spatial pixel domain [49] rather than the persistence diagram, it is fundamentally exempt from the Lipschitz constraints of the Total Computational Stability Bound (Theorem 4.1). While mathematically invariant to rigid motion, Φ_{geo} relies strictly on absolute spatial integration. Consequently, it is inherently vulnerable to radiometric perturbations, as uniform intensity drift irreversibly alters the underlying mass integrals [43].

Radiometric Control: Summary Persistence Statistics.

For the radiometric control, we select summary persistence statistics that are devoid of spatial information. For computational efficiency, we evaluate Top- k Persistence Sums and Truncated Total Persistence.

Top- k Persistence Sum. Let $\mathcal{D}$ be a persistence diagram. Let $P = (p_1, p_2, \dots)$ be the sequence of finite persistences $p_i = d_i - b_i$ for all features $(b_i, d_i) \in \mathcal{D}$, sorted in descending order such that $p_1 \geq p_2 \geq \dots \geq 0$.

If the cardinality of the diagram is strictly less than k ($|\mathcal{D}| < k$), the sequence is padded by setting $p_i = 0$ for all $i > |\mathcal{D}|$. The *Top- k persistence sum* is defined as:

$$S_k(\mathcal{D}) := \sum_{i=1}^k p_i.$$

Truncated Total Persistence. Let $\mathcal{D}$ be a persistence diagram. Let $p(b, d) = d - b$ denote the persistence of a finite feature $(b, d) \in \mathcal{D}$. For a threshold $\tau \geq 0$, the τ -truncated persistence is defined via the continuous retraction $p^\tau(b, d) := \max(0, p(b, d) - \tau)$. The τ -truncated total persistence is defined as the sum of these truncated values over all finite features:

$$E_{\text{tot}}^\tau(\mathcal{D}) := \sum_{(b,d) \in \mathcal{D}, d < \infty} p^\tau(b, d).$$

The theoretical guarantees of the Computational Stability Bound apply strictly at the diagrammatic level (Proposition 3.13). To invoke the Total Computational Stability Bound (Theorem 4.1), these two vectorisation maps must be Lipschitz continuous.

Lemma 4.4 (STABILITY OF THE TOP- k PERSISTENCE SUM). *Let $\mathcal{D}$ and $\mathcal{D}'$ be persistence diagrams such that $d_B(\mathcal{D}, \mathcal{D}') \leq \delta$. Let $p_1, \dots, p_k$ and $p'_1, \dots, p'_k$ be the persistences of the k most persistent features of $\mathcal{D}$ and $\mathcal{D}'$, respectively. Then,*

$$|S_k(\mathcal{D}) - S_k(\mathcal{D}')| \leq 2k\delta.$$

Hence, S_k is $2k$ -Lipschitz.

Proof. Let $\gamma : \mathcal{D} \cup \Delta \rightarrow \mathcal{D}' \cup \Delta$ be an optimal bottleneck matching such that $\|x - \gamma(x)\|_\infty \leq \delta$ for all x . For any matched pair $x = (b, d)$ and $y = \gamma(x) = (b', d')$ strictly off the diagonal, the variation in persistence $p(x) = d - b$ is bounded via the triangle inequality:

$$|p(x) - p(y)| = |(d - b) - (d' - b')| \leq |d - d'| + |b - b'| \leq 2\delta.$$

If a feature $x = (b, d)$ is matched to its projection on the diagonal Δ , the ℓ_∞ -distance is strictly $\frac{d-b}{2} \leq \delta$, implying $p(x) \leq 2\delta$. Since its matched diagonal feature possesses a trivial persistence of zero, the bound

$$|p(x) - 0| \leq 2\delta$$

holds globally across the entire matching. Because γ constitutes a bijection between the augmented diagrams where no matched pair deviates by more than 2δ in persistence, the stability of order statistics under bounded bijections guarantees that the descending sorted sequences of persistences $P(\mathcal{D}) = (p_1, p_2, \dots)$ and $P(\mathcal{D}') = (p'_1, p'_2, \dots)$ must satisfy the pointwise inequality $|p_i - p'_i| \leq 2\delta$ for all rank indices i . Applying this pointwise bound to the summation yields:

$$|S_k(\mathcal{D}) - S_k(\mathcal{D}')| = \left| \sum_{i=1}^k p_i - \sum_{i=1}^k p'_i \right| \leq \sum_{i=1}^k |p_i - p'_i| \leq 2k\delta.$$

That S_k is $2k$ -Lipschitz immediately follows. $\square$

Remark 4.5 (THEORETICAL VS. OPERATIONAL LIPSCHITZ BOUNDS) For a standard 2048×2048 fundus image, the ambient complex cardinality $N_{\text{max}} = |K(\Omega)| \gg 10^7$ yields a catastrophically large theoretical Lipschitz constant. This worst-case bound assumes every local extremum survives truncation. Operationally, setting the truncation parameter τ above the optical noise floor annihilates transient generators ("topological dust"), reducing the effective feature multiplicity from N_{max} to a clinically negligible integer. Consequently, while S_k guarantees stability algebraically via a strict $2k$ bound (Lemma 4.4), E_{tot}^τ achieves stability operationally by driving the noise-driven feature count toward zero.

Lemma 4.6 (LIPSCHITZ CONTINUITY OF TRUNCATED TOTAL PERSISTENCE). *Let $\mathcal{D}$ and $\mathcal{D}'$ be persistence diagrams defined on a fixed discrete domain Ω , such that $d_B(\mathcal{D}, \mathcal{D}') \leq \delta$. Then, the difference in their truncated total persistence is strictly bounded by:*

$$|E_{tot}^\tau(\mathcal{D}) - E_{tot}^\tau(\mathcal{D}')| \leq 2N_{\max}\delta \quad \text{where} \quad N_{\max} = |K(\Omega)|.$$

Proof. Let $\gamma : \mathcal{D} \cup \Delta \rightarrow \mathcal{D}' \cup \Delta$ be an optimal bottleneck matching. For any matched pair $x = (b, d)$ and $y = \gamma(x)$, the variation in raw persistence $p(x) = d - b$ satisfies $|p(x) - p(y)| \leq 2\delta$ via the triangle inequality. Because the truncation operator $f(p) = \max(0, p - \tau)$ is globally 1-Lipschitz, this bound is strictly preserved, i.e.

$$|p^\tau(x) - p^\tau(y)| \leq 2\delta.$$

On the finite discrete domain Ω , the total number of homological generators is strictly bounded by the number of faces of all dimensions within the ambient complex, $N_{\max} = |K(\Omega)|$. We represent the truncated persistences of $\mathcal{D}$ and $\mathcal{D}'$ as descending sorted sequences of length $N_{\max}$, padded with zeros:

$$P^\tau(\mathcal{D}) = (p_1^\tau, \dots, p_{N_{\max}}^\tau) \quad \text{and} \quad P^\tau(\mathcal{D}') = (p_1'^\tau, \dots, p_{N_{\max}}'^\tau).$$

Because γ constitutes a bijection where no matched pair deviates by more than 2δ , the stability of order statistics enforces the pointwise inequality $|p_i^\tau - p_i'^\tau| \leq 2\delta$ for all rank indices i . Applying this pointwise bound to the summation yields:

$$|E_{tot}^\tau(\mathcal{D}) - E_{tot}^\tau(\mathcal{D}')| \leq \sum_{i=1}^{N_{\max}} |p_i^\tau - p_i'^\tau| \leq 2N_{\max}\delta.$$

Thus, the vectorisation map $E_{tot}^\tau : (\mathcal{D}, d_B) \rightarrow (\mathbb{R}, d_2)$ is $2N_{\max}$ -Lipschitz. $\square$

Each fundus image $(\Omega, \mathcal{I})$ generates three distinct persistence diagrams: $\mathcal{D}_0$ representing $\mathbf{H}_0$ sublevel features (dark connected components), $\mathcal{D}_1$ representing $\mathbf{H}_1$ sublevel features (dark vascular loops), and $\mathcal{D}_S$ representing $\mathbf{H}_0$ superlevel features (bright components, such as exudates)³.

Definition 4.7 (TOPOLOGICAL FEATURE VECTOR). Let $\mathcal{D}_0, \mathcal{D}_1$, and $\mathcal{D}_S$ be the persistence diagrams generated from an image $(\Omega, \mathcal{I})$ via the defined sublevel and superlevel filtrations. The *topological feature vector* $\mathbf{v}_{\text{topo}} \in \mathbb{R}^6$ is defined as

$$\mathbf{v}_{\text{topo}} := (S_k(\mathcal{D}_0), E_{tot}^\tau(\mathcal{D}_0), S_k(\mathcal{D}_1), E_{tot}^\tau(\mathcal{D}_1), S_k(\mathcal{D}_S), E_{tot}^\tau(\mathcal{D}_S))^\top.$$

This yields a 6-D radiometric control⁴, which is structurally comparable to the 7-D geometric control.

Proposition 4.8 (LIPSCHITZ CONTINUITY OF THE TOPOLOGICAL VECTOR). *Under the conditions of Lemmata 4.4 and 4.6, the joint vectorisation map $\Phi_{\text{topo}} : (\mathcal{D}, d_B)^3 \rightarrow (\mathbb{R}^6, d_2)$ generating the concatenated feature vector $\mathbf{v}_{\text{topo}}$ in Definition 4.7 is*

$$2\sqrt{3(k^2 + N_{\max}^2)}\text{-Lipschitz.}$$

Proof. Let $\delta = \max_{* \in \{0,1,S\}} d_B(\mathcal{D}_*, \mathcal{D}'_*)$. By Lemmata 4.4 and 4.6, the scalar components satisfy:

$$|S_k(\mathcal{D}_*) - S_k(\mathcal{D}'_*)| \leq 2k\delta \quad \text{and} \quad |E_{tot}^\tau(\mathcal{D}_*) - E_{tot}^\tau(\mathcal{D}'_*)| \leq 2N_{\max}\delta.$$

³To preclude notational ambiguity, the subscript S is employed for the superlevel filtration to distinguish it from the sublevel $\mathcal{D}_0$.

⁴While recent literature [50–52] cites Alexander Duality to justify treating sublevel and superlevel filtrations as redundant, this assumes continuous topology. On discrete lattices, this assumption breaks down. Bleile et al. [53] demonstrate that strict persistent duality requires alternating between dual pixel connectivity models and compactifying image boundaries. Under standard uniform 8-connectivity, this duality is violated (the Rosenfeld Paradox [54]). Furthermore, even if algebraically dual, forcing a model to identify a bright localised exudate ($\mathbf{H}_S$ component) via its dual representation as a dark global ring ($\mathbf{H}_1$ loop) fundamentally destroys the physical interpretability of the diagnostic signal.

Let $\mathbf{v}_{\text{topo}}$ and $\mathbf{v}'_{\text{topo}}$ be the vectorisations corresponding to $(\mathcal{D}_0, \mathcal{D}_1, \mathcal{D}_S)$ and $(\mathcal{D}'_0, \mathcal{D}'_1, \mathcal{D}'_S)$, respectively. Then,

$$\begin{aligned} \|\mathbf{v}_{\text{topo}} - \mathbf{v}'_{\text{topo}}\|_2 &= \sqrt{\sum_{* \in \{0,1,S\}} \left(|S_k(\mathcal{D}_*) - S_k(\mathcal{D}'_*)|^2 + |E_{\text{tot}}^\tau(\mathcal{D}_*) - E_{\text{tot}}^\tau(\mathcal{D}'_*)|^2 \right)} \\ &\leq \sqrt{3(2k\delta)^2 + 3(2N_{\max}\delta)^2} \\ &= 2\sqrt{3(k^2 + N_{\max}^2)}\delta. \end{aligned}$$

Thus, the joint vectorisation map is $2\sqrt{3(k^2 + N_{\max}^2)}$ -Lipschitz. $\square$

Classifier: The Linear Probe

The Total Computational Stability Bound (Theorem 4.1) dictates that mechanical perturbations (λ) and radiometric perturbations (ϵ_{rad}) inflate the error bounds asymmetrically depending on the chosen invariant. To validate the functional orthogonality upon which the Differential Failure Matrix relies, we formalise this dichotomy as a testable hypothesis.

Robustness decay rate. Let $\mathcal{J}$ be the space of retinal images and let $\Theta_\epsilon : \mathcal{J} \rightarrow \mathcal{J}$ be a degradation operator parameterised by magnitude $\epsilon \geq 0$. For a generic vectorisation map $\Phi : \mathcal{J} \rightarrow \mathbb{R}^n$, we define the *Robustness Decay Rate* over an operational envelope $E = [0, \epsilon_{\max}]$ as the normalised integral of the performance degradation of the linear probe:

$$\mathcal{R}(\Phi) := \frac{1}{\epsilon_{\max}} \int_0^{\epsilon_{\max}} \frac{\text{Acc}(\Phi \circ \Theta_0) - \text{Acc}(\Phi \circ \Theta_\epsilon)}{\epsilon} d\epsilon.$$

Definition 4.9 (THE COARSE VECTORISATION HYPOTHESIS). Let $\Phi_{\text{geo}} : \mathcal{J} \rightarrow \mathbb{R}^7$ be the geometric vectorisation map and $\Phi_{\text{topo}} : \mathcal{J} \rightarrow \mathbb{R}^6$ be the joint topological vectorisation map. The hypothesis posits an **Orthogonal Stability Profile** consistent with the Total Computational Stability Bound (Theorem 4.1):

1. **Radiometric Regime:** Where interpolation error $\lambda \rightarrow 0$ but radiometric error $\epsilon_{\text{rad}} > 0$, topology is expected to exhibit a slower decay profile: $\mathcal{R}(\Phi_{\text{topo}}) < \mathcal{R}(\Phi_{\text{geo}})$.
2. **Mechanical Regime:** Where radiometric error $\epsilon_{\text{rad}} \rightarrow 0$ but interpolation error $\lambda > 0$, geometry is expected to exhibit a slower decay profile: $\mathcal{R}(\Phi_{\text{geo}}) < \mathcal{R}(\Phi_{\text{topo}})$.

Remark 4.10 (AN ANALYTICAL NOTE ON THE DECAY INTEGRAL) Because our empirical evaluation relies on a finite validation cohort, the measured accuracy $\text{Acc}(\epsilon)$ is strictly a discrete step function. The integral defined in $\mathcal{R}(\Phi)$ represents the continuous limit of this discrete process. By evaluating the limit as $\epsilon \rightarrow 0$ in the denominator, this formulation deliberately places a logarithmic penalty on models that fail early in the perturbation envelope. This satisfies the strict engineering requirement to aggressively penalise acute structural fragility.

The Linear Probe. To evaluate the Robustness Decay Rate, we isolate feature quality from classifier capacity. Complex non-linear models, such as kernel support vector machines or random forests, can compensate for degraded feature spaces by learning more elaborate decision boundaries [55]. To reduce that confounding effect, and to ensure that any observed performance difference is attributable to the topological or geometric signal itself, we use a **linear probe** [56] implemented via logistic regression.

By restricting the hypothesis class to linear hyperplanes, the classifier is denied the flexibility to learn around the degradation operators. If the Coarse Vectorisation Hypothesis holds, the respective mapping $\Phi(\mathcal{J})$ must effectively unroll the complex data manifold into a flat vector space where a hyperplane can successfully separate the classes despite the injection of noise. Consequently, the performance of the linear probe serves as a direct proxy for the linear separability (and hence, the inherent accessibility and robustness) of the underlying vectorisations.

Model Specification. Let $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ denote the binary-labelled dataset. We model the probability of Diabetic Retinopathy $P(y = 1|\mathbf{x})$ using the logistic sigmoid function $\sigma(\mathbf{w}^\top \mathbf{x} + b)$. The optimisation objective is the standard negative log-likelihood with **Tikhonov (ℓ_2) regularisation** [57]. This penalty is mathematically required to manage inherent multicollinearity: the *Top- k Persistence Sum* (S_k) is a partial sum of, and naturally covaries with, the *Truncated Total Persistence* (E_{tot}^τ). Both must be retained as they encode distinct clinical realities— S_k measures the acute severity of dominant lesions, whilst E_{tot}^τ captures the global density of widespread pathology. Ridge (ℓ_2) regularisation stabilises parameter estimation under this strict correlation without performing variable selection (as Lasso would), thereby preserving the complete topological signature.

We now proceed to Chapter 5, which serves as the deployment of this proof-of-concept architecture on the FIVES dataset.

Chapter 5

Experiments

This chapter tests the diagnostic framework developed in Chapter 4. The computational stability bound predicts which perturbations should preserve topological information and which should not, but the point at which a probe becomes operationally unreliable is an empirical question. The experiments therefore subject the geometric and topological probes to a structured sensitivity audit on retinal fundus images.

Reproducibility The experimental pipeline is reproducible from the accompanying repository:

<https://github.com/rejosephnieto/ph>

Repository documentation records the software environment, data-access requirements, and rebuild procedure for the analysis.

5.1 Cohort and Probe Definition

We validate the framework on the FIVES (Fundus Image Vessel Segmentation) dataset [1], restricting attention to the Normal and Diabetic Retinopathy cohorts. For the sensitivity audit, the official partitions are pooled into a balanced cohort ($N = 400$, with 200 Normal and 200 DR images) and evaluated under Stratified 5-Fold Cross-Validation, so that degradation effects are measured against a common baseline rather than against pre-existing split effects. The baseline image is taken to be the green channel and is processed with Contrast Limited Adaptive Histogram Equalisation (CLAHE) [58], which keeps the intensity range controlled for the stability framework of Chapter 3.

Probe Instantiation. To construct the parallel probes, we use simple dimensional projections rather than high-fidelity vectorisations. Discrete features such as Betti numbers are excluded because feature counts are highly discontinuous with respect to pixel noise and therefore violate the Lipschitz continuity required by the Total Computational Stability Bound.

Radiometric Probe (Φ_{topo}): A simple $\mathbb{R}^6$ projection defined by the combined vector $\mathbf{H}_0\mathbf{H}_1\mathbf{H}_S$. To ground the model in clinical reality, the Top- k persistence sum (S_5) tracks $k = 5$ generators, pre-empting the NHS referral threshold of 8 lesions [6]. A strict truncation threshold of $\tau = 5.0$ is applied to the total persistence (E_{tot}^5) to systematically prune high-frequency stochastic noise ("topological dust").

Mechanical Probe (Φ_{geo}): An $\mathbb{R}^7$ projection composed of the Log-Hu Moments ($\phi_1 \dots \phi_7$).

A preliminary univariate audit supports the choice of probes: the topological Top-5 Persistence Sum (S_5) emerges as the strongest pathological signal, while the geometric invariants provide weaker but more globally integrated baselines. This is consistent with the Orthogonality Condition: Hu moments are engineered for global shape invariance, whereas early retinal pathology is largely textural and stochastic.

Implementation and Isolation Controls. The linear probes are trained exclusively on the Baseline folds (F_{train}) and evaluated on the perturbed counterparts F_j^c without retraining. Feature vectors are standardised using training statistics alone, and the fitted scaler is then applied to validation partitions to prevent data leakage. Models are instantiated via `LogisticRegression` from the `scikit-learn` library [59]. To account for slight class imbalance in the training folds, we apply `class_weight='balanced'`, which scales penalties

inversely to class frequencies. The solver is set to `liblinear` for numerical stability on the restricted dataset size ($N = 400$), with the fixed `GLOBAL_SEED` used throughout the pipeline for reproducibility.

5.2 Evaluation Methodology

The validation objective is to measure stability (invariance under degradation) rather than strictly optimising accuracy on pristine data. We adopt the "Train on Clean, Test on Corrupt" paradigm [60].

Isolation Controls. As specified in Section 5.1, both probes are trained only on the clean baseline folds and then evaluated on perturbed counterparts without retraining. The purpose is to ensure that any observed drop is attributable to feature instability under Θ_ϵ , rather than to adaptation of the classifier itself.

The Sensitivity Audit Protocol. The cohort is subjected to ten distinct perturbation regimes (Θ_ϵ), restricted to the valid retinal field. The parameter ranges extend into severe degradation to forcibly discover the mathematical breaking points at which the parallel probes lose structural integrity.

Protocol	Range (ϵ)	Implementation	Protocol	Range (ϵ)	Implementation
Radiometric Stress			Standard Baseline		
Gamma	$\gamma \in [0.1, 3.0]$	$I_{out} = I_{in}^\gamma$	Standard	—	No perturbation
Contrast	$\alpha \in [0.5, 3.0]$	Linear scale	Mechanical Stress		
Drift	$\delta \in [-150, 150]$	Offset clip	Rotation	$\pm 10^\circ$	Bi-linear interpolation
Gaussian	$\eta \in [0.0, 0.2]$	Correlated noise	Blur	$\sigma \in [0.0, 1.5]$	Gaussian kernel
Poisson	$s \in [0.0, 0.2]$	Shot + read noise	Resolution	$64px \dots 2048px$	Area resize
S & P	$p \leq 0.025$	Impulse noise			
Bit Depth	$k \in \{2, \dots, 8\}$	Uniform quantisation			

Table 5.1: Summary of perturbation regimes (Θ_ϵ). Radiometric protocols simulate failures of the sensor hardware. Mechanical protocols simulate failures in physical alignment and optical focus.

Primary Metric: Robustness Gap. The *Robustness Gap* $\Delta(\epsilon)$ at perturbation magnitude ϵ is defined as the difference in mean cross-validation accuracy between the Radiometric Control and the Mechanical Control:

$$\Delta(\epsilon) := \text{Acc}(\Phi_{\text{topo}} \circ \Theta_\epsilon) - \text{Acc}(\Phi_{\text{geo}} \circ \Theta_\epsilon).$$

Monitoring the sign and magnitude of this gap is the exact mathematical mechanism that routes a silent failure into a specific diagnostic quadrant. A strictly positive gap ($\Delta > 0$) during a domain shift isolates a radiometric issue, whereas an inversion to a negative gap ($\Delta < 0$) mathematically isolates a mechanical alignment issue.

The Operational Envelope (Pass / Pass) The baseline discriminative utility of the respective probes on pristine data ($\epsilon = 0$) is established. This dictates the **Operational Envelope (Pass / Pass)** quadrant of the Differential Failure Matrix.

Feature	Components	Dim	Accuracy	Gap (Δ)
Mechanical Control Probe				
Geometry	Hu Moments ($\phi_1 \dots \phi_7$)	7	70.70%	0.0%
Topological Sub-components				
H₁	Vascular Loops (S_5, E_{tot}^τ)	2	70.30%	−0.4%
H₀	Dark Lesions (S_5, E_{tot}^τ)	2	81.70%	+11.0%
H_S	Bright Lesions (S_5, E_{tot}^τ)	2	85.70%	+15.0%
Radiometric Control Probe				
H₀H₁H_S	Combined Topological Summary	6	86.70%	+16.0%

Table 5.2: The Operational Envelope (Baseline). Performance of the parallel probes before degradation ($\epsilon = 0$). Accuracy is reported as the mean over 5 stratified folds.

The Mechanical Probe achieves an accuracy of 70.70%, while the full Radiometric Probe reaches 86.70%. Consequently, the baseline Robustness Gap is strictly positive at $\Delta(0) = +16.0\%$. Any significant deviation from these thresholds under the degradation operators Θ_ϵ triggers a transition into the failure quadrants.

5.3 Results and Analysis

This section maps the comparative stability curves derived from the Sensitivity Audit directly into the quadrants of the Differential Failure Matrix.

The Mechanical Issue Quadrant (Geo Pass / Topo Fail) The Mechanical Stress tests evaluate vulnerability to spatial and grid-level perturbations. In this regime, the hypothesis predicts a **negative gap** ($\Delta < 0$), where the global integration of the Mechanical Probe outlasts the brittle grid-dependency of the Radiometric Probe. The results (Figure 5.1) empirically validate this theoretical asymmetry.

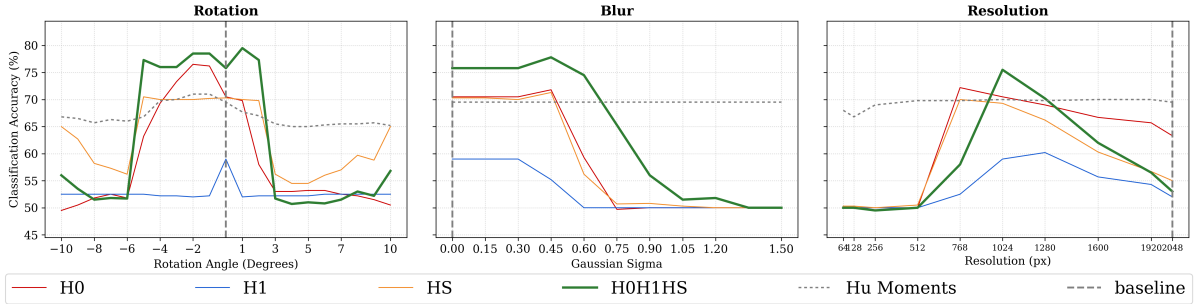


Figure 5.1: Triggering the **Mechanical Issue Quadrant**. Performance of the Radiometric Probe (solid lines) and Mechanical Probe (dotted grey). **Left (Rotation):** The Mechanical Probe is invariant, whilst the Radiometric Probe degrades rapidly under sub-pixel interpolation ($\Delta < 0$). **Middle (Blur):** Geometry remains stable under global smoothing. **Right (Resolution):** Geometry exhibits continuous scale invariance, whereas Topology is sensitive to sampling density.

Rotation. The rotation panel shows the practical importance of registration. The Mechanical Probe remains largely invariant ($\approx 68\%$ accuracy) across the $\pm 10^\circ$ range, consistent with continuous moment theory. The Radiometric Probe, by contrast, falls toward near-random performance ($\approx 55\%$) for rotations as small as $|\theta| = 3^\circ$. The resulting divergence matches the intended diagnostic logic: isolated topological failure points to loss of spatial registration rather than loss of pathological signal.

Blur. The blur panel shows the difference between global integral descriptors and local gradient-based summaries. The Mechanical Probe remains comparatively stable under Gaussian smoothing, whereas the Radiometric Probe degrades rapidly once σ exceeds 0.6 and fails by $\sigma = 0.9$. As blur merges local maxima into the background, the persistent generators disappear from the filtration.

Resolution. The resolution panel reveals an interior optimum for the Radiometric Probe at approximately 1024px. Performance deteriorates at low resolutions (< 512 px), where physical features are aliased, and again at the native resolution (2048 px), where high-frequency sensor variation re-enters the summary. In this setting, moderate downsampling acts as a useful low-pass filter.

The Radiometric Issue Quadrant (Geo Fail / Topo Pass) The Radiometric Stress tests evaluate resilience against sensor illumination and exposure anomalies. Here, the hypothesis predicts a **positive gap** ($\Delta > 0$), where the structural connectivity of the Radiometric Probe outlasts the rigid intensity integrals of the Mechanical Probe. The results (Figure 5.2) confirm this functional asymmetry.

Drift. As global illumination increases, the Mechanical Probe degrades rapidly ($\delta > 20$) because the absolute intensity integrals are distorted. The Radiometric Probe maintains near-peak performance ($> 75\%$) well into the saturation regime ($\delta \approx 100$). Since the relevant topological features depend on relative critical pairings rather than on absolute intensity, the structural signal remains intact under substantial positive drift. Under sufficiently strong negative drift ($\delta < -50$), however, both probes collapse as the image gradients themselves disappear.

Gamma and Contrast. In the Gamma regime, the Radiometric Probe exhibits a convex performance boost for $\gamma > 1.0$, indicating that non-linear contrast stretching isolates the critical points within the per-

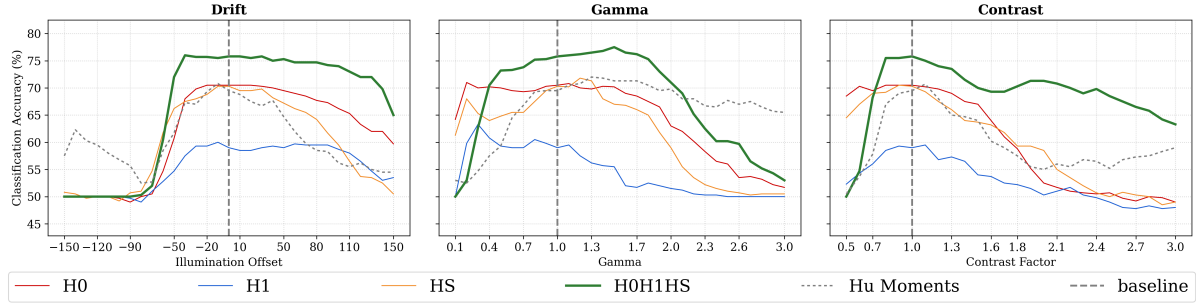


Figure 5.2: Triggering the Radiometric Issue Quadrant. **Left (Drift):** Positive drift reveals topological robustness against saturation, triggering the quadrant. Extreme negative drift causes mutual collapse. **Middle (Gamma):** Topology effectively exploits non-linear contrast enhancement. **Right (Contrast):** Topology maintains a significant performance lead over geometry ($\Delta > 0$) in low-contrast environments.

sistence diagram. In the Contrast regime, the Radiometric Probe maintains a consistent positive gap ($\Delta \approx +10\% \dots +15\%$). While Hu moments fail in the low-contrast limit ($\alpha < 0.7$) where stochastic noise dominates the integrals, the topological reliance on critical pair existence ensures faint structural lesions remain detectable.

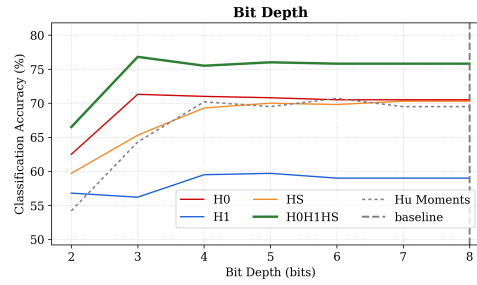


Figure 5.3: Bit Depth Stability (The Terrace Effect). The Radiometric Probe exhibits a strict stability plateau ($\approx 76\%$) from 8-bit down to 3-bit.

Bit Depth. The bit-depth ablation (Figure 5.3) shows a clear stability plateau. Consistent with the Quantisation Stability Bound (Proposition 3.9), the Radiometric Probe maintains essentially constant accuracy from 8-bit down to 3-bit depth. Reducing bit depth converts continuous gradients into flat intensity terraces, but persistent homology depends on relative critical pairings rather than on terrace height itself. The probe collapses only at 2-bit quantisation, where the quantisation step exceeds the intrinsic contrast of the lesions.

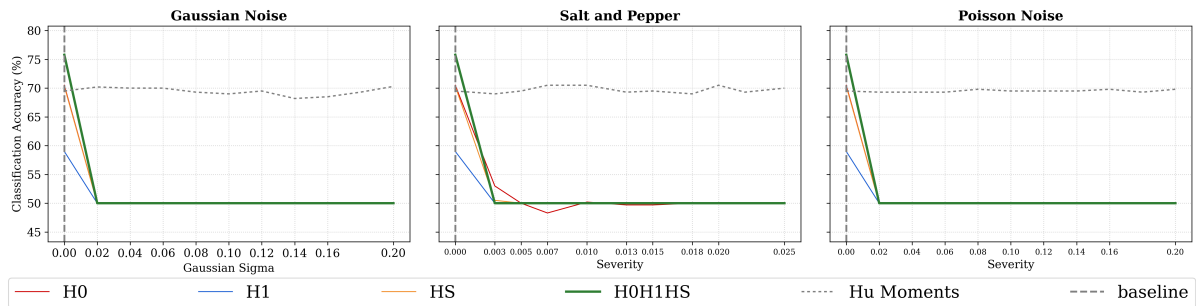


Figure 5.4: Stochastic Stability Curves. Topological features degrade rapidly toward random chance ($\approx 50\%$) under zero-mean stochastic noise, whilst the Mechanical Control remains mathematically invariant.

Stochastic Noise: A Mechanical Analogue Stochastic noise behaves differently from global radiometric drift. Although it is radiometric in origin, its effect on the persistence summaries is mechanically similar: it perturbs local pixel ordering, injects artificial extrema, and rapidly inverts the robustness gap.

The Mechanical Probe remains stable ($\approx 70\%$) across all noise protocols because Hu moments average zero-mean fluctuations over the global grid. By contrast, the Radiometric Probe degrades sharply ($\Delta \ll 0$), reaching near-random performance by $\sigma = 0.02$. In the language of the Differential Failure Matrix, this is classified as a Mechanical Issue because the noise acts through local ordering disruption rather than through a coherent global intensity shift. Operationally, the result indicates that edge-preserving denoising is a prerequisite if topological summaries are to remain useful on clinical sensor data.

The Empirical Failure Matrix The Sensitivity Audit operationalises the theoretical framework. By mapping the performance curves against the baseline Operational Envelope, the Differential Failure Matrix is populated with concrete, empirically derived thresholds (Table 5.3).

	Topology Pass (Stable) Acc $\geq 80\%$	Topology Fail (Unstable) Acc $< 80\%$
Geometry Pass (Stable) Acc $\geq 65\%$	Operational Envelope • Clean Baseline ($\epsilon = 0$) • Quantisation (≥ 3-bit)	Mechanical Issue (Aim) • Rotation ($\theta \geq 3^\circ$) • Stochastic Noise ($\sigma \geq 0.01$) • Gaussian Blur ($\sigma \geq 0.6$)
Geometry Fail (Unstable) Acc $< 65\%$	Radiometric Issue (Shoot) • Positive Drift ($\delta \geq 30$) • Low Contrast ($\alpha \leq 0.7$)	Catastrophic Failure • Negative Drift ($\delta \leq -50$) • Extreme Blur ($\sigma \geq 1.5$)

Table 5.3: The Empirical Differential Failure Matrix. The theoretical quadrants populated with the approximate numerical degradation thresholds discovered during the Sensitivity Audit.

Chapter 6

Conclusion

This thesis has argued that the apparent instability of cubical persistent homology in retinal imaging is best understood as a sampled-acquisition problem rather than as a failure of the underlying stability theory. The central result is the Aim-and-Shoot decomposition, which separates topological error into mechanical and radiometric contributions and explains why topology and geometry respond differently to acquisition shift.

The experiments support that decomposition. In the present setting, geometric descriptors behave as effective mechanical controls: they are comparatively stable under affine misalignment yet vulnerable to radiometric drift. The topological probe behaves as a radiometric control: it remains comparatively stable under exposure drift and quantisation, but fails under interpolation-heavy transformations and stochastic corruption.

That asymmetry is the basis of the Differential Failure Matrix. The point of the framework is not to replace geometric descriptors with topological ones, but to use their disagreement as a diagnostic signal for fault isolation in degraded acquisition settings.

6.1 Synthesis

The Differential Failure Matrix By monitoring the mechanical and radiometric controls, the pipeline can invert the failure modes identified during the sensitivity audit to infer the physical origin of an acquisition error. Table 6.1 synthesises the operational envelope of the system into an analytical heuristic.

Mechanical Control (Geometry)	Radiometric Control (Topology)	Diagnostic Condition	Corrective Action
PASS (Stable)	PASS (Stable)	Limit of Diagnostic Resolution Baseline macro-integrity intact; domain shift is subtle or orthogonal.	Investigate Unmodelled Covariates
PASS (Stable)	FAIL (Unstable)	Mechanical Issue (Aim) Grid shattering (e.g., rotation, spatial variance, stochastic noise).	Examine Focus / Registration
FAIL (Unstable)	PASS (Stable)	Radiometric Issue (Shoot) Absolute integration distortion (e.g., saturation, uniform drift).	Examine Exposure / Sensor
FAIL (Unstable)	FAIL (Unstable)	Compound / Unidentifiable Error Single-mode assumption violated; catastrophic information loss.	System Audit / Reject Image

Table 6.1: The Differential Failure Matrix. An analytical heuristic derived from the functional orthogonality of geometric and topological feature collapse.

The matrix translates the stability analysis into an operational heuristic, but only at the coarse level for which the probes were designed. Because Hu moments are comparatively stable under rigid motion, an isolated collapse of the topological signal points to grid-level mechanical variance. Conversely, survival of the topological probe alongside failure of the geometric probe points to a radiometric distortion in which relative structure is preserved but absolute intensity is not.

The two extreme states mark the limits of the framework. A Pass/Pass outcome during a known classifier failure suggests either a subtler shift than these coarse controls can detect or a confounder outside the present model. A Fail/Fail outcome indicates either compound degradation or a loss of structural information severe enough that fault isolation is no longer reliable.

6.2 Summary of Empirical Observations

The experiments isolate four recurring features of the framework's behaviour, all of them governed by the interaction between the discrete image gradient ($L_{\mathcal{I}}$) and the perturbation operator. First, the topological probe remains stable under strong positive drift and moderate contrast loss even as the geometric control deteriorates. This is consistent with the fact that the relevant filtrations depend on relative intensity ordering rather than on absolute global intensity.

Second, that same structural robustness extends to coarse quantisation. The Radiometric Control retains baseline discriminative power down to 3-bit depth, supporting the predicted Terrace Effect and suggesting that persistent connectivity can survive aggressive compression when lesion contrast remains above the quantisation scale.

Third, spatial sampling exhibits a genuine optimum rather than monotonic improvement. Performance peaks at an intermediate resolution (1024px) and weakens both below that scale, where lesions are aliased, and above it, where high-frequency sensor variation re-enters the filtration. In this sense, controlled down-sampling acts as a low-pass filter on the discrete gradient.

Fourth, the topological summaries are markedly fragile under interpolation-heavy transformations and stochastic noise. Grid misalignment, blur, and variance-driven extrema all disrupt the local ordering on which persistence depends. The practical consequence is that registration and edge-preserving denoising are not optional refinements but operational preconditions for stable topological analysis on retinal images.

Theoretical Implications: Isolating Physical Tolerances These empirical results are consistent with the Total Computational Stability Bound derived in Chapter 3. One recurring limitation of algebraic stability in the discrete setting is that standard theorems guarantee stability only for perturbations that are called "small" without tying that language to the geometry of a pixel grid. Our derivation addresses that limitation by decomposing the error term into explicit physical components:

$$d_B(\mathcal{D}(\mathcal{I}), \mathcal{D}(T_S \circ T_A \circ \mathcal{I})) \leq \lceil \lambda \rceil L_{\mathcal{I}} + \epsilon_{rad}$$

The value of this bound lies in making the relevant tolerances explicit. It turns the abstract stability statement into a trade-off between spatial displacement (λ), local image gradient ($L_{\mathcal{I}}$), and radiometric corruption (ϵ_{rad}). In that form, the Aim-and-Shoot model is no longer just a heuristic interpretation of the experiments; it becomes the practical consequence of the bound itself. Stable topological analysis requires spatial misalignment to be controlled before radiometric variation can be interpreted reliably.

Remark 6.1 (THEORETICAL FOUNDATIONS) While the main text focuses on the operational utility of this bound, its validity rests upon the rigorous algebraic structure of persistence modules. For the complete derivation of the functorial pipeline, we refer the reader to **Appendix A**.

6.3 Future Work

Deployment and Remote Diagnostics. The empirical resolution optimum near 1024px suggests that the framework may be compatible with lower-cost retinal hardware, including handheld systems used in resource-constrained settings [61]. More broadly, the Aim-and-Shoot decomposition may be useful whenever remote or high-volume imaging systems cannot be inspected manually. The natural next question is

whether the Differential Failure Matrix can support automated triage by separating operator-induced mechanical faults from sensor-induced radiometric faults in deployed environments.

Transfer and Model Design. Because both cubical homology and spatial moments operate on discrete grids rather than on a modality-specific physics model, the framework may extend beyond visible-light fundus imaging. It would be natural to test whether the same orthogonal failure profiles appear in other imaging modalities, including thermal, hyperspectral, X-ray, or acoustic arrays. A related question is whether those complementary profiles can be incorporated directly into predictive models, so that orthogonal geometric and topological features improve robustness under compound domain shift.

Mathematical and Algorithmic Extensions. The present framework is intentionally coarse: it uses low-dimensional summaries to preserve interpretability and computational control. Future theoretical work could tighten the computational stability bound beyond the discrete Lipschitz heuristic $L_{\mathcal{I}}$, perhaps by incorporating operator-theoretic or energy-based estimates. On the algorithmic side, persistent cohomology may offer a route from global fault detection to spatial localisation, allowing radiometric defects to be identified rather than merely flagged.

Appendix A

Functoriality and Triangulation

This appendix formalises the categorical transfer of stability theorems from the simplicial domain [10, 28] to the cubical domain intrinsic to the integer lattice $\mathbb{Z}^n$. Functoriality guarantees the strict preservation of internal data relationships; if image A is a geometric sub-region of image B , a functorial pipeline ensures the topological features of A map deterministically into the algebraic features of B . This structural certification is mathematically required to prevent unpredictable algorithm failure under radiometric or mechanical domain shifts.

Prerequisite Algebraic Structures

The functorial setup strictly follows Nanda [10, Chapters 1 and 4], with deeper structural definitions derived from Mac Lane [62]. Working over $\mathbb{F}_2$, we inherently eliminate the alternating signs associated with standard simplicial orientation.

Categories and Functors. A *category* $\mathcal{C}$ consists of:

- C1.** A collection of *objects* $\mathcal{C}_0$;
- C2.** For any $x, y \in \mathcal{C}_0$, a collection $\mathcal{C}(x, y)$ of *morphisms* $f : x \rightarrow y$;
- C3.** An associative *composition law* $\circ : \mathcal{C}(x, y) \times \mathcal{C}(y, z) \rightarrow \mathcal{C}(x, z)$;
- C4.** For each object x , an *identity morphism* id_x such that $f \circ \text{id}_x = f = \text{id}_y \circ f$ for any $f : x \rightarrow y$.

If $\mathcal{D}$ is also a category, a **functor** $\Phi : \mathcal{C} \rightarrow \mathcal{D}$ assigns every object $x \in \mathcal{C}_0$ to an object $\Phi(x) \in \mathcal{D}_0$, and every morphism $f : x \rightarrow y$ to a morphism $\Phi(f) : \Phi(x) \rightarrow \Phi(y)$, satisfying:

- F1. Preservation of Identity:** For every object $x \in \mathcal{C}$, $\Phi(\text{id}_x) = \text{id}_{\Phi(x)}$.
- F2. Preservation of Composition:** For all morphisms $f : x \rightarrow y$ and $g : y \rightarrow z$, $\Phi(g \circ f) = \Phi(g) \circ \Phi(f)$.

NOTATION To preclude ambiguity across geometric and algebraic domains, we define the following foundational categories and distinguish simplicial from cubical operators using superscripts ($\triangle$ and $\square$):

- **SimpCx**: The category of abstract simplicial complexes and simplicial maps.
- **CubeCx**: The category of finite cubical complexes $K \subset \mathbb{Z}^n$ and cubical maps.
- **ChainCx**: The category of chain complexes and chain maps.
- **Vect $_{\mathbb{F}_2}$** : The category of vector spaces over $\mathbb{F}_2$ and linear maps.

Compositional Transfer to the Simplicial Domain Because the diagnostic pipeline operates natively on discrete digital grids, we formalise the initial geometric category. A cubical map $f : K \rightarrow L$ in **CubeCx** assigns every elementary k -cube $\kappa \in K$ to an elementary m -cube $f(\kappa) \in L$ (where $m \leq k$), strictly preserving the face relation: $\tau \preceq \kappa \in K \implies f(\tau) \preceq f(\kappa) \in L$.

Proposition A.1 (FUNCTORIALITY OF CUBICAL CHAINS). *Let $f : K \rightarrow L$ be a cubical map. The induced $\mathbb{F}_2$ -linear map on k -chains, $\mathbf{C}_k f : \mathbf{C}_k(K) \rightarrow \mathbf{C}_k(L)$, defined on basis cubes $\sigma \in K$ by $\mathbf{C}_k f(\sigma) = f(\sigma)\mathbb{I}(\dim f(\sigma) = k)$, defines a rigorous functor $\mathbf{C}_\bullet^\square : \mathbf{CubeCx} \rightarrow \mathbf{ChainCx}$.*

Proof. To establish functoriality, $\mathbf{C}_\bullet^\square$ must assign valid morphisms in $\mathbf{ChainCx}$ and preserve categorical structure. Let $f_\# = \mathbf{C}_\bullet^\square.f$.

Chain Map Condition: For $f_\#$ to be a valid morphism, it must commute with the cubical boundary operator $\partial^\square$. Because f is a cubical map, it inherently maps the geometrical boundaries of any elementary n -cube $c \in K$ to the corresponding boundaries of $f(c) \in L$. Extending linearly over $\mathbb{F}_2$ yields $\partial_n^\square \circ f_\#(c) = f_\# \circ \partial_n^\square(c)$. Thus, $f_\#$ is a valid chain map.

F1 (Identity): Let $\text{id}_K : K \rightarrow K$. By definition, $(\text{id}_K)_\#(c) = c$. Extending linearly across the chain complex yields $(\text{id}_K)_\# = \text{id}_{\mathbf{C}_\bullet^\square(K)}$.

F2 (Composition): Let $f : K \rightarrow L$ and $g : L \rightarrow M$ be cubical maps. Evaluating the generators directly yields $(g \circ f)_\#(c) = g(f(c)) = g_\#(f_\#(c))$. Linear extension gives $(g \circ f)_\# = g_\# \circ f_\#$.

Because $\mathbf{C}_\bullet^\square$ assigns well-defined chain maps, preserves identities, and distributes over composition, it satisfies all functorial axioms. $\square$

Simplicial Homology Pipeline. Following Nanda [10], the standard simplicial homology pipeline is formulated as shown in Figure A.1:

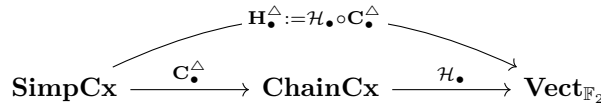


Figure A.1: The standard simplicial homology pipeline represented as a composition of functors.

where $\mathbf{H}_\bullet^\Delta$ explicitly denotes the composite homology functor. To exploit this pre-built mathematical infrastructure, we map the cubical domain into this setup. We require the CFK functor $\mathbf{T} : \mathbf{CubeCx} \rightarrow \mathbf{SimpCx}$ to act as a structural bridge, extending the sequence as shown in Figure A.2:

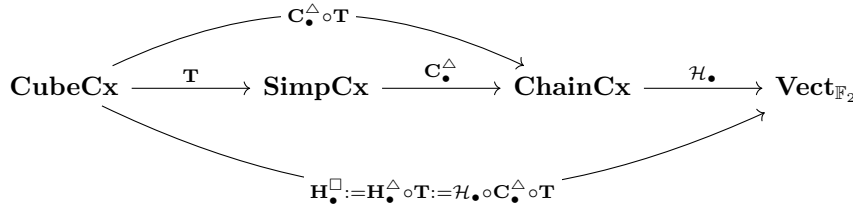


Figure A.2: Extension of the homological pipeline to the cubical domain via the CFK functor $\mathbf{T}$.

This composition ensures the rigorous application of simplicially grounded persistent homology theory directly to the discrete digital grid.

Coxeter-Freudenthal-Kuhn Triangulation

The Coxeter-Freudenthal-Kuhn (CFK) triangulation provides a canonical decomposition of the cubical domain into a simplicial complex without introducing new vertices. This geometric construction establishes the basis for the algebraic triangulation map $\mathbb{T}$, which operates on chain complexes. To prevent notational overloading, $\mathbb{T}$ is strictly distinguished from the categorical triangulation functor $\mathbf{T}$, which is developed subsequently to bridge the $\mathbf{CubeCx}$ and $\mathbf{SimpCx}$ categories.

GEOMETRIC DEFINITION: HYPERPLANE ARRANGEMENT [63] Let K be a cubical complex embedded in $\mathbb{R}^n$. The CFK triangulation K^Δ is the simplicial complex obtained by slicing the domain along the following affine hyperplanes for all indices $1 \leq i < j \leq n$ and integers $k \in \mathbb{Z}$:

1. **Grid cuts** ($x_i = k$): Defining the boundaries of the elementary cubes.
2. **Diagonal cuts** ($x_i - x_j = k$): Subdividing the interior of the cubes.

In dimension $n = 2$, this restricts to splitting each square pixel along the diagonal $x_1 - x_2 = k$.

COMBINATORIAL REALISATION [64] Algorithmic implementation relies on combinatorial sorting. The restriction of the CFK triangulation to the standard unit n -cube $[0, 1]^n$ yields exactly $n!$ simplices. Each simplex σ_π corresponds to a unique permutation π on n elements, defined strictly by the coordinate ordering:

$$0 \leq x_{\pi(1)} \leq \dots \leq x_{\pi(n)} \leq 1.$$

The vertices $v_0, \dots, v_n$ of σ_π are the cumulative sums of the permuted basis vectors $v_k = \sum_{j=1}^k \mathbf{e}_{\pi(j)}$ with $v_0 = \mathbf{0}$. By translation invariance, the local triangulation of any arbitrary elementary cube located at $\mathbf{z} \in \mathbb{Z}^n$ is the push-forward of these reference simplices via the translation $T_{\mathbf{z}}(x) = x + \mathbf{z}$.

Definition A.2 (2D TRIANGULATION MAP). Let $K \in \mathbf{CubeCx}$ be a 2-dimensional cubical complex and K^Δ its CFK triangulation. The *triangulation map* $\mathbb{T} : \mathbf{C}_\bullet(K) \rightarrow \mathbf{C}_\bullet(K^\Delta)$ is defined on basis elements by dimension:

- **0-cubes:** $\mathbb{T}(\kappa) = \kappa$.
- **1-cubes:** $\mathbb{T}(\kappa) = e(\kappa)$, the 1-simplex spanned by its endpoints.
- **2-cubes:** For $\kappa = [i, i+1] \times [j, j+1]$ with vertices $a = (i, j)$, $b = (i, j+1)$, $c = (i+1, j+1)$, and $d = (i+1, j)$,

$$\mathbb{T}(\kappa) = \sigma^+(\kappa) + \sigma^-(\kappa),$$

where the *upper triangle* $\sigma^+(\kappa)$ is spanned by $\{a, b, c\}$ and the *lower triangle* $\sigma^-(\kappa)$ is spanned by $\{a, c, d\}$ (See Figure A.3).

Linear extension ensures the map is well-defined over all chains in $\mathbf{C}_\bullet(K)$.

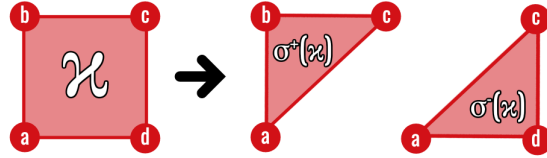


Figure A.3: The Triangulation Map. Applying the canonical CFK triangulation to a 2-cube κ . The diagonal cut splits the square into an upper triangle $\sigma^+(\kappa)$ and a lower triangle $\sigma^-(\kappa)$, such that $\mathbb{T}(\kappa) = \sigma^+ + \sigma^-$.

Proposition A.3. The triangulation map $\mathbb{T}$ strictly commutes with the boundary operators:

$$\partial_k^\Delta \circ \mathbb{T}_k = \mathbb{T}_{k-1} \circ \partial_k^\square.$$

Proof. Over the field $\mathbb{F}_2$, this is verified explicitly for $k = 2$. Let κ be a unit square. Applying $\mathbb{T} \circ \partial^\square$ yields the topological perimeter of the square as four 1-simplices. Conversely, evaluating $\partial^\Delta \circ \mathbb{T}$ on $\sigma^+(\kappa) + \sigma^-(\kappa)$ generates the perimeter edges alongside exactly two instances of the internal diagonal $[a, c]$. Because $2 \equiv 0 \pmod{2}$ in $\mathbb{F}_2$, the internal diagonal precisely vanishes, proving strict commutativity. $\square$

Proposition A.4 (HOMOLOGICAL EQUIVALENCE). For any $K \in \mathbf{CubeCx}$ and $k \geq 0$, $\mathbf{H}_k(K) \cong \mathbf{H}_k(K^\Delta)$.

Proof. The CFK construction partitions elementary cubes without altering the underlying point set topology. Consequently, the geometric realisations are strictly homeomorphic ($|K| \cong |K^\Delta|$), guaranteeing isomorphic homology groups. $\square$

Filtered Categories and the Triangulation Functor

To evaluate cross-domain transformations dynamically, we establish that the conversion from cubes to simplices preserves morphisms across entire filtrations.

Theorem A.5 (THE TRIANGULATION FUNCTOR). Let $\mathbf{CubeCx}_{\leq}$ denote the subcategory of cubical complexes restricted to order-preserving cubical maps. The assignment of the CFK triangulation to objects ($K \mapsto K^{\Delta}$) and the induced simplicial map to morphisms ($f \mapsto f^{\Delta}$) defines a strictly structure-preserving functor: $\mathbf{T} : \mathbf{CubeCx}_{\leq} \rightarrow \mathbf{SimpCx}$.

Proof. The assignment on objects is well-defined. For morphisms, if f is strictly order-preserving, the image of any valid vertex chain in K^{Δ} forms a valid chain in the target cube, spanning a valid simplex in L^{Δ} . Since simplicial maps are uniquely determined by vertex assignments, the identity ($\mathbf{T}(\text{id}_K) = \text{id}_{K^{\Delta}}$) and composition ($\mathbf{T}(g \circ f) = \mathbf{T}(g) \circ \mathbf{T}(f)$) axioms are trivially satisfied. $\square$

Because filtration inclusion maps ($\iota(x) = x$) strictly preserve the coordinate partial order, $\mathbf{T}$ naturally lifts to map of filtered sequences.

Category of Persistence Modules [10]. The category $\mathbf{PersMod}$ consists of persistence modules $(V_{\bullet}, a_{\bullet})$ over $\mathbb{F}$. A morphism $\phi_{\bullet} : (V_{\bullet}, a_{\bullet}) \rightarrow (W_{\bullet}, b_{\bullet})$ is a sequence of linear maps where $b_t \circ \phi_t = \phi_{t+1} \circ a_t$ for all t .

Filtered Categories and Filtered Functors. The category $\mathbf{CubeFilt}$ consists of constantly extended filtrations $\mathbf{F}_{\bullet} = (K_t)_{t \in \mathbb{Z}}$ where $K_s \subseteq K_t$ for $s \leq t$. Morphisms are families of compatible cubical maps. $\mathbf{ChainFilt}$ is defined analogously for sequences of chain complexes. By applying the standard functors component-wise, we define the **Filtered Chain Functor** $\mathbf{C}_{\bullet}^{\text{filt}}$ and the **Filtered Homology Functor** $\mathcal{H}_k^{\text{filt}}$. Their composition provides the direct translation from geometric filtrations to persistence.

Persistent Homology Functor Let $k \geq 0$. We define the *k-th persistent homology functor* as

$$\mathbf{PH}_k := \mathcal{H}_k^{\text{filt}} \circ \mathbf{C}_{\bullet}^{\text{filt}}.$$

Remark A.6 (ABUSE OF NOTATION: THE PERSISTENT HOMOLOGY OPERATOR) We intentionally overload the symbol $\mathbf{PH}$. In the categorical context, $\mathbf{PH}_k := \mathcal{H}_k^{\text{filt}} \circ \mathbf{C}_{\bullet}^{\text{filt}}$ denotes the functor mapping a filtered sequence of cubical complexes to a persistence module in $\mathbf{PersMod}$. Conversely, in the discrete algebraic context, $\mathbf{PH}_{i \rightarrow j}$ denotes the specific persistent homology group (a vector space in $\mathbf{Vect}_{\mathbb{F}_2}$) defined by the image of the transition map. This abuse of notation is deliberate: a distinct symbol solely for the functor would inflate the notational burden without yielding operational clarity.

This construction yields a unified commutative framework, visualised in Figure A.4, wherein the triangulation functor $\mathbf{T}$ acts as a categorical bridge, formally lifting native cubical definitions into the established simplicial domain.

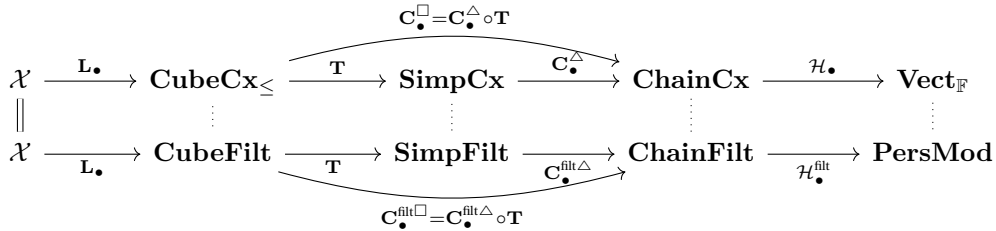


Figure A.4: Commutative diagram illustrating the functorial pipeline mapping static and filtered cubical complexes to homology vector spaces and persistence modules.

The composition of functors is itself a functor [62], we may mathematically apply the entire stability machinery of simplicial persistent homology developed by Nanda [10] directly to our discrete $\mathcal{O}(N)$ cubical pipeline.

A Functorial Proof of Algebraic Stability

To quantify stability, we require a metric on $\mathbf{PersMod}$. Standard metrics are unsuitable because perturbations shift features in time (index) rather than magnitude. We formalise the "slide rule" intuition via the shift functor.

The Shift Functor [24] and Structure Morphism [65]. For $k \in \mathbb{Z}_{\geq 0}$, the **shift functor** $(\bullet)[k] : \mathbf{PersMod} \rightarrow \mathbf{PersMod}$ advances the index of a constantly extended persistence module $\mathbf{V}$ by k steps: $(\mathbf{V}[k])_n := \mathbf{V}_{n+k}$. Internal structure maps shift accordingly:

$$\rho[k]_{i \rightarrow j} := \rho_{i+k \rightarrow j+k}.$$

The internal flow of information during a shift is captured by the **structure morphism** $\mathbf{S}(\mathbf{V}, k) : \mathbf{V} \rightarrow \mathbf{V}[k]$, defined component-wise as the transition map

$$\rho_{n \rightarrow n+k} : V_n \rightarrow V_{n+k}.$$

Equivalent definition for discrete interleavings. Two modules $\mathbf{V}$ and $\mathbf{W}$ are k -**interleaved** if there exist morphisms $\phi : \mathbf{V} \rightarrow \mathbf{W}[k]$ and $\psi : \mathbf{W} \rightarrow \mathbf{V}[k]$ satisfying Triangle Relations. Specifically, as illustrated in Figure A.5, mapping to the alternate module and back ($\psi \circ \phi$ and $\phi \circ \psi$) must algebraically equal the internal structure morphism of a $2k$ shift.

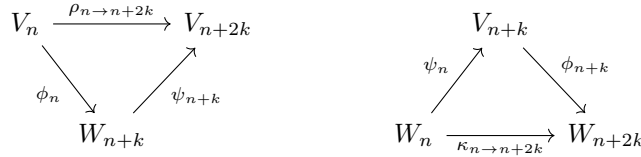


Figure A.5: Triangle Relations: mapping to the alternate module and back ($\psi \circ \phi$ and $\phi \circ \psi$) must algebraically equal the internal structure morphism of a $2k$ shift.

The shift functor and structure morphisms formalise discrete interleavings categorically, facilitating a functorial proof of the Algebraic Stability Theorem (Theorem 2.13).

Functorial Proof for Algebraic Stability. We apply the Isometry Theorem to establish the robustness of our pipeline¹. The functorial construction $(\mathbf{PH}_\bullet = \mathcal{H}_\bullet^{\text{filt}} \circ \mathbf{C}_\bullet^{\text{filt}})$ maps geometric interleavings directly to algebraic interleavings. Rather than complex topological arguments, we observe that functors map geometric interleavings directly to algebraic interleavings. Since $\|\mathcal{I} - \mathcal{J}\|_\infty \leq k$, pixel values differ by at most k , so sublevel sets are nested with a shift of k indices:

$$X_n(\mathcal{I}) \subseteq X_{n+k}(\mathcal{J}) \quad \text{and} \quad X_n(\mathcal{J}) \subseteq X_{n+k}(\mathcal{I}) \quad \forall n.$$

While these asynchronous inclusions do not constitute standard morphisms in $\mathbf{CubeFilt}$ (which requires index preservation $t \rightarrow t$), they induce well-defined linear maps on homology that target the *shifted* persistence modules. This construction follows the interpolation lemma strategy of Chazal et al. [66]. Specifically, the inclusion $\iota : X_n(\mathcal{I}) \hookrightarrow X_{n+k}(\mathcal{J})$ induces the linear map:

$$\phi_n : \mathbf{H}_\bullet(X_n(\mathcal{I})) \longrightarrow \mathbf{H}_\bullet(X_{n+k}(\mathcal{J})) = (\mathbf{PH}(\mathcal{J})[k])_n.$$

The collection $\phi_\bullet = \{\phi_n\}_n$ forms a valid morphism $\phi : \mathbf{PH}(\mathcal{I}) \rightarrow \mathbf{PH}(\mathcal{J})[k]$ in $\mathbf{PersMod}$. By symmetry, we construct $\psi : \mathbf{PH}(\mathcal{J}) \rightarrow \mathbf{PH}(\mathcal{I})[k]$. These morphisms satisfy Triangle Relations because the geometric identity inclusions on the underlying grid induce them. Thus, $\mathbf{PH}(\mathcal{I})$ and $\mathbf{PH}(\mathcal{J})$ are algebraically k -interleaved. By the Isometry Theorem (Theorem 2.12), this algebraic distance corresponds exactly to the bottleneck distance:

$$d_B(\mathbf{Bar}(\mathcal{I}), \mathbf{Bar}(\mathcal{J})) = d_I(\mathbf{PH}(\mathcal{I}), \mathbf{PH}(\mathcal{J})) \leq k.$$

□

¹This result adapts the general Stability Theorem of Cohen-Steiner et al. [28] to the discrete cubical domain.

A Stable and Dynamic Functorial Pipeline

We consolidate our constructions into a unified perspective linking discrete sensor geometry directly to clinical safety guarantees. In the context of fundus imagery, we restrict our attention to dimensions $k = 0$ (connected components, identifying haemorrhages and exudates) and $k = 1$ (loops, identifying vascular topology).

Let K^- and K^+ denote the ambient cubical complexes associated with the inverted (dark) and standard (bright) intensity functions, respectively. The resulting parallel workflow is visualised below:

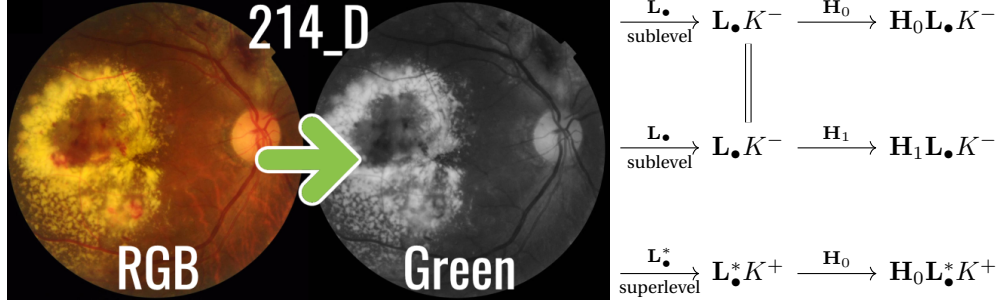
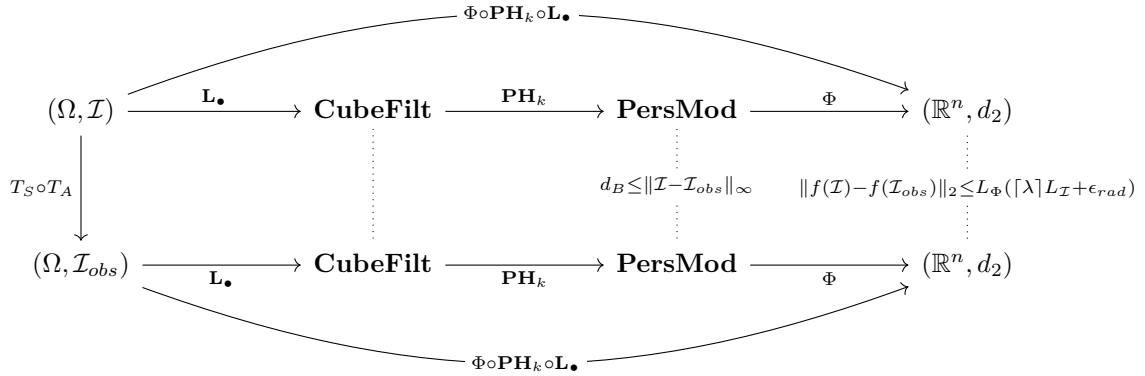


Figure A.6: The Retinal Topology Pipeline. Evaluating three parallel homology sequences for structural biomarkers.

Because superlevel filtrations are algebraically equivalent to sublevel filtrations on the inverted intensity function, these parallel streams represent initialised instances of a single, high-level functorial mapping $\text{PH}_k \circ \text{L}_\bullet$.

The complete clinical pipeline, mapping a raw sensor image $(\Omega, \mathcal{I})$ down to a stable machine-learning feature vector, acts as a continuous mapping sequence bounded by defined Lipschitz constants. The diagram below visualises this stability propagation for both a ground-truth image and its perturbed clinical observation $(\Omega, \mathcal{I}_{obs})$, demonstrating that radiometric noise is strictly mathematically confined mathematically prior to reaching downstream diagnostic models.



While technically a diagram of discrete functional layers, the unified operator $\Phi \circ \text{PH}_k \circ \text{L}_\bullet$ conceptually serves as a direct, stable pipeline from noisy pixel grids to robust vector representations.

Appendix B

Coefficient Field Choice

While Nanda [10] and Kaczynski et al. [13] define homology over arbitrary coefficient fields, planar digital image analysis does not require torsion information. We only need to detect the presence or absence of components and loops, so throughout this thesis we work over the field of two elements, denoted $\mathbb{F}_2$.¹

This choice is standard in computational topology [23, 28, 67, 68] and is particularly natural for image-based TDA, where binary arithmetic maps directly to efficient machine-level operations.

Working over $\mathbb{F}_2$ yields several simplifications:

1. Orientation Independence. Since $-1 = 1$ in characteristic 2, the alternating signs in the boundary operator disappear. The boundary of a cube is therefore just the unweighted sum of its facets, as in Definition 1.8.

2. Binary Arithmetic. Incidence matrices reduce from entries in $\{-1, 0, +1\}$ to $\{0, 1\}$. Matrix reduction can therefore be implemented using binary operations such as XOR.

3. Geometric Interpretation of Chains. An n -chain is identified with the subset of n -cubes having coefficient 1 [13]. Addition corresponds to symmetric difference of sets.

4. Immediate Boundary Cancellation. The identity $\partial \circ \partial = 0$ becomes arithmetically transparent: every $(n - 2)$ -face appears twice, and $1 + 1 = 0$ in $\mathbb{F}_2$.

5. Vector Space Structure. Over $\mathbb{F}_2$, homology groups are vector spaces, so they are completely characterised by their dimensions. The Betti number β_k is simply $\dim \mathbf{H}_k$.

6. Simpler Matrix Reduction. Nanda [10] uses Smith Normal Form to handle integral coefficients and torsion. Over a field such as $\mathbb{F}_2$, Gaussian elimination suffices [69], avoiding unnecessary algebraic overhead.

These simplifications make $\mathbb{F}_2$ the natural choice for the present setting: it preserves the Betti numbers relevant to retinal image analysis while discarding torsion information that plays no role in our clinical interpretation of components and loops.

¹In group theory, the cyclic group of order 2 is often written $\mathbb{Z}_2$, but in algebraic number theory $\mathbb{Z}_2$ denotes the ring of 2-adic integers. We therefore use $\mathbb{F}_2$ to emphasise the field structure and avoid ambiguity.

Appendix C

Detailed Numerical Results

This appendix contains the full tabulation of classification accuracy across all 10 perturbation protocols. The **Baseline** condition (zero perturbation) is highlighted in green for reference.

Rotation ($^{\circ}$)					
Param	H0	H1	H _S	Comb.	Hu
-10	0.495	0.525	0.650	0.560	0.668
-9	0.505	0.525	0.627	0.535	0.665
-8	0.518	0.525	0.582	0.515	0.657
-7	0.525	0.525	0.573	0.518	0.663
-6	0.518	0.525	0.562	0.517	0.660
-5	0.632	0.525	0.705	0.773	0.668
-4	0.693	0.522	0.700	0.760	0.698
-3	0.733	0.522	0.700	0.760	0.700
-2	0.765	0.520	0.700	0.785	0.710
-1	0.762	0.522	0.702	0.785	0.710
0	0.705	0.590	0.703	0.758	0.695
1	0.698	0.520	0.700	0.795	0.677
2	0.580	0.522	0.698	0.773	0.670
3	0.530	0.522	0.562	0.517	0.655
4	0.530	0.522	0.545	0.507	0.650
5	0.532	0.522	0.545	0.510	0.650
6	0.532	0.525	0.560	0.508	0.653
7	0.525	0.525	0.570	0.515	0.655
8	0.522	0.525	0.597	0.530	0.655
9	0.515	0.525	0.588	0.522	0.657
10	0.505	0.525	0.650	0.568	0.652

Blur (σ)					
Param	H0	H1	H _S	Comb.	Hu
0.0	0.705	0.590	0.703	0.758	0.695
0.15	0.705	0.590	0.703	0.758	0.695
0.3	0.705	0.590	0.700	0.758	0.695
0.45	0.718	0.552	0.713	0.778	0.695
0.6	0.593	0.500	0.562	0.745	0.695
0.75	0.497	0.500	0.507	0.653	0.695
0.9	0.500	0.500	0.508	0.560	0.695
1.05	0.500	0.500	0.503	0.515	0.695
1.2	0.500	0.500	0.500	0.518	0.695
1.35	0.500	0.500	0.500	0.500	0.695
1.5	0.500	0.500	0.500	0.500	0.695

Resolution (px)					
Param	H0	H1	H _S	Comb.	Hu
64	0.500	0.500	0.503	0.500	0.680
128	0.500	0.500	0.503	0.500	0.668
256	0.500	0.500	0.500	0.495	0.690
512	0.500	0.500	0.505	0.500	0.698
768	0.722	0.525	0.700	0.580	0.698
1024	0.705	0.590	0.693	0.755	0.700
1280	0.690	0.602	0.662	0.702	0.698
1600	0.667	0.557	0.603	0.620	0.700
1920	0.657	0.543	0.567	0.565	0.700
2048	0.633	0.520	0.550	0.530	0.695

Bit Depth					
Param	H0	H1	H _S	Comb.	Hu
2	0.625	0.568	0.597	0.665	0.542
3	0.713	0.562	0.653	0.768	0.643
4	0.710	0.595	0.693	0.755	0.702
5	0.708	0.597	0.700	0.760	0.695
6	0.705	0.590	0.698	0.758	0.707
7	0.705	0.590	0.703	0.758	0.695
8	0.705	0.590	0.703	0.758	0.695

Table C.1: Classification accuracy for Mechanical and Discrete perturbations.

Drift (px)						Gamma						Contrast					
Param	H0	H1	H _S	Comb.	Hu	Param	H0	H1	H _S	Comb.	Hu	Param	H0	H1	H _S	Comb.	Hu
-150	0.500	0.500	0.508	0.500	0.575	0.1	0.642	0.500	0.613	0.500	0.530	0.5	0.685	0.523	0.645	0.500	0.505
-140	0.500	0.500	0.505	0.500	0.623	0.2	0.710	0.598	0.680	0.530	0.525	0.6	0.703	0.543	0.670	0.547	0.537
-130	0.500	0.500	0.497	0.500	0.603	0.3	0.700	0.633	0.653	0.627	0.547	0.7	0.695	0.560	0.690	0.682	0.578
-120	0.500	0.500	0.500	0.500	0.595	0.4	0.702	0.608	0.640	0.705	0.575	0.8	0.705	0.585	0.692	0.755	0.670
-110	0.500	0.500	0.500	0.500	0.580	0.5	0.700	0.593	0.648	0.732	0.593	0.9	0.705	0.593	0.705	0.755	0.690
-100	0.497	0.500	0.492	0.500	0.568	0.6	0.695	0.590	0.655	0.733	0.645	1.0	0.705	0.590	0.703	0.758	0.695
-90	0.490	0.497	0.507	0.500	0.557	0.7	0.693	0.590	0.655	0.738	0.668	1.1	0.702	0.595	0.693	0.750	0.707
-80	0.500	0.490	0.510	0.503	0.525	0.8	0.695	0.605	0.675	0.752	0.693	1.2	0.698	0.568	0.675	0.740	0.680
-70	0.505	0.510	0.547	0.520	0.527	0.9	0.703	0.598	0.695	0.753	0.695	1.3	0.690	0.573	0.660	0.735	0.650
-60	0.547	0.528	0.618	0.605	0.585	1.0	0.705	0.590	0.703	0.758	0.695	1.4	0.675	0.565	0.640	0.715	0.647
-50	0.607	0.547	0.662	0.720	0.617	1.1	0.708	0.595	0.702	0.760	0.705	1.5	0.670	0.540	0.637	0.700	0.640
-40	0.680	0.575	0.675	0.760	0.673	1.2	0.700	0.575	0.718	0.762	0.708	1.6	0.640	0.537	0.632	0.693	0.602
-30	0.698	0.593	0.680	0.757	0.670	1.3	0.698	0.562	0.713	0.765	0.720	1.7	0.610	0.525	0.618	0.693	0.590
-20	0.705	0.593	0.688	0.757	0.693	1.4	0.703	0.557	0.680	0.768	0.718	1.8	0.588	0.522	0.593	0.703	0.575
-10	0.705	0.600	0.703	0.755	0.708	1.5	0.702	0.555	0.670	0.775	0.713	1.9	0.552	0.515	0.593	0.713	0.555
0	0.705	0.590	0.703	0.758	0.695	1.6	0.690	0.520	0.668	0.765	0.713	2.0	0.525	0.503	0.585	0.713	0.550
10	0.705	0.585	0.695	0.758	0.688	1.7	0.685	0.517	0.660	0.762	0.713	2.1	0.517	0.510	0.550	0.708	0.560
20	0.705	0.585	0.695	0.755	0.675	1.8	0.675	0.525	0.650	0.753	0.705	2.2	0.510	0.517	0.535	0.700	0.555
30	0.703	0.590	0.698	0.758	0.667	1.9	0.665	0.520	0.618	0.730	0.695	2.3	0.507	0.503	0.520	0.690	0.568
40	0.700	0.593	0.682	0.750	0.677	2.0	0.630	0.515	0.590	0.710	0.698	2.4	0.505	0.498	0.507	0.698	0.565
50	0.695	0.590	0.672	0.753	0.647	2.1	0.620	0.512	0.555	0.690	0.680	2.5	0.507	0.490	0.500	0.685	0.552
60	0.690	0.597	0.662	0.747	0.620	2.2	0.602	0.505	0.535	0.652	0.680	2.6	0.497	0.480	0.508	0.675	0.568
70	0.685	0.595	0.655	0.747	0.598	2.3	0.583	0.503	0.522	0.625	0.667	2.7	0.492	0.478	0.503	0.665	0.573
80	0.677	0.595	0.642	0.747	0.585	2.4	0.565	0.503	0.515	0.602	0.665	2.8	0.500	0.483	0.500	0.658	0.575
90	0.673	0.595	0.617	0.742	0.583	2.5	0.560	0.500	0.510	0.602	0.677	2.9	0.498	0.478	0.485	0.642	0.583
100	0.662	0.587	0.595	0.740	0.562	2.6	0.535	0.500	0.507	0.597	0.670	3.0	0.490	0.480	0.490	0.633	0.590
110	0.653	0.580	0.565	0.730	0.555	2.7	0.537	0.500	0.503	0.565	0.675	Salt & Pepper					
120	0.633	0.565	0.537	0.720	0.562	2.8	0.532	0.500	0.505	0.552	0.665	Param	H0	H1	H _S	Comb.	Hu
130	0.620	0.547	0.535	0.720	0.550	2.9	0.522	0.500	0.505	0.543	0.657	0.0	0.705	0.590	0.703	0.758	0.695
140	0.620	0.530	0.525	0.698	0.545	3.0	0.517	0.500	0.505	0.530	0.655	0.0025	0.530	0.500	0.505	0.500	0.690
150	0.597	0.535	0.505	0.650	0.545	Poisson Noise						0.005	0.500	0.500	0.500	0.500	0.695
Gaussian Noise						Param	H0	H1	H _S	Comb.	Hu	0.0075	0.483	0.500	0.500	0.500	0.705
Param	H0	H1	H _S	Comb.	Hu	0.0	0.705	0.590	0.703	0.758	0.695	0.01	0.502	0.500	0.500	0.500	0.705
0.0	0.705	0.590	0.703	0.758	0.695	0.02	0.500	0.500	0.500	0.500	0.693	0.0125	0.497	0.500	0.500	0.500	0.693
0.02	0.500	0.500	0.500	0.500	0.702	0.04	0.500	0.500	0.500	0.500	0.693	0.015	0.497	0.500	0.500	0.500	0.695
0.04	0.500	0.500	0.500	0.500	0.700	0.06	0.500	0.500	0.500	0.500	0.693	0.0175	0.500	0.500	0.500	0.500	0.690
0.06	0.500	0.500	0.500	0.500	0.700	0.08	0.500	0.500	0.500	0.500	0.698	0.02	0.500	0.500	0.500	0.500	0.705
0.08	0.500	0.500	0.500	0.500	0.693	0.1	0.500	0.500	0.500	0.500	0.695	0.0225	0.500	0.500	0.500	0.500	0.693
0.1	0.500	0.500	0.500	0.500	0.690	0.12	0.500	0.500	0.500	0.500	0.695	0.025	0.500	0.500	0.500	0.500	0.700
0.12	0.500	0.500	0.500	0.500	0.695	0.14	0.500	0.500	0.500	0.500	0.695						
0.14	0.500	0.500	0.500	0.500	0.682	0.16	0.500	0.500	0.500	0.500	0.698						
0.16	0.500	0.500	0.500	0.500	0.685	0.18	0.500	0.500	0.500	0.500	0.693						
0.18	0.500	0.500	0.500	0.500	0.693	0.2	0.500	0.500	0.500	0.500	0.698						
0.2	0.500	0.500	0.500	0.500	0.703												

Table C.2: Classification accuracy for Radiometric and Noise perturbations.

Appendix D

A Proof of the Structure Theorem

Let K be a finite cubical complex and $\mathcal{I} : K \rightarrow \mathbb{R}$ be a face-monotone intensity function. A **compatible ordering** is a total ordering of the elementary cubes in K , denoted $\kappa_1, \kappa_2, \dots, \kappa_{|K|}$, satisfying

$$\mathcal{I}(\kappa_i) < \mathcal{I}(\kappa_j) \Rightarrow i < j \quad \text{and} \quad \kappa_i \preceq \kappa_j \Rightarrow i < j.$$

A compatible ordering defines a **refined filtration** $\mathbf{F}_\bullet^*$ of length $|K| + 1$ where $\mathbf{F}_i^* K = \{\kappa_1, \dots, \kappa_i\}$. This effectively performs a "time dilation", expanding the instantaneous transition between integer thresholds into a sequence of atomic steps. For each step $i \in [\mathcal{M}]$, the original sublevel set $\mathbf{F}_i$ coincides with the refined complex $\mathbf{F}_{t_i}^*$, where $t_i = \max \{r \mid \mathcal{I}(\kappa_r) \leq i\}$. By serialising the addition of cubes, we guarantee that topological events occur individually. Consequently, the topology changes by at most one Betti number per step.

Lemma D.1. *Let $\mathbf{F}_\bullet^*$ be a refined filtration. The addition of the k -cube κ_i in the transition $\mathbf{F}_{i-1}^* K \hookrightarrow \mathbf{F}_i^* K$ triggers exactly one of the following events:*

1. **Birth:** The dimension of $\mathbf{H}_k$ increases by 1 ($\beta_k \rightarrow \beta_k + 1$).
2. **Death:** The dimension of $\mathbf{H}_{k-1}$ decreases by 1 ($\beta_{k-1} \rightarrow \beta_{k-1} - 1$).

Proof. Let $F_j = \mathbf{F}_j^* K$. By topological validity, $\partial \kappa_i \in F_{i-1}$. Since $(\partial \circ \partial)(\kappa_i) = 0$, the boundary is a cycle: $\partial \kappa_i \in Z_{k-1}(F_{i-1})$. We distinguish two mutually exclusive cases:

Case 1 (Death): If $\partial \kappa_i \notin B_{k-1}(F_{i-1})$, then $\partial \kappa_i$ represents a non-trivial cycle in $\mathbf{H}_{k-1}(F_{i-1})$. In F_i , κ_i fills this cycle, meaning $[\partial \kappa_i] = 0$ in $\mathbf{H}_{k-1}(F_i)$. This annihilates exactly one homology class, reducing β_{k-1} by 1. No new k -cycles are created.

Case 2 (Birth): If $\partial \kappa_i \in B_{k-1}(F_{i-1})$, there exists a chain $c \in \mathbf{C}_k(F_{i-1})$ such that $\partial c = \partial \kappa_i$. Define the k -chain $z = \kappa_i - c$. Then $\partial z = \partial \kappa_i - \partial c = 0$, making z a cycle in F_i . Since $\kappa_i \notin F_{i-1}$, z is a new cycle. Furthermore, z cannot be a boundary in F_i because F_i contains no $(k+1)$ -cubes. Thus, z generates a new homology class, increasing β_k by 1. $\square$

Proof of the Structure Theorem. Let $\kappa_1, \dots, \kappa_N$ be a compatible ordering of K . We proceed by induction on the step t . The base case ($t = 0$) holds trivially for the empty complex.

For the inductive step, assume the homology at $t - 1$ is uniquely described by a valid multiset $\mathcal{B}_{t-1}$. We add the elementary cube κ_t . By Lemma D.1, this triggers exactly one topological event.

If a new class is born, we add a new interval starting at the current intensity

$$\mathcal{B}_t = \mathcal{B}_{t-1} \cup \{[\mathcal{I}(\kappa_t), \infty)\}.$$

If a class dies, we pair the destruction with an existing open interval using the Elder Rule. If the feature corresponding to $[b, \infty) \in \mathcal{B}_{t-1}$ is killed, we close the interval

$$\mathcal{B}_t = (\mathcal{B}_{t-1} \setminus \{[b, \infty)\}) \cup \{[b, \mathcal{I}(\kappa_t))\}.$$

Because each atomic step deterministically updates the multiset without pairing ambiguity, the unique interval decomposition follows by induction up to $t = N$. $\square$

Notation

Symbol	Definition	Symbol	Definition
General & Set Theory			
$A := B$	Define A by B	$ S $	The cardinality of a set S
$\emptyset$	The empty set	$\subseteq$	Subset inclusion
$\setminus$	Set difference	$f _A$	Restriction of function f to subset A
$\hookrightarrow$	Inclusion map	$\cong$	Isomorphism (algebraic or structural)
$\simeq$	Homotopy equivalence	$\sim$	Homotopic relation
$\circ$	Function composition	id_X	Identity map on set/space X
$[n]$	The set $\{0, 1, \dots, n-1\}$	$\mathbb{I}(\bullet)$	The Indicator function
$\mathcal{T}$	A metric-induced topology	$\mathbf{x}$	Vector notation
Fields & Metric Spaces			
$\mathbb{F}_2$	The field of two elements (bits)	$\mathbb{R}$	The field of real numbers
$\overline{\mathbb{R}}$	The extended real line	$\mathbb{R}^n$	n -dimensional Euclidean space
$\mathbb{Z}^n$	Integer lattice in n dimensions	d_X	Metric on space X
$\ \bullet\ _p$	ℓ_p norm ($p = 2$ or ∞)	$\mathcal{M}$	Maximum discrete intensity
$\mathcal{I}$	Discrete intensity function	$L_{\mathcal{I}}$	Discrete Graph Lipschitz constant
Linear Algebra			
$\text{span}_{\mathbb{F}} S$	Vector space over $\mathbb{F}$ spanned by S	V/W	Quotient vector space
$V \oplus W$	Direct sum of vector spaces	$\text{Im } \phi$	Image of a linear map ϕ
$\ker \phi$	Kernel of a linear map ϕ	$\dim V$	Dimension of vector space V
$\text{rank } M$	Rank of matrix M		
Graph Theory			
$G = (V, E)$	Graph with vertex set V and edge E	G_{∞}	Moore-based Graph (8-connectivity)
uv	Edge connecting vertices u and v	$\mathcal{G}[\mathcal{D}_1, \mathcal{D}_2]$	Diagram pairing graph
Cubical Geometry			
κ	An elementary cube (e.g., pixel)	$\dim \kappa$	Dimension of cube κ
$\preceq$	Face relation	$I = [k, k+1]$	Elementary interval (non-degenerate)
$I = [k]$	Elementary interval (degenerate)	$\delta_i^{\pm} \kappa$	Upper (+) or lower (−) i -th facet
$K(\Omega)$	Ambient cubical complex on Ω	K_k	The set of k -cubes in K
$(\Omega, \mathcal{I})$	Digital image with domain Ω	h	Mesh spacing (resolution)
$\hat{\mathcal{I}}$	Extended intensity function	$\tilde{\mathcal{I}}$	Inverted intensity function
Homology			
∂_k	Boundary operator on k -chains	$\mathbf{C}_k(K)$	k -th Chain group of K
$Z_k(K)$	k -th Cycle group ($\ker \partial_k$)	$B_k(K)$	k -th Boundary group ($\text{Im } \partial_{k+1}$)
$\mathbf{H}_k(K)$	k -th Homology group (Z_k/B_k)	β_k	k -th Betti number ($\dim \mathbf{H}_k$)
$[z]$	Homology class of cycle z		

Continued on next page

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Symbol	Definition	Symbol	Definition
Persistence			
$\mathbf{F}_\bullet$	A generic filtration	$\mathbf{F}_\bullet^*$	A refined filtration
$\mathbf{L}_\bullet$	Sublevel set filtration functor	$\mathbf{L}_\bullet^*$	Superlevel set filtration functor
$a_{i \rightarrow j}$	Structure map ($V_i \rightarrow V_j$)	$\mathbf{PH}_{i \rightarrow j}$	Persistent Homology Group
$\beta_{i \rightarrow j}$	Persistent Betti Number ($\dim \mathbf{PH}_{i \rightarrow j}$)	$\mathbf{Bar}(V_\bullet)$	Persistence barcode
$\mathcal{B}$	Alias for barcode / multiset	$\mathcal{D}(V_\bullet)$	Persistence Diagram
Δ	The diagonal set in a diagram	$I_\bullet^{b,d}$	Interval Module supported on $[b, d]$
$\mu(b, d)$	Multiplicity of interval $[b, d]$	$\pi(z)$	Persistence of feature z ($d - b$)
d_I	Interleaving distance	d_B	Bottleneck distance
Stability and Feature Extraction			
T_S	Sensor Operator (Radiometric)	T_A	Alignment Operator (Mechanical)
$\mathcal{I}_{aim}$	Aligned Image ($T_A \circ \mathcal{I}$)	$\mathcal{I}_{obs}$	Observed Realisation ($T_S \circ T_A \circ \mathcal{I}$)
ρ	Deterministic transfer function	η	Stochastic noise field
ϵ_{rad}	Radiometric noise bound	λ	Geometric displacement magnitude
Φ	Vectorisation map ($\mathbf{Bar} \rightarrow \mathbb{R}^n$)	Δ	Quantisation step size
L_Φ	Lipschitz constant of map	$N_{\max}$	Absolute combinatorial bound ($ K(\Omega) $)
S_k	Top- k Persistence Sum	τ	Truncation threshold
E_{tot}^τ	Truncated Total Persistence	$\mathbf{H}_0$	Sublevel topological feature
$\mathbf{H}_S$	Superlevel topological feature	$\mathbf{H}_1$	Sublevel loop feature
$\mathbf{v}_{\text{topo}}$	Combined topological vector ($\in \mathbb{R}^6$)	ϕ_i	Hu Invariant Moment
h_i	Log-transformed Hu Moment	$\mathbf{v}_{\text{geo}}$	Geometric feature vector ($\in \mathbb{R}^7$)
$\Delta(\epsilon)$	Robustness Gap at ϵ	θ	Rotation angle (perturbation)
Category Theory (Appendix)			
$\mathbf{Vect}_{\mathbb{F}}$	Category of vector spaces over $\mathbb{F}$	$\mathbf{CubeCx}$	Category of cubical complexes
$\mathbf{CubeCx}_{\leq}$	Order-preserving cubical complexes	$\mathbf{SimpCx}$	Category of simplicial complexes
$\mathbf{ChainCx}$	Category of chain complexes	$\mathbf{PersMod}$	Category of Persistence Modules
$\mathbf{CubeFilt}$	Filtered Cubical Complexes	$\mathbf{ChainFilt}$	Filtered Chain Complexes
$\mathbf{SimpFilt}$	Filtered Simplicial Complexes	$\mathbf{C}_\bullet^\square$	The Cubical Chain Functor
$\mathbf{C}_\bullet^\Delta$	The Simplicial Chain Functor	$\mathbf{C}_\bullet^{\text{filt}}$	The Filtered Chain Functor
$\mathcal{H}_k$	Algebraic Homology Functor	$\mathcal{H}_k^{\text{filt}}$	Filtered Algebraic Homology Functor
$\mathbf{PH}_k$	Persistent Homology Functor	$(\bullet)[k]$	The Shift Functor
$\mathbf{S}$	The Structure Morphism		
Simplicial Geometry (Appendix)			
σ	A geometric simplex	K^Δ	CFK Triangulation
$ K^\Delta $	Geometric realisation of K^Δ	$\sigma^\pm(\kappa)$	Simplicial decomp. of a 2-cube
$\mathbb{T}$	The Triangulation Map	$\mathbf{T}$	The Triangulation Functor

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